C. Create a secondary Amazon Elastic Block Store (Amazon EBS) volume on the EC2 instance for the data.

This option is incorrect because Amazon EBS is a block-level storage service that is designed for use with Amazon EC2 instances. It is not suitable for storing large amounts of data that need to be accessed by multiple EC2 instances, like in the case of the NFS-based file system on the on-premises SFTP server. Instead, you should use Amazon EFS, which is a fully managed, scalable, and distributed file system that can be accessed by multiple EC2 instances concurrently.

upvoted 9 times

Buruguduystunstugudunstuy 3 years, 5 months ago

D. Manually use an operating system copy command to push the data to the EC2 instance.

This option is not wrong, but it is not the most efficient or automated way to transfer the data from the on-premises SFTP server to the EC2 instance with the EFS file system. Manually transferring the data using an operating system copy command would require manual intervention and would not scale well for large amounts of data. It would also not provide the same level of performance and reliability as a fully managed service like AWS DataSync.

upvoted 8 times

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: AB

A. Launch the EC2 instance into the same Availability Zone as the EFS file system.
Makes sense to have the instance in the same AZ the EFS storage is.
B. Install an AWS DataSync agent in the on-premises data center.
The DataSync with move the data to the EFS, which already uses the EC2 instance (see the info provided). No more things are required...
C. Create a secondary Amazon Elastic Block Store (Amazon EBS) volume on the EC2 instance for the data.
This secondary EBS volume isn't required... the data should be move on to EFS...
D. Manually use an operating system copy command to push the data to the EC2 instance.
Potentially possible (instead of A), BUT the "automate this task" premise goes against any "manually" action. So, we should keep A.
E. Use AWS DataSync to create a suitable location configuration for the on-premises SFTP server.
I don't get the relationship between DataSync and the configuration for SFTP "on-prem"! Nonsense.
So, anwers are A&B

upvoted 62 times

Cizzla7049 3 years, 6 months ago

E is correct
https://aws.amazon.com/blogs/storage/migrating-storage-with-aws-datasync/

upvoted 5 times

happpieee 2 years, 3 months ago

Use AWS Transfer Family instead of DataSync for SFTP. So E seems incorrect.

When do I use AWS DataSync and when do I use AWS Transfer Family?

A: If you currently use SFTP to exchange data with third parties, AWS Transfer Family provides a fully managed SFTP, FTPS, FTP, and AS2 transfer directly into and out of Amazon S3, while reducing your operational burden.

If you want an accelerated and automated data transfer between NFS servers, SMB file shares, Hadoop clusters, self-managed or cloud object storage, AWS Snowcone, Amazon S3, Amazon EFS, and Amazon FSx, you can use AWS DataSync. DataSync is ideal for customers who need online migrations for active data sets, timely transfers for continuously generated data, or replication for business continuity.

upvoted 1 times

RBSK 3 years, 6 months ago

will A,B work without E?

upvoted 4 times

Lalo 3 years, 3 months ago

CORRECT ANSWER: B&E
Steps 4 &5
https://aws.amazon.com/datasync/getting-started/?nc1=h_ls

upvoted 16 times

attila9778 3 years, 6 months ago

Can someone explain why A is correct?
EFS is spread across Availability Zones in a region, as per https://aws.amazon.com/blogs/gametech/gearbox-entertainment-goes-remote-with-aws-and-perforce/
My question then is whether it makes sense to launch EC2 instances in the *same Availability Zone as the EFS file system* ?

upvoted 7 times

lovelazur 3 years, 2 months ago

However, launching the EC2 instance in the same AZ as the EFS file system can provide some performance benefits, such as reduced network latency and improved throughput. Therefore, it may be a best practice to launch the EC2 instance in the same AZ as the EFS file system if performance is a concern.

upvoted 2 times

BlueVolcano1 3 years, 4 months ago

Yes exactly, that's why A doesn't make sense. I voted for B and E.

upvoted 4 times

ln2718 Most Recent 9 months, 2 weeks ago

Selected Answer: BE

Revision on my previous answer...
B. Install an AWS DataSync agent in the on-premises data center.
Required for automated transfers from on-prem NFS to AWS.

E. Use AWS DataSync to create a suitable location configuration for the on-premises SFTP server. Configure the on-prem NFS storage (behind the SFTP server) as the source location and EFS as the destination, then run a DataSync task.

upvoted 1 times

ln2718 9 months, 4 weeks ago

Selected Answer: AB

Chat says A,B

upvoted 1 times

kaleido 12 months ago

Selected Answer: BE

I hate it when the questions are no longer about AWS and become more like a logic test lol. The EC2 instance with an EFS is "implied" to exist because of the way the question is posed, so the answer becomes B and E.

upvoted 1 times

Vandaman 1 year, 3 months ago

Selected Answer: BE

I was initially going with AB, but I questioned why would you need to launch an EC2 Instance for?

upvoted 2 times

satyaammm 1 year, 5 months ago

Selected Answer: BE

Since DataSync needs to me installed locally on on-premises and also we need DataSync to use SFTP.

upvoted 2 times

Gizmo2022 1 year, 5 months ago

I'm going to go with AB because AWS DataSync does not support SFTP location configuration.

upvoted 1 times

tom_cruise 1 year, 6 months ago

Selected Answer: BE

A is wrong because EFS is across AZ.

upvoted 1 times

0de7d1b 1 year, 6 months ago

Selected Answer: AB

A. Launch the EC2 instance into the same Availability Zone as the EFS file system:
Amazon EFS file systems are tied to specific Availability Zones (AZs) within a region, and for optimal performance and availability, the EC2 instance should be launched in the same AZ as the EFS file system. This ensures low-latency access to the EFS data.

B. Install an AWS DataSync agent in the on-premises data center:
AWS DataSync is a service that automates data transfer between on-premises storage and AWS. To transfer the data from the on-premises server to EFS, you would need to install the DataSync agent on your on-premises system. The agent securely handles data transfer, ensuring the process is efficient and reliable.

upvoted 2 times

ChymKuBoy 1 year, 7 months ago

Selected Answer: AB

A, B for sure

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: AB

Ans A,B - hint: "The server must be hosted on an Amazon EC2 instance that uses an Amazon Elastic File System (Amazon EFS) file system" then use DataSync on-prem

upvoted 2 times

XXXXXlNN 1 year, 9 months ago

option E is not valid because AWS DataSync does not support SFTP as a location configuration. Instead, you would need to use AWS Transfer Family for SFTP transfers.

upvoted 3 times

SaurabhTiwari1 1 year, 9 months ago

Selected Answer: BE

BE is correct

upvoted 1 times

KTEgghead 1 year, 10 months ago

Selected Answer: BE

let DataSync work it all out - AWS DataSync can create a suitable location configuration for an on-premises SFTP server.

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: BE

B and E is the most logical solution, launching the instance in the same AZ as the EFS is not cruicial, C and D negate the automation part and the par where it says that we need to use EFS.

upvoted 2 times

freedafeng 1 year, 11 months ago

I don't think E is correct. You create an EC2, and DataSync to migrate the NFS to EFS. That's it. You don't need to migrate anything on the on-presmise sftp server

upvoted 1 times

Question #103 *Topic 1*

A company has an AWS Glue extract, transform, and load (ETL) job that runs every day at the same time. The job processes XML data that is in an Amazon S3 bucket. New data is added to the S3 bucket every day. A solutions architect notices that AWS Glue is processing all the data during each run.
What should the solutions architect do to prevent AWS Glue from reprocessing old data?

A. Edit the job to use job bookmarks. **Most Voted**

B. Edit the job to delete data after the data is processed.

C. Edit the job by setting the NumberOfWorkers field to 1.

D. Use a FindMatches machine learning (ML) transform.

Correct Answer: A

Community vote distribution

A (100%)

Comments

123jhl0 Highly Voted 3 years, 1 month ago

Selected Answer: A

This is the purpose of bookmarks: "AWS Glue tracks data that has already been processed during a previous run of an ETL job by persisting state information from the job run. This persisted state information is called a job bookmark. Job bookmarks help AWS Glue maintain state information and prevent the reprocessing of old data."
https://docs.aws.amazon.com/glue/latest/dg/monitor-continuations.html

upvoted 51 times

cookieMr Highly Voted 2 years, 5 months ago

Selected Answer: A

A. Job bookmarks in Glue allow you to track the last-processed data in a job. By enabling job bookmarks, Glue keeps track of the processed data anc automatically resumes processing from where it left off in subsequent job runs.

B. Results in the permanent removal of the data from the S3, making it unavailable for future job runs. This is not desirable if the data needs to be retained or used for subsequent analysis.

C.It would only affect the parallelism of the job but would not address the issue of reprocessing old data. It does not provide a mechanism to track the processed data or skip already processed data.

D. It is not directly related to preventing Glue from reprocessing old data. The FindMatches transform is used for identifying and matching duplicate or matching records in a dataset. While it can be used in data processing pipelines, it does not address the specific requirement of avoiding reprocessing old data in this scenario.

upvoted 12 times

awsgeek75 Most Recent 1 year, 11 months ago

Selected Answer: A

B: Glue can delete DataSet but this option is too vague to consider or too open to mean anything
C: Won't help with repeated ETL. This property affects parallelism
D: Too vague

upvoted 1 times

Ruffyit 2 years, 1 month ago

https://docs.aws.amazon.com/glue/latest/dg/monitor-continuations.html

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

The best solution is to edit the AWS Glue job to use job bookmarks.

Job bookmarks allow AWS Glue ETL jobs to track which data has already been processed during previous runs. This prevents reprocessing of old data.

Deleting the data after processing would cause the data to be lost and unavailable for future processing. Reducing the number of workers may improve performance but does not prevent reprocessing of old data. Using a FindMatches ML transform is used for record matching, not preventing reprocessing.

So the solutions architect should enable job bookmarks in the AWS Glue job configuration. This will allow the ETL job to keep track of processed data and only transform the new data added since the last run.

upvoted 1 times

bedwal2020 2 years, 7 months ago

Selected Answer: A

Job bookmark to make sure that the glue job will not process already processed files.

upvoted 1 times

Heric 2 years, 7 months ago

Selected Answer: A

Job bookmarks are used in AWS Glue ETL jobs to keep track of the data that has already been processed in a previous job run. With bookmarks enabled, AWS Glue will read the bookmark information from the previous job run and will only process the new data that has been added to the data source since the last job run. This saves time and reduces costs by eliminating the need to reprocess old data.

Therefore, a solutions architect should edit the AWS Glue ETL job to use job bookmarks so that it will only process new data added to the S3 bucket since the last job run.

upvoted 2 times

linux_admin 2 years, 8 months ago

Selected Answer: A

Job bookmarks enable AWS Glue to track the data that has been processed in a previous run of the job. With job bookmarks enabled, AWS Glue will only process new data that has been added to the S3 bucket since the previous run of the job, rather than reprocessing all data every time the job runs.

upvoted 2 times

gustavtd 2 years, 11 months ago

Delete files in S3 freely is not good. so B is not correct,

upvoted 1 times

techhb 2 years, 11 months ago

Selected Answer: A

A is correct

upvoted 1 times

Buruguduystunstugudunstuy 2 years, 11 months ago

Selected Answer: A

Option A. Edit the job to use job bookmarks.

Job bookmarks in AWS Glue allow the ETL job to track the data that has been processed and to skip data that has already been processed. This can prevent AWS Glue from reprocessing old data and can improve the performance of the ETL job by only processing new data. To use job bookmarks, the solutions architect can edit the job and set the "Use job bookmark" option to "True". The ETL job will then use the job bookmark to track the data that has been processed and skip data that has already been processed in subsequent runs.

upvoted 3 times

career360guru 2 years, 11 months ago

Selected Answer: A

Option A

upvoted 1 times

SilentMilli 2 years, 12 months ago

Selected Answer: A

It's obviously A. Bookmarks serve this purpose

upvoted 1 times

Wpcorgan 3 years ago

A is correct

upvoted 2 times

LeGloupier 3 years, 1 month ago

Selected Answer: A

A
https://docs.aws.amazon.com/glue/latest/dg/monitor-continuations.html
upvoted 3 times

Question #104 *Topic 1*

A solutions architect must design a highly available infrastructure for a website. The website is powered by Windows web servers that run on Amazon EC2 instances. The solutions architect must implement a solution that can mitigate a large-scale DDoS attack that originates from thousands of IP addresses. Downtime is not acceptable for the website.
Which actions should the solutions architect take to protect the website from such an attack? (Choose two.)

A. Use AWS Shield Advanced to stop the DDoS attack. **Most Voted**

B. Configure Amazon GuardDuty to automatically block the attackers.

C. Configure the website to use Amazon CloudFront for both static and dynamic content. **Most Voted**

D. Use an AWS Lambda function to automatically add attacker IP addresses to VPC network ACLs.

E. Use EC2 Spot Instances in an Auto Scaling group with a target tracking scaling policy that is set to 80% CPU utilization.

Correct Answer: AC

Community vote distribution

AC (89%) 6%

Comments

alvarez100 Highly Voted 3 years, 7 months ago

Selected Answer: AC

I think it is AC, reason is they require a solution that is highly available. AWS Shield can handle the DDoS attacks. To make the solution HA you can use cloud front. AC seems to be the best answer imo.
AB seem like redundant answers. How do those answers make the solution HA?

upvoted 26 times

attila9778 3 years, 6 months ago

A - AWS Shield Advanced
C - (protecting this option) IMO: AWS Shield Advanced has to be attached. But it can not be attached directly to EC2 instances.
According to the docs: https://aws.amazon.com/shield/
It requires to be attached to services such as CloudFront, Route 53, Global Accelerator, ELB or (in the most direct way using) Elastic IP (attached to the EC2 instance)

upvoted 34 times

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago

Selected Answer: AC

Option A. Use AWS Shield Advanced to stop the DDoS attack.

It provides always-on protection for Amazon EC2 instances, Elastic Load Balancers, and Amazon Route 53 resources. By using AWS Shield Advanced, the solutions architect can help protect the website from large-scale DDoS attacks.

Option C. Configure the website to use Amazon CloudFront for both static and dynamic content.

CloudFront is a content delivery network (CDN) that integrates with other Amazon Web Services products, such as Amazon S3 and Amazon EC2, to deliver content to users with low latency and high data transfer speeds. By using CloudFront, the solutions architect can distribute the website's content across multiple edge locations, which can help absorb the impact of a DDoS attack and reduce the risk of downtime for the website.

upvoted 23 times

satyaammm Most Recent 1 year, 5 months ago

Selected Answer: AC

AWS Shield Advanced protects form DDos attacks while CloudFront deals with the downtime.

upvoted 1 times

XXXXXlNN 1 year, 9 months ago

Note great options for us to select but AC seem make more sense comparing to others

upvoted 1 times

KTEgghead 1 year, 10 months ago

Selected Answer: AC

CoPilot - "No, you do not need Amazon CloudFront to implement AWS Shield Advanced. AWS Shield Advanced provides protection for several AWS services, including Amazon EC2, Elastic Load Balancing (ELB), AWS Global Accelerator, and Amazon Route 53 resources, in addition to CloudFront distributions1. It's designed to offer more sophisticated protection against Distributed Denial of Service (DDoS) attacks, regardless of the AWS servic being used1. However, it's important to note that while CloudFront is not a requirement, using AWS Shield Advanced with CloudFront can enhance your application's security by providing additional DDoS protection."

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: AC

A and C is the most logical combination, we implement cloudfront so we can use shield advanced. Both of these options mitigate the impact of a DDOS attack.

upvoted 2 times

jatric 1 year, 11 months ago

Selected Answer: AC

AC is more close to meet the requirenment

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: AC

A: For DDoS attakcs
C: For scalable available site

B: Irrelevant
D: How would Lambda identify the attacker IP even if this was possible (ACL has a limit of 40 rules each way)
E: Scaling is not an issue here

upvoted 5 times

xdkonorek2 2 years, 7 months ago

Selected Answer: AC

A - use aws shield advanced for DDoS protection, but it cannot be used with EC2 instace if it's not using EIP, which is not mentioned
C - but it can be used with cloudfront distribution
thus AC is the answer

upvoted 3 times

Ruffyit 2 years, 7 months ago

DDoS attack will choose the AWS Shield Advanced
Cloudfront have attached the WAF

upvoted 2 times

Devsin2000 2 years, 8 months ago

Selected Answer: AE

A - no brainer
E = "must design a highly available infrastructure". I am not sure if CloudFront addresses this requirement.

upvoted 1 times

pentium75 2 years, 5 months ago

Is CloudFront not HA? Answer E uses Spot instances which might be unavailable, thus are NEVER an option for HA.

upvoted 4 times

sidharthwader 2 years, 3 months ago

You are right if it was On demand instances we could think of E

upvoted 2 times

LoXoL 2 years, 5 months ago

pentium75 is right.

upvoted 1 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: AC

Mitigate a large-scale DDoS attack = AWS Shield Advanced
Downtime is not acceptable for the website = high availability = Amazon CloudFront

upvoted 3 times

mtmayer 2 years, 9 months ago

Selected Answer: D

yeah , AWS Shield Advanced can be used directly on EC2.....
https://docs.aws.amazon.com/waf/latest/developerguide/ddos-protections-by-resource-type.html
upvoted 1 times

pentium75 2 years, 5 months ago
Why D then?
upvoted 1 times

Guru4Cloud 2 years, 10 months ago
Selected Answer: AC
Cloud front supports SHIELD ADVANCED integration
upvoted 3 times

diabloexodia 2 years, 11 months ago
Cloud front supports SHIELD ADVANCED integration
upvoted 2 times

Aash24 2 years, 11 months ago
Selected Answer: D
D should be the one here
upvoted 3 times

pentium75 2 years, 5 months ago
"Use an AWS Lambda function to automatically add attacker IP addresses to VPC network ACLs"?????
upvoted 2 times

cookieMr 2 years, 11 months ago
Selected Answer: AC
A. AWS Shield Advanced provides advanced DDoS protection for AWS resources, including EC2. It includes features such as real-time threat intelligence, automatic protection, and DDoS cost protection.

C. CloudFront is a CDN service that can help mitigate DDoS attacks. By routing traffic through CloudFront, requests to the website are distributed across multiple edge locations, which can absorb and mitigate DDoS attacks more effectively. CloudFront also provides additional DDoS protection features, such as rate limiting, SSL/TLS termination, and custom security policies.

B. While GuardDuty can detect and provide insights into potential malicious activity, it is not specifically designed for DDoS mitigation.

D. Network ACLs are not designed to handle high-volume traffic or DDoS attacks efficiently.

E. Spot Instances are a cost optimization strategy and may not provide the necessary availability and protection against DDoS attacks compared to using dedicated instances with DDoS protection mechanisms like Shield Advanced and CloudFront.
upvoted 4 times

Question #105 *Topic 1*

A company is preparing to deploy a new serverless workload. A solutions architect must use the principle of least privilege to configure permissions that will be used to run an AWS Lambda function. An Amazon EventBridge (Amazon CloudWatch Events) rule will invoke the function.
Which solution meets these requirements?

A. Add an execution role to the function with lambda:InvokeFunction as the action and * as the principal.

B. Add an execution role to the function with lambda:InvokeFunction as the action and Service: lambda.amazonaws.com as the principal.

C. Add a resource-based policy to the function with lambda:* as the action and Service: events.amazonaws.com as the principal.

D. Add a resource-based policy to the function with lambda:InvokeFunction as the action and Service: events.amazonaws.com as the principal. **Most Voted**

Correct Answer: D

Community vote distribution

D (97%) 3%

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: D

Best way to check it... The question is taken from the example shown here in the documentation:
https://docs.aws.amazon.com/eventbridge/latest/userguide/eb-use-resource-based.html#eb-lambda-permissions

upvoted 38 times

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago

Selected Answer: D

The correct solution is D. Add a resource-based policy to the function with lambda:InvokeFunction as the action and Service: events.amazonaws.com as the principal.

The principle of least privilege requires that permissions are granted only to the minimum necessary to perform a task. In this case, the Lambda function needs to be able to be invoked by Amazon EventBridge (Amazon CloudWatch Events). To meet these requirements, you can add a resource-based policy to the function that allows the InvokeFunction action to be performed by the Service: events.amazonaws.com principal. This will allow Amazon EventBridge to invoke the function, but will not grant any additional permissions to the function.

upvoted 30 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Why other options are wrong

Option A is incorrect because it grants the lambda:InvokeFunction action to any principal (*), which would allow any entity to invoke the function and goes beyond the minimum permissions needed.

Option B is incorrect because it grants the lambda:InvokeFunction action to the Service: lambda.amazonaws.com principal, which would allow any Lambda function to invoke the function and goes beyond the minimum permissions needed.

Option C is incorrect because it grants the lambda:* action to the Service: events.amazonaws.com principal, which would allow Amazon EventBridge to perform any action on the function and goes beyond the minimum permissions needed.

upvoted 23 times

huaze_lei Most Recent 1 year, 9 months ago

Selected Answer: D

Following the principle of least privilege, you should not grant Events the * privilege. Just enough to perform its job will do.

Also, you need a resource-based policy to attach to the function, for Events to be able to execute the function

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: D

D is the correct answer

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

This is a good example article with nice learning material.
https://docs.aws.amazon.com/eventbridge/latest/userguide/eb-run-lambda-schedule.html

upvoted 2 times

chasingsummer 2 years, 6 months ago

Selected Answer: D

Good explanation from ChatGPT:

In order to adhere to the principle of least privilege when configuring permissions for an AWS Lambda function invoked by an Amazon EventBridge (CloudWatch Events) rule, the most appropriate solution would be:

D. Add a resource-based policy to the function with lambda:InvokeFunction as the action and Service: events.amazonaws.com as the principal.

This solution involves attaching a resource-based policy to the Lambda function. It specifies that the only entity allowed to invoke the Lambda function is the Amazon EventBridge service (represented by the principal events.amazonaws.com) and restricts the action to only invoking the function (lambda:InvokeFunction). This aligns with the principle of least privilege by granting the necessary permissions explicitly to the service that needs them, without providing overly permissive access.

upvoted 3 times

MiniYang 2 years, 6 months ago

Selected Answer: B

Is anyone can explain why B is can't be a good choice? The option adds the execution role to the function, with lambda:InvokeFunction as the action and Service: lambda.amazonaws.com as the body. This restricts the Lambda function to only the Lambda service, providing an effective layer of security. and fully complies with the principle of least privilege

upvoted 3 times

pentium75 2 years, 5 months ago

Because the question is about the permission 'to run the function' (permission for the administrator to invoke it), while B is about execution permissions (permission for the function to access resources).

upvoted 1 times

Evonne_HY 2 years, 8 months ago

why not choose B, an execution role is attached to lambda and a policy is attached to an execution role

upvoted 1 times

Georgeyp 2 years, 8 months ago

B would be the wrong choice as the both roles are granted to lambda, however the question requires Eventbridge to call the Lambda function.

upvoted 1 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: D

lambda:InvokeFunction is the action needed to invoke the Lambda function.
Service: events.amazonaws.com is the principal (the AWS service) that is allowed to invoke the Lambda function. In this case, you're explicitly allowinc CloudWatch Events to invoke the function.

upvoted 2 times

MNotABot 2 years, 11 months ago

D
* is BIG NO. And we are talking about policy --> hence D

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: D

In this solution, a resource-based policy is added to the Lambda function, which allows the specified principal (events.amazonaws.com) to invoke the function. The lambda:InvokeFunction action provides the necessary permission for the Amazon EventBridge rule to trigger the Lambda function.

Option A is incorrect because it assigns the lambda:InvokeFunction action to all principals (*), which grants permission to invoke the function to any entity, which is broader than necessary.

Option B is incorrect because it assigns the lambda:InvokeFunction action to the specific principal "lambda.amazonaws.com," which is the service principal for AWS Lambda. However, the requirement is for the EventBridge service principal to invoke the function.

Option C is incorrect because it assigns the lambda:* action to the specific principal "events.amazonaws.com," which is the service principal for Amazon EventBridge. However, it grants broader permissions than necessary, allowing any Lambda function action, not just lambda:InvokeFunction.
upvoted 4 times

Abrar2022 3 years ago
Option C is incorrect, the reason is that, firstly, lambda:* allows Amazon EventBridge to perform any action on the function and this is beyond the minimum permissions needed.
upvoted 2 times

Rahulbit34 3 years, 1 month ago
Since its for Lamda which is a resource, resource policy is the trick
upvoted 2 times

bdp123 3 years, 4 months ago
Selected Answer: D
https://docs.aws.amazon.com/eventbridge/latest/userguide/resource-based-policies-eventbridge.html#lambda-permissions
upvoted 1 times

gustavtd 3 years, 5 months ago
Selected Answer: D
The definition scope of D is the smallest, so is it
upvoted 1 times

techhb 3 years, 5 months ago
Selected Answer: D
events.amazonaws.com is principal for eventbridge
upvoted 1 times

career360guru 3 years, 5 months ago
Selected Answer: D
Option D
upvoted 1 times

Question #106 Topic 1

A company is preparing to store confidential data in Amazon S3. For compliance reasons, the data must be encrypted at rest. Encryption key usage must be logged for auditing purposes. Keys must be rotated every year.
Which solution meets these requirements and is the MOST operationally efficient?

A. Server-side encryption with customer-provided keys (SSE-C)

B. Server-side encryption with Amazon S3 managed keys (SSE-S3)

C. Server-side encryption with AWS KMS keys (SSE-KMS) with manual rotation

D. Server-side encryption with AWS KMS keys (SSE-KMS) with automatic rotation Most Voted

Correct Answer: D

Community vote distribution

D (93%) 7%

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: D

The MOST operationally efficient one is D.
Automating the key rotation is the most efficient.
Just to confirm, the A and B options don't allow automate the rotation as explained here:
https://aws.amazon.com/kms/faqs/#:~:text=You%20can%20choose%20to%20have%20AWS%20KMS%20automatically%20rotate%20KMS,KMS%20cu
stom%20key%20store%20feature

upvoted 22 times

vadiminski_a 3 years, 5 months ago

In addition you cannot log key usage in B, for A I am not certain

upvoted 2 times

ocbn3wby 3 years, 6 months ago

Thank you for the explanation.

upvoted 1 times

cookieMr Highly Voted 2 years, 11 months ago

Selected Answer: D

SSE-KMS provides a secure and efficient way to encrypt data at rest in S3. SSE-KMS uses KMS to manage the encryption keys securely. With SSE-KMS, encryption keys can be automatically rotated using KMS key rotation feature, which simplifies the key management process and ensures compliance with the requirement to rotate keys every year.

Additionally, SSE-KMS provides built-in audit logging for encryption key usage through CloudTrail, which captures API calls related to the management and usage of KMS keys. This meets the requirement for logging key usage for auditing purposes.

Option A (SSE-C) requires customers to provide their own encryption keys, but it does not provide key rotation or built-in logging of key usage.
Option B (SSE-S3) uses Amazon S3 managed keys for encryption, which simplifies key management but does not provide key rotation or detailed key usage logging.
Option C (SSE-KMS with manual rotation) uses AWS KMS keys but requires manual rotation, which is less operationally efficient than the automatic key rotation available with option D.

upvoted 10 times

satyaammm Most Recent 1 year, 5 months ago

Selected Answer: D

Automating the key rotation is the most important here and hence D is the most suitable option here.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans D - just the Amazon provided service with key automatic key rotation
upvoted 1 times

MehulKapadia 2 years, 2 months ago

Selected Answer: D

Correct Answer: D
Automatic Key Rotation = KMS, hence Option A & B are not correct answer.
Hence Possible answer is Option C or D. Now mentioned in the requirement that key rotation solution must be automated. So Option C is not the correct answer.
Correct Answer: D - SSE with KMS which support automatic key rotation.
upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

I'll go for D as SSS-S3 has unpublished scheduled of rotation which may or may not be "each year".
https://docs.aws.amazon.com/AmazonS3/latest/userguide/UsingKMSEncryption.html
upvoted 2 times

rcptryk 2 years, 6 months ago

Selected Answer: B

SSE-S3 can be used for logging in cloudtrail since January 5, 2023
https://docs.aws.amazon.com/AmazonS3/latest/userguide/UsingServerSideEncryption.html
upvoted 2 times

pentium75 2 years, 5 months ago

But "keys must be rotated every year". I understand that SSE-S3 rotates the keys "regularly" but you have no influence on the schedule.
upvoted 2 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: D

The correct answer is D. Server-side encryption with AWS KMS keys (SSE-KMS) with automatic rotation.

SSE-KMS is the most secure way to encrypt data in Amazon S3. It uses AWS KMS, which is a highly secure key management service that is managed by AWS. AWS KMS logs all key usage, so the company can meet its compliance requirements. AWS KMS also rotates keys automatically, so the company does not have to worry about manually rotating keys.
upvoted 4 times

SilentMilli 3 years, 5 months ago

Selected Answer: D

Server-side encryption with AWS KMS keys (SSE-KMS) with automatic rotation meets the requirements and is the most operationally efficient solution. This option allows you to use AWS KMS to automatically rotate the keys every year, which simplifies key management. In addition, key usage is logged for auditing purposes, and the data is encrypted at rest to meet compliance requirements.
upvoted 3 times

Zerotn3 3 years, 5 months ago

Selected Answer: B

mazon API Gateway is a fully managed service that makes it easy to create, publish, maintain, monitor, and secure APIs at any scale. You can use API Gateway to create a REST API that exposes the location data as an API endpoint, allowing you to access the data from your analytics platform.

AWS Lambda is a serverless compute service that lets you run code in response to events or HTTP requests. You can use Lambda to write the code that retrieves the location data from your data store and returns it to API Gateway as a response to API requests. This allows you to scale the API to handle a large number of requests without the need to provision or manage any infrastructure.
upvoted 2 times

pentium75 2 years, 5 months ago

This question is about server-side encryption, not API Gateway
upvoted 1 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: D

The most operationally efficient solution that meets the requirements listed would be option D: Server-side encryption with AWS KMS keys (SSE-KMS) with automatic rotation.

SSE-KMS allows you to use keys that are managed by the AWS Key Management Service (KMS) to encrypt your data at rest. KMS is a fully managed service that makes it easy to create and control the encryption keys used to encrypt your data. With automatic key rotation enabled, KMS will automatically create a new key for you on a regular basis, typically every year, and use it to encrypt your data. This simplifies the key rotation process and reduces the operational burden on your team.

In addition, SSE-KMS provides logging of key usage through AWS CloudTrail, which can be used for auditing purposes.
upvoted 3 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Why other options are wrong

Option A: Server-side encryption with customer-provided keys (SSE-C) would require you to manage the encryption keys yourself, which can be more operationally burdensome.

Option B: Server-side encryption with Amazon S3 managed keys (SSE-S3) does not allow for key rotation or logging of the key usage.

Option C: Server-side encryption with AWS KMS keys (SSE-KMS) with manual rotation would require you to manually initiate the key rotation process, which can be more operationally burdensome compared to automatic rotation.

upvoted 4 times

career360guru 3 years, 5 months ago

Selected Answer: D

Option D

upvoted 1 times

Berny 3 years, 5 months ago

You can choose to have AWS KMS automatically rotate KMS keys every year, provided that those keys were generated within AWS KMS HSMs. Automatic key rotation is not supported for imported keys, asymmetric keys, or keys generated in a CloudHSM cluster using the AWS KMS custom key store feature. If you choose to import keys to AWS KMS or asymmetric keys or use a custom key store, you can manually rotate them by creating a new KMS key and mapping an existing key alias from the old KMS key to the new KMS key.

upvoted 2 times

PavelTech 3 years, 6 months ago

Can anybody correct me if I'm wrong, KMS does not offer automatic rotations but SSE-KMS only allows automatic rotation once in 3 years thus if we want rotation every year we need to rotate it manually?

upvoted 2 times

JayBee65 3 years, 5 months ago

You're wrong :) "All AWS managed keys are automatically rotated every year. You cannot change this rotation schedule."
https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#customer-cmk

upvoted 2 times

PS_R 3 years, 7 months ago

Selected Answer: D

Agree Also, SSE-S3 cannot be audited.

upvoted 2 times

Question #107 Topic 1

A bicycle sharing company is developing a multi-tier architecture to track the location of its bicycles during peak operating hours. The company wants to use these data points in its existing analytics platform. A solutions architect must determine the most viable multi-tier option to support this architecture. The data points must be accessible from the REST API.
Which action meets these requirements for storing and retrieving location data?

A. Use Amazon Athena with Amazon S3.

B. Use Amazon API Gateway with AWS Lambda. **Most Voted**

C. Use Amazon QuickSight with Amazon Redshift.

D. Use Amazon API Gateway with Amazon Kinesis Data Analytics.

Correct Answer: B

Community vote distribution

B (48%) D (47%) 6%

Comments

ArielSchivo Highly Voted 3 years, 7 months ago

Selected Answer: B

API Gateway is needed to get the data so option A and C are out.
"The company wants to use these data points in its existing analytics platform" so there is no need to add Kynesis. Option D is also out.
This leaves us with option B as the correct one.

upvoted 111 times

alfonso_ciampa 2 years, 11 months ago

You are right, but it clearly say "store data".
AWS Lambda don't store data, Kinesis could.

upvoted 12 times

FlyingHawk 1 year, 4 months ago

AWS Lambda provides a serverless way to process incoming requests and interact with a storage backend.
This solution can store location data in a service like Amazon DynamoDB or Amazon S3 and use Lambda functions to retrieve and serve this data through the API Gateway. This is cost-effective, scalable, and straightforward for real-time queries.

upvoted 2 times

ces26015 3 years, 4 months ago

i dont understand the use of a lambda function here, maybe if there would be need to transform the data, can you explain?

upvoted 6 times

Tsige 1 year, 7 months ago

While API Gateway and Lambda can work together for processing requests, this setup doesn't efficiently handle continuous real-time data streaming and analysis, which is required for tracking the bicycle locations. Lambda is stateless and is better suited for on-demand execution, not for real-time analytics of streaming data. So, the correct option is D

upvoted 10 times

MutiverseAgent 2 years, 11 months ago

B might work but D works better. B requieres API gateway + lambda for data input & output, whereas D is a broader solution, as Kinesis Data Analytics APIs can be used to extract and process data better that API Gateway + Lambdas. Also, Kinesis is highly recommended for telemetry data which is the question scenario. @See Kinesys flexible API (https://aws.amazon.com/documentation-overview/kinesis-data-analytics/)

upvoted 12 times

MutiverseAgent 2 years, 11 months ago

Also by using kinesis the analytics platform will have a storing buffer to take & process data through the kinesys API. The lambda aproach in the B scenario is to wide and leaves many loose ends.

upvoted 5 times

Six_Fingered_Jose Highly Voted 3 years, 7 months ago

Selected Answer: D

I dont understand why you will vote B?
how are you going to store data with just lambda?
> Which action meets these requirements for storing and retrieving location data

In this use case there will obviously be a ton of data and you want to get real-time location data of the bicycles, and to analyze all these info kinesis i the one that makes most sense here.

upvoted 82 times

a070112 3 years, 5 months ago

Lambda isn't storing the data themselves. It's triggering the data store to the company's "existing data analytics platform"

upvoted 10 times

kmliuy73 3 years, 5 months ago

Real-time analytics on Kinesis Data Streams & Firehose using SQL, not store db ...

upvoted 5 times

Six_Fingered_Jose 3 years, 7 months ago

https://aws.amazon.com/blogs/aws/real-time-hotspot-detection-in-amazon-kinesis-analytics/

upvoted 5 times

UWSFish 3 years, 7 months ago

I don't think you need to worry about storing data. The question states their is an existing platform.

upvoted 4 times

Tim2026 Most Recent 1 month, 2 weeks ago

Selected Answer: B

You need a REST API + real-time processing of location data → API Gateway exposes the REST endpoints, Lambda handles ingestion/logic.

upvoted 1 times

TEC65 1 year, 1 month ago

Selected Answer: D

better for streaming data

upvoted 1 times

Vivobook11 1 year, 2 months ago

Selected Answer: D

D. Use Amazon API Gateway with Amazon Kinesis Data Analytics.
Why?
Real-time Data Processing: Amazon Kinesis Data Analytics can process streaming data (e.g., bicycle locations) in real-time.

REST API Access: Amazon API Gateway can provide a secure REST API for accessing the processed data.

Scalability & Performance: Kinesis efficiently ingests large volumes of data during peak hours.

Integration with Analytics: The processed data can be stored in Amazon S3, DynamoDB, or Redshift for further analytics.

upvoted 2 times

Faraz999 1 year, 2 months ago

Selected Answer: B

API Gateway is needed to get the data so option A and C are out.
"The company wants to use these data points in its existing analytics platform" so there is no need to add Kynesis. Option D is also out.
This leaves us with option B as the correct one.

upvoted 1 times

CloudExpert01 1 year, 2 months ago

Selected Answer: B

when choosing between B and D:
"The company wants to use these data points in its existing analytics platform." so need for Lambda to integrate data points in its existing analytics platform

upvoted 2 times

Faraz999 1 year, 2 months ago

Selected Answer: B

API Gateway is needed to get the data so option A and C are out.
"The company wants to use these data points in its existing analytics platform" so there is no need to add Kynesis. Option D is also out.
This leaves us with option B as the correct one.

upvoted 1 times

Faraz999 1 year, 2 months ago

Selected Answer: B

API Gateway is needed to get the data so option A and C are out.
"The company wants to use these data points in its existing analytics platform" so there is no need to add Kynesis. Option D is also out.
This leaves us with option B as the correct one.

upvoted 2 times

Saloniip 1 year, 3 months ago

Selected Answer: B

lambda can store data in db which is not mentioned but can be implemented, wherein kinesis data analytics would be more complex to implement just for simple location tracking. Also, a data analytics tool is already there. Hence, B is more suitable.

upvoted 2 times

tch 1 year, 4 months ago

Selected Answer: D

Amazon Kinesis Data Analytics allows you to build more complex analytics applications that support flexible processing choices and robust fault-tolerance with exactly-once processing without duplicates, and analytics that can be performed over an entire data stream across multiple logical partitions. With KDA, you can analyze data over multiple types of aggregation windows (tumbling window, stagger window, sliding window, session window) using either the event time or the processing tim

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: D

Lambda is not most viable, or real time

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: B

Given that the company already has an existing analytics platform and needs a solution for storing and retrieving location data that can be accessed through a REST API, the best choice is:

B. Use Amazon API Gateway with AWS Lambda

upvoted 3 times

FlyingHawk 1 year, 4 months ago

D. Amazon API Gateway with Amazon Kinesis Data Analytics: Kinesis Data Analytics is for continuous real-time stream processing. If the requiremen focuses only on storing and retrieving (not streaming) data, this adds unnecessary complexity.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: D

Kinesis is specifically designed for telemetry data, making it a better fit for tracking bicycle locations.

upvoted 3 times

FlyingHawk 1 year, 4 months ago

I changed my mind to B now as there is existing analytic platform already, the data will be stored in the existing analytic platform, so you can use Lambda to process and store data in the existing analytic platform.

upvoted 1 times

Rcosmos 1 year, 5 months ago

Selected Answer: D

B. Amazon API Gateway com AWS Lambda:

Essa combinação é útil para fornecer APIs REST, mas o Lambda não é otimizado para ingestão contínua de dados de localização em tempo real. Ele seria mais adequado para processos eventuais ou tarefas específicas.

upvoted 2 times

hashepsut 1 year, 5 months ago

Selected Answer: B

the best answer is B: Use Amazon API Gateway with AWS Lambda

Reasoning:

API Gateway provides the required REST API endpoint
Lambda can handle real-time processing of location data
This combination is cost-effective and scalable

Can easily integrate with other AWS services for analytics
Provides the necessary multi-tier architecture (API Layer, Processing Layer, Storage Layer)

upvoted 1 times

EllenLiu 1 year, 5 months ago

Selected Answer: D

A common use is the real-time aggregation of data followed by loading the aggregate data into a data warehouse or map-reduce cluster.
https://docs.aws.amazon.com/streams/latest/dev/introduction.html#using-the-service

upvoted 1 times

Question #108 *Topic 1*

A company has an automobile sales website that stores its listings in a database on Amazon RDS. When an automobile is sold, the listing needs to be removed from the website and the data must be sent to multiple target systems.
Which design should a solutions architect recommend?

A. Create an AWS Lambda function triggered when the database on Amazon RDS is updated to send the information to an Amazon Simple Queue Service (Amazon SQS) queue for the targets to consume. **Most Voted**

B. Create an AWS Lambda function triggered when the database on Amazon RDS is updated to send the information to an Amazon Simple Queue Service (Amazon SQS) FIFO queue for the targets to consume.

C. Subscribe to an RDS event notification and send an Amazon Simple Queue Service (Amazon SQS) queue fanned out to multiple Amazon Simple Notification Service (Amazon SNS) topics. Use AWS Lambda functions to update the targets.

D. Subscribe to an RDS event notification and send an Amazon Simple Notification Service (Amazon SNS) topic fanned out to multiple Amazon Simple Queue Service (Amazon SQS) queues. Use AWS Lambda functions to update the targets.

Correct Answer: A

Community vote distribution

A (50%) D (46%) 2%

Comments

romko Highly Voted 3 years, 6 months ago

Selected Answer: A

Interesting point that Amazon RDS event notification doesn't support any notification when data inside DB is updated.
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_Events.overview.html
So subscription to RDS events doesn't give any value for Fanout = SNS => SQS

B is out because FIFO is not required here.

A is left as correct answer

upvoted 91 times

Jiang_aws1 3 years, 6 months ago

D is connect
RDS event notification by RDS stream or advance audit DML so it is possible

upvoted 2 times

Jiang_aws1 3 years, 6 months ago

The key is "Fanned out" due to "Multiple target systems" need to update

upvoted 4 times

JayBee65 3 years, 5 months ago

Please provide reference for this claim: " event notification by RDS stream or advance audit DML"

upvoted 4 times

ruqui 3 years ago

I don't think A is a valid solution ... how do you send the data to multiple targets using a single SQS?

upvoted 22 times

Vic_d_gr8 3 years, 6 months ago

Amazon RDS uses the SNS to provide notification when an Amazon event occurs
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_Events.html

upvoted 4 times

studynoplay 3 years, 1 month ago

Wow, great find romko. Didn't realize that Event notification doesn't notify when the data is changed, it notifies when major changes at DB level occur like settings etc

upvoted 2 times

ksolovyov Highly Voted 3 years, 5 months ago

Selected Answer: A

RDS events only provide operational events such as DB instance events, DB parameter group events, DB security group events, and DB snapshot events. What we need in the scenario is to capture data-modifying events (INSERT, DELETE, UPDATE) which can be achieved thru native functions or stored procedures.

upvoted 15 times

BlueVolcano1 3 years, 4 months ago

I agree with it requiring a native function or stored procedure, but can they in turn invoke a Lambda function? I have only seen this being possible with Aurora, but not RDS - and I'm not able to find anything googling for it either. I guess it has to be possible, since there's no other option that fits either.

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Integrating.Lambda.html

upvoted 1 times

BlueVolcano1 3 years, 4 months ago

To add to that though, A also states to only use SQS (no SNS to SQS fan-out), which doesn't seem right as the message needs to go to multiple targets?

upvoted 13 times

39f3859 Most Recent 3 weeks, 2 days ago

Selected Answer: D

Per CHAT GPT:

Subscribe to an RDS event notification and send an Amazon Simple Notification Service topic fanned out to multiple Amazon Simple Queue Service queues. Use AWS Lambda functions to update the targets.
Why D is Correct
The key requirement is:
"data must be sent to multiple target systems"
This is a classic fanout architecture.
The ideal AWS pattern is:

upvoted 1 times

46e1a46 1 month, 1 week ago

Selected Answer: D

A and B describe a Lambda function pushing data to a single SQS queue for "the targets (plural) to consume". An SQS queue delivers each message to exactly one consumer. If three target systems are reading from the same queue then:

- Target 1 picks up message -> marks it as processed -> message is gone
- Target 2 and 3 never see the message
- Only one target gets the data

Hence a fan out architecture works best here

upvoted 1 times

MrBret077 4 months, 2 weeks ago

Selected Answer: D

RDS Event Notification: Can notify you when a change occurs in the database (like a deletion or modification).
SNS Topic: Acts as a pub-sub mechanism to fan out the event.
SQS Queues: Each target system can consume the data from its own queue, ensuring decoupled, scalable architecture.
Lambda: Each SQS queue can trigger a Lambda function to process and forward the data to its specific target system.

upvoted 1 times

bmmbmm 6 months, 1 week ago

Selected Answer: D

FanOut(SNS->SQS) to distribute to multiple systems, generating multiple messages

upvoted 1 times

Jess94 8 months, 2 weeks ago

Selected Answer: D

Why D is correct

RDS event notifications can detect changes or actions in the database, such as updates or deletes.

Amazon SNS allows you to fan out messages to multiple subscribers (in this case, multiple SQS queues).

SQS queues then decouple the delivery, ensuring each target system processes the message independently and reliably.

AWS Lambda functions can process messages from each SQS queue and update the target systems without affecting others.

This design provides scalability, fault isolation, and loose coupling, which are best practices for event-driven architectures.
upvoted 1 times

LucV3 9 months, 2 weeks ago

Selected Answer: D

None is perfectly correct. The fan-out pattern in D (SNS → SQS) is right, but the trigger must not be an RDS event. The correct design is "app publishes to SNS," which isn't one of the options.

If you're on Aurora, you could do: DB trigger → Lambda → SNS → SQS, which aligns with the fan-out part while keeping least-privilege and decoupling.
upvoted 1 times

aszd 10 months, 2 weeks ago

Selected Answer: D

A & B: These suggest triggering Lambda directly from RDS updates, which is not natively supported. Also, a single SQS queue (standard or FIFO) does not support multiple consumers efficiently—fan-out is needed.

C: This reverses the correct pattern—SQS queues cannot fan out to SNS topics. The correct approach is SNS fanning out to SQS queues.

D: Correctly uses RDS Event Notifications → SNS (fan-out) → Multiple SQS queues → Lambda for processing.
upvoted 1 times

a0c03ac 11 months, 1 week ago

Selected Answer: D

D. Subscribe to an RDS event notification and send an Amazon Simple Notification Service (Amazon SNS) topic fanned out to multiple Amazon Simple Queue Service (Amazon SQS) queues. Use AWS Lambda functions to update the targets.
✅ Explanation:
This solution provides a scalable, decoupled, and event-driven architecture that meets the requirement of sending data to multiple target systems when a listing is sold:

RDS event notifications can be configured to publish to an SNS topic.
SNS supports fan-out to multiple SQS queues, allowing each target system to receive the event independently.
Lambda functions can be triggered from each SQS queue to process and forward the data to the respective target systems.
This design ensures:

Loose coupling between components.
Scalability and fault tolerance.
Asynchronous processing for better performance and reliability
upvoted 1 times

Charlesvg 12 months ago

Selected Answer: A

i really hesitated between A & D.
But it is really not posssible for RDS event notifications to send messages about a delete row.
An option A only says to use a SQS, event if multiple targets can not consume from one SQS message, you can set up an SNS with the SQS queue to implement the fan out
upvoted 1 times

kaleido 12 months ago

Selected Answer: D

I prefer answer D because "data needs to be sent to targets". SNS is more efficient.
upvoted 1 times

bora4motion 1 year ago

Selected Answer: D

going with d as you need sns to fan out to multiple systems
upvoted 1 times

Kp002 1 year, 1 month ago

Selected Answer: D

Why not A or B: Lambda cannot be directly triggered by Amazon RDS data changes unless you're using Amazon Aurora with the advanced Aurora MySQL features, which support Aurora database triggers to invoke Lambda. If it's standard RDS, you'd need to build this logic into the application itself or poll the database — not ideal.
• FIFO queues (Option B) are only needed if message order is critical, which isn't indicated in the question.
upvoted 1 times

c6ee0dd 1 year, 1 month ago

Selected Answer: D

D : event publishes message to a topic making it available to multiple consumers
A : Its a single SQS queue which can lead to problems when multiple consumers are consuming the data concurrently

upvoted 1 times

TEC65 1 year, 1 month ago

Selected Answer: D

D - RDS cannot send notification to lambda

upvoted 1 times

network_enthusiast 1 year, 1 month ago

Selected Answer: B

B is the correct answer

upvoted 1 times

Question #109 | Topic 1

A company needs to store data in Amazon S3 and must prevent the data from being changed. The company wants new objects that are uploaded to Amazon S3 to remain unchangeable for a nonspecific amount of time until the company decides to modify the objects. Only specific users in the company's AWS account can have the ability 10 delete the objects.
What should a solutions architect do to meet these requirements?

A. Create an S3 Glacier vault. Apply a write-once, read-many (WORM) vault lock policy to the objects.

B. Create an S3 bucket with S3 Object Lock enabled. Enable versioning. Set a retention period of 100 years. Use governance mode as the S3 bucket's default retention mode for new objects.

C. Create an S3 bucket. Use AWS CloudTrail to track any S3 API events that modify the objects. Upon notification, restore the modified objects from any backup versions that the company has.

D. Create an S3 bucket with S3 Object Lock enabled. Enable versioning. Add a legal hold to the objects. Add the s3:PutObjectLegalHold permission to the IAM policies of users who need to delete the objects. Most Voted

Correct Answer: D

Community vote distribution

D (77%) B (23%)

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: D

A - No as "specific users can delete"
B - No as "nonspecific amount of time"
C - No as "prevent the data from being change"
D - The answer: "The Object Lock legal hold operation enables you to place a legal hold on an object version. Like setting a retention period, a legal hold prevents an object version from being overwritten or deleted. However, a legal hold doesn't have an associated retention period and remains ir effect until removed." https://docs.aws.amazon.com/AmazonS3/latest/userguide/batch-ops-legal-hold.html

upvoted 38 times

PassNow1234 3 years, 5 months ago

The Object Lock legal hold operation enables you to place a legal hold on an object version. Like setting a retention period, a legal hold prevents ar object version from being overwritten or deleted. However, a legal hold doesn't have an associated retention period and remains in effect until removed.

Correct

upvoted 2 times

Chunsli Highly Voted 3 years, 7 months ago

typo -- 10 delete the objects => TO delete the objects

upvoted 19 times

oddnoises 2 years, 8 months ago

they were trying to speak in binary lol

upvoted 7 times

reviewmine 2 years, 3 months ago

HAHAHA

upvoted 1 times

46e1a46 Most Recent 1 month, 1 week ago

Selected Answer: D

Legal hold on the objects makes objects immutable indefinitely, with no expiration data. The objects stay protected until someone (who is authorized) explicitly removes the legal hold.

By granting only specific users the s3:PutObjectLegalHold permission, only those users can lift the protection.
upvoted 1 times

a0c03ac 11 months, 1 week ago

Selected Answer: B

B. Create an S3 bucket with S3 Object Lock enabled. Enable versioning. Set a retention period of 100 years. Use governance mode as the S3 bucket's default retention mode for new objects.
✅ Explanation:
To meet the requirements of immutability and controlled deletion, Amazon S3 Object Lock is the ideal solution:

S3 Object Lock allows you to store objects using a write-once-read-many (WORM) model.
Governance mode lets you protect objects from being overwritten or deleted, but authorized users can override the lock if needed.
Versioning must be enabled to use Object Lock.
Setting a long retention period (e.g., 100 years) ensures objects remain unchangeable until explicitly modified.
Only users with the necessary IAM permissions (like s3:BypassGovernanceRetention) can delete or modify objects.
upvoted 1 times

jorgemenegaz 1 year ago

Selected Answer: B

Here's why option B is suitable:

S3 Object Lock: This feature allows you to prevent objects from being deleted or changed for a specified retention period. In governance mode, ever if a user has delete permissions, they cannot delete the objects until the retention period has expired.

Versioning: Enabling versioning ensures that multiple versions of an object are stored. This provides an additional layer of data protection, allowing you to restore data if an object is accidentally modified or deleted.

Retention Period: By setting a long retention period (like 100 years), you ensure compliance with data retention policies beyond short-term needs.
upvoted 1 times

c12ab95 1 year ago

Selected Answer: B

Why This Works
S3 Object Lock with Governance Mode:

Immutable by Default: New objects inherit a 100-year retention period, ensuring they cannot be modified or deleted during this time.

Flexible Override: Authorized users (with s3:BypassGovernanceRetention permissions) can shorten the retention period or delete objects before 100 years, aligning with the requirement for nonspecific retention until the company decides.

Versioning:

Protects against accidental overwrites or deletions by maintaining multiple object versions.

Governance Mode Benefits:

Allows controlled exceptions for authorized users while blocking unauthorized changes.

Why Other Options Fail
Option Shortcoming
A Glacier vaults are for archival, not active S3 storage.
C CloudTrail is reactive (audits changes but doesn't prevent them).
D Legal holds are indefinite but require manual application per object, which is not scalable for new uploads.
upvoted 1 times

CristiaNNN 1 year, 4 months ago

Selected Answer: D

Option D: Subscribe to an RDS event notification and send an Amazon Simple Notification Service (Amazon SNS) topic fanned out to multiple Amazon Simple Queue Service (Amazon SQS) queues. Use AWS Lambda functions to update the targets.

Correct Choice: This approach leverages RDS event notifications and SNS for fanning out updates to multiple SQS queues. Each target can then independently process updates using Lambda functions, making it scalable and modular.
upvoted 1 times

satyaammm 1 year, 5 months ago

Selected Answer: D

S3 Bucket Versioning helps deal with modifications while the legal hold coveys no deletes or changes until removed.
upvoted 1 times

aatikah 1 year, 5 months ago

Selected Answer: B

D is wrong
A legal hold can make objects immutable, but the question specifies new objects must remain unchangeable by default. Legal holds must be applied manually to individual objects, so they are not practical for this use case.
upvoted 1 times

PaulGa 1 year, 8 months ago
Ans D - "The Object Lock legal hold operation enables you to place a legal hold on an object version. Like setting a retention period, a legal hold prevents an object version from being overwritten or deleted. However, a legal hold doesn't have an associated retention period and remains in effect until removed."
https://docs.amazonaws.cn/en_us/AmazonS3/latest/userguide/batch-ops-legal-hold.html
The other options do not make sense for the situation in hand.
upvoted 2 times

huaze_lei 1 year, 9 months ago
Selected Answer: D
The Object Lock legal hold operation enables you to place a legal hold on an object version. Like setting a retention period, a legal hold prevents an object version from being overwritten or deleted. However, a legal hold doesn't have an associated retention period and remains in effect until removed.

You can use S3 Batch Operations with Object Lock to add legal holds to many Amazon S3 objects at once. You can do this by listing the target objects in your manifest and submitting that list to Batch Operations. Your S3 Batch Operations job with Object Lock legal hold runs until completion until cancellation, or until a failure state is reached.
upvoted 2 times

jaradat02 1 year, 10 months ago
Selected Answer: D
D is the best choice
upvoted 3 times

professorx123 2 years, 1 month ago
Selected Answer: B
Adding legal holds to objects and managing permissions for users to delete objects does not provide the same level of data immutability and retention control as S3 Object Lock.
upvoted 2 times

awsgeek75 2 years, 4 months ago
Selected Answer: B
https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.html
A: WORM doesn't allow delete by some users
C: Irrelevant
D: Permission only allows putting legal hold on objects. Not a complete solution
B: Closest apart from 100 years as question is asking for indefinite. Governance allows modification by some users
upvoted 2 times

pentium75 2 years, 5 months ago
Selected Answer: B
"With governance mode, you protect objects against being deleted by most users, but you can still grant some users permission to alter the retention settings or delete the objects if necessary."

D is wrong because it applies Object Lock AND Legal Hold, which are two different things that achieve similar results. 'Adding the s3:PutObjectLegalHold permission' to user's policies would allow them to remove the Legal Hold but NOT the Object Lock. (Also, it would probably make more sense to add the permissions to the bucket policy, not the "IAM policies of users".)
upvoted 4 times

LoXoL 2 years, 5 months ago
Isn't Legal Hold a subcategory of Object Lock? Object Lock itself doesn't imply anything imho: you should go either for a Retention Mode OR Legal Hold. Why would you go for B if they ask "for a nonspecific amount of time"? Open to change my mind.
upvoted 1 times

Abitek007 2 years, 8 months ago
Selected Answer: D
I only picked this because of restricted users who can delete, and the easiest way of achieving this is them assuming the role
upvoted 2 times

TariqKipkemei 2 years, 9 months ago
Selected Answer: D
"The company wants new objects that are uploaded to Amazon S3 to remain unchangeable for a nonspecific amount of time until the company decides to modify the objects" = A legal hold prevents an object version from being overwritten or deleted. However, a legal hold doesn't have an associated retention period and remains in effect until removed.
s3:PutObjectLegalHold permission is required in your IAM role to add or remove legal hold from objects.

upvoted 3 times

Question #110 Topic 1

A social media company allows users to upload images to its website. The website runs on Amazon EC2 instances. During upload requests, the website resizes the images to a standard size and stores the resized images in Amazon S3. Users are experiencing slow upload requests to the website.
The company needs to reduce coupling within the application and improve website performance. A solutions architect must design the most operationally efficient process for image uploads.
Which combination of actions should the solutions architect take to meet these requirements? (Choose two.)

A. Configure the application to upload images to S3 Glacier.

B. Configure the web server to upload the original images to Amazon S3.

C. Configure the application to upload images directly from each user's browser to Amazon S3 through the use of a presigned URL **Most Voted**

D. Configure S3 Event Notifications to invoke an AWS Lambda function when an image is uploaded. Use the function to resize the image. **Most Voted**

E. Create an Amazon EventBridge (Amazon CloudWatch Events) rule that invokes an AWS Lambda function on a schedule to resize uploaded images.

Correct Answer: CD

Community vote distribution

CD (54%) BD (45%)

Comments

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago

Selected Answer: CD

To meet the requirements of reducing coupling within the application and improving website performance, the solutions architect should consider taking the following actions:

C. Configure the application to upload images directly from each user's browser to Amazon S3 through the use of a pre-signed URL. This will allow the application to upload images directly to S3 without having to go through the web server, which can reduce the load on the web server and improve performance.

D. Configure S3 Event Notifications to invoke an AWS Lambda function when an image is uploaded. Use the function to resize the image. This will allow the application to resize images asynchronously, rather than having to do it synchronously during the upload request, which can improve performance.

upvoted 60 times

jdr75 3 years, 2 months ago

presigned URL is for download the data from S3, not for uploads, so the user does not upload anything. C is no correct.

upvoted 16 times

mauroicardi 2 years, 2 months ago

A user who does not have AWS credentials to upload a file can use a presigned URL to perform the upload.

https://boto3.amazonaws.com/v1/documentation/api/latest/guide/s3-presigned-urls.html

upvoted 7 times

Vandaman 1 year, 3 months ago

This makes sense. I will change my answers to CD

upvoted 1 times

allanm 10 months, 1 week ago

Read the link you have posted -

"A user who does not have AWS credentials or permission to access an S3 object can be granted temporary access by using a presigned URL.

A presigned URL is generated by an AWS user who has access to the object. The generated URL is then given to the unauthorized user. The presigned URL can be entered in a browser or used by a program or HTML webpage. The credentials used by the presigned URL are those of the AWS user who generated the URL."

You don't need to have access to an S3 object in this case. You just need the server to upload into S3. C is for downloading something from S3, not a "put" function so C is incorrect.

upvoted 1 times

EricYu2023 3 years, 1 month ago

Presigned URL can be use for upload.

upvoted 14 times

AF_1221 3 years, 1 month ago

but for temporary purpose not for permanent

upvoted 4 times

tuso 2 years, 4 months ago

So? You only need a presigned URL for the moment you upload the image, not forever

upvoted 2 times

AnhNguyen99 1 year, 11 months ago

that's right

upvoted 1 times

PoisonBlack 3 years, 1 month ago

https://docs.aws.amazon.com/AmazonS3/latest/userguide/PresignedUrlUploadObject.html

upvoted 9 times

AF_1221 3 years, 1 month ago

preassigned URL is for upload or download for temporary time and for specific users outside the company

upvoted 5 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Why other options are wrong
Option A, Configuring the application to upload images to S3 Glacier, is not relevant to improving the performance of image uploads.

Option B, Configuring the webserver to upload the original images to Amazon S3, is not a recommended solution as it would not reduce coupling within the application or improve performance.

Option E, Creating an Amazon EventBridge (Amazon CloudWatch Events) rule that invokes an AWS Lambda function on a schedule to resize uploaded images, is not a recommended solution as it would not be able to resize images in a timely manner and would not improve performance.

upvoted 5 times

MutiverseAgent 2 years, 11 months ago

About your comments regarding option B)... But if images are being saved directly to S3 instead of the EBS/SSD storage of E2 instances as they originally were, the new approach will reduce coupling and improve performance. Also you have to consider the security concerns about presign URLs as the question does not mention if users are public or private.

upvoted 2 times

Yelizaveta 3 years, 3 months ago

Here it means to decouple the processes, so that the web server don't have to do the resizing, so it doesn't slow down. The customers access the web server, so the web server have to be involved in the process, and how the others already wrote, the pre-signed URL is not the right solution because, of the explanation you can read in the other comments.

And additional! "Configure the application to upload images directly from EACH USER'S BROWSER to Amazon S3 through the use of a pre-signed URL"

I am not an expert, but I can't imagine that you can store an image that an user uploads in his browser etc.

upvoted 7 times

fkie4 Highly Voted 3 years, 3 months ago

Selected Answer: BD

Why would anyone vote C? signed URL is for temporary access. also, look at the vote here:
https://www.examtopics.com/discussions/amazon/view/82971-exam-aws-certified-solutions-architect-associate-saa-c02/

upvoted 36 times

sheilawu 2 years ago

I agree with you. C seems a temporary solution if a download or upload demend is urgent.

upvoted 1 times

46e1a46 Most Recent 1 month, 1 week ago

Selected Answer: CD

C lets users upload directly to S3 from their browser, bypassing the EC2 web server entirely which eliminates the bottleneck
D uses S3 event notifications to trigger a Lambda function that resizes images automatically

upvoted 1 times

0bf2fba 10 months, 3 weeks ago

Selected Answer: CD

Option C removes the server from the equation, it reduces latency.
Once the image sits on s3, use lambda to resize

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: CD

see https://aws.amazon.com/blogs/compute/uploading-large-objects-to-amazon-s3-using-multipart-upload-and-transfer-acceleration/
B separate concerns and has the best performance

upvoted 1 times

salman7540 1 year, 5 months ago

Selected Answer: BD

BD looks accurate. Presigned URLs provide short time access to users to upload or download objects to/from S3. This approach is not feasible to renew URL every time and give it to every random users who plans to access website.
Presigned URL generally used for occasional sharing of private files.

upvoted 1 times

Tjazz04 1 year, 6 months ago

Selected Answer: CD

CD is the more appropriate solution bec. when the user uploads the images, it will directly uploaded to the S3 while if BD, when the user uploads the images, it will first go to the web server then to the S3 bucket and This can cause a slow upload process since the web server is processing the download from the user, then upload to the s3 bucket.

upvoted 2 times

architect_kags 1 year, 6 months ago

Selected Answer: BE

While C can reduce web server load, it doesn't address image resizing, which is a critical requirement. So B & D would be the answers.

upvoted 1 times

Abdullah2004 1 year, 7 months ago

Selected Answer: CD

It's very clear C D

upvoted 2 times

Abdullah2004 1 year, 7 months ago

I'm sorry it's B D

upvoted 1 times

ChymKuBoy 1 year, 7 months ago

Selected Answer: CD

C, D for sure

upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: BD

Ans B, D -
1st: Upload the original images to Amazon S3
2nd: Configure S3 Event Notifications to invoke an AWS Lambda function when an image is uploaded to resize the image.

upvoted 3 times

bignatov 1 year, 9 months ago

Selected Answer: CD

I am voting for C and D.
B and D could also work, but when you upload the images to the EC2 and then to S3 it could cause additional performance and network traffic load.

upvoted 2 times

Moo 1 year, 10 months ago

I don't understand why the expiration time of presigned urls is being questioned. You can certainly design an upload flow that uses the SDK to create a new presigned url before uploading.

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: BD

B and D is the way to go

upvoted 3 times

creamymangosauce 1 year, 11 months ago

Selected Answer: CD

CD - No point having the instance do extra work when we can use pre signed URLs and let the user directly upload to S3, hence B is not an operationally efficient option. Furthermore B results in more traffic through the instance which is inefficient.

upvoted 2 times

jatric 1 year, 11 months ago

Selected Answer: BD

C is not a valid option to upload the image is to have presigned url to retrieve the file/image from S3

upvoted 1 times

MomenAWS 1 year, 12 months ago

Selected Answer: BD

BD is more reasonable than CD

upvoted 3 times

Question #111 *Topic 1*

A company recently migrated a message processing system to AWS. The system receives messages into an ActiveMQ queue running on an Amazon EC2 instance. Messages are processed by a consumer application running on Amazon EC2. The consumer application processes the messages and writes results to a MySQL database running on Amazon EC2. The company wants this application to be highly available with low operational complexity.
Which architecture offers the HIGHEST availability?

A. Add a second ActiveMQ server to another Availability Zone. Add an additional consumer EC2 instance in another Availability Zone. Replicate the MySQL database to another Availability Zone.

B. Use Amazon MQ with active/standby brokers configured across two Availability Zones. Add an additional consumer EC2 instance in another Availability Zone. Replicate the MySQL database to another Availability Zone.

C. Use Amazon MQ with active/standby brokers configured across two Availability Zones. Add an additional consumer EC2 instance in another Availability Zone. Use Amazon RDS for MySQL with Multi-AZ enabled.

D. Use Amazon MQ with active/standby brokers configured across two Availability Zones. Add an Auto Scaling group for the consumer EC2 instances across two Availability Zones. Use Amazon RDS for MySQL with Multi-AZ enabled. **Most Voted**

Correct Answer: D

Community vote distribution

D (98%) 2%

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: D

Answer is D as the "HIGHEST available" and less "operational complex"
The "Amazon RDS for MySQL with Multi-AZ enabled" option excludes A and B
The "Auto Scaling group" is more available and reduces operational complexity in case of incidents (as remediation it is automated) than just adding one more instance. This excludes C.

C and D to choose from based on
D over C since is configured

upvoted 23 times

satyaammm Most Recent 1 year, 5 months ago

Selected Answer: D

Using RDS Multi-AZ is suitable for high availability and using EC2 Auto Scaling Groups is suitable for operational overhead.

upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans D - extends the architecture rather than complicating it: Amazon MQ with active/standby brokers configured across two Availability Zones; Auto Scaling group for the consumer EC2 instances across two Availability Zones; Amazon RDS for MySQL with Multi-AZ enabled.

upvoted 3 times

huaze_lei 1 year, 9 months ago

Selected Answer: D

- Active MQ with active/standby
- Auto scaling across 2 AZs
- RDS with Multi AZ.

These provide the highest availability and least complexity (using Amazon MQ, auto scaling and RDS, all managed services)

upvoted 3 times

jaradat02 1 year, 10 months ago

Selected Answer: D

D is obviously the highest available.

upvoted 2 times

effiecancode 1 year, 11 months ago

Definitely it's D

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

D: Managed and auto-scaling, resilient and HA service for each tier. This is well-architected too.

upvoted 3 times

Ruffyit 2 years, 7 months ago

Using Amazon MQ with active/standby brokers provides highly available message queuing across AZs.

Adding an Auto Scaling group for consumer EC2 instances across 2 AZs provides highly available processing.

Using RDS MySQL with Multi-AZ provides a highly available database.

This architecture provides high availability for all components of the system - queue, processing, and database.

upvoted 3 times

prabhjot 2 years, 8 months ago

Ans is C - C. Option C uses Amazon MQ with active/standby brokers, adds an additional consumer EC2 instance, and uses Amazon RDS for MySQL with Multi-AZ enabled. Amazon RDS Multi-AZ automatically replicates your database to another AZ and provides automated failover. This ensures high availability for both the messaging system and the database. Option D- bring More scalabilty rather HA

upvoted 3 times

pentium75 2 years, 5 months ago

D automates replacement of failed instances, thus it has higher availability than C.

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: D

HIGHEST availability. Definitely option D.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

The key reasons are:

Amazon MQ active/standby brokers across AZs for queue high availability
Auto Scaling group with consumer EC2 instances across AZs for redundant processing
RDS MySQL with Multi-AZ for database high availability
This combines the HA capabilities of MQ, EC2 and RDS to maximize fault tolerance across all components. The auto scaling also provides flexibility tc scale processing capacity as needed.

upvoted 3 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: D

D is the correct answer.

Using Amazon MQ with active/standby brokers provides highly available message queuing across AZs.

Adding an Auto Scaling group for consumer EC2 instances across 2 AZs provides highly available processing.

Using RDS MySQL with Multi-AZ provides a highly available database.

This architecture provides high availability for all components of the system - queue, processing, and database.

upvoted 3 times

james2033 2 years, 10 months ago

Selected Answer: D

Keyword Amazon RDS, has C and D. Then D has "Auto Scaling group", choose D.

upvoted 3 times

MNotABot 2 years, 11 months ago

D

With 3 options with Amazon MQ --> A is odd one out / Then ASG with M-AZ was an easy choice

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: D

Amazon MQ with active/standby brokers configured across two AZ ensures high availability for the message broker. In case of a failure in one AZ, the other AZ's broker can take over seamlessly.

Adding an ASG for the consumer EC2 instances across two AZ provides redundancy and automatic scaling based on demand. If one consumer instance becomes unavailable or if the message load increases, the ASG can automatically launch additional instances to handle the workload.

Using RDS for MySQL with Multi-AZ enabled ensures high availability for the database. Multi-AZ automatically replicates the database to a standby instance in another AZ. If a failure occurs, RDS automatically fails over to the standby instance without manual intervention.

This architecture combines high availability for the message broker (Amazon MQ), scalability and redundancy for the consumer EC2 instances (ASG), and high availability for the database (RDS Multi-AZ). It offers the highest availability with low operational complexity by leveraging managed services and automated failover mechanisms.

upvoted 3 times

Kostya 2 years, 11 months ago

Selected Answer: D

Correct answer D

upvoted 1 times

Bmarodi 3 years ago

Selected Answer: D

to achieve ha + low operational complexity, the solution architect has to choose option D, which fulfill these requirements.

upvoted 1 times

Question #112 *Topic 1*

A company hosts a containerized web application on a fleet of on-premises servers that process incoming requests. The number of requests is growing quickly. The on-premises servers cannot handle the increased number of requests. The company wants to move the application to AWS with minimum code changes and minimum development effort.
Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS Fargate on Amazon Elastic Container Service (Amazon ECS) to run the containerized web application with Service Auto Scaling. Use an Application Load Balancer to distribute the incoming requests. **Most Voted**

B. Use two Amazon EC2 instances to host the containerized web application. Use an Application Load Balancer to distribute the incoming requests.

C. Use AWS Lambda with a new code that uses one of the supported languages. Create multiple Lambda functions to support the load. Use Amazon API Gateway as an entry point to the Lambda functions.

D. Use a high performance computing (HPC) solution such as AWS ParallelCluster to establish an HPC cluster that can process the incoming requests at the appropriate scale.

Correct Answer: A

Community vote distribution

A (100%)

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: A

Less operational overhead means A: Fargate (no EC2), move the containers on ECS, autoscaling for growth and ALB to balance consumption.
B - requires configure EC2
C - requires add code (developpers)
D - seems like the most complex approach, like re-architecting the app to take advantage of an HPC platform.

upvoted 20 times

satyaammm Most Recent 1 year, 5 months ago

Selected Answer: A

AWS Fargate is a serverless compute engine and is fully managed by AWS. So it is more suitable here as it provides less operational overhead.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: A

And A - its containers so use FarGate, add some auto-scaling...

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: A

A makes sense

upvoted 1 times

cosmiccliff 2 years, 7 months ago

Selected Answer: A

key = LEAST operational overhead

Fargate a serverless service fully managed by aws
https://docs.aws.amazon.com/AmazonECS/latest/userguide/what-is-fargate.html#:~:text=AWS%20Fargate%20is,optimize%20cluster%20packing.

upvoted 3 times

Ruffyit 2 years, 7 months ago

Less operational overhead means A: Fargate (no EC2), move the containers on ECS, autoscaling for growth and ALB to balance consumption.

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: A

LEAST operational overhead = AWS Fargate

upvoted 2 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: A

A is the best solution to meet the requirements with the least operational overhead. The key reasons are:

AWS Fargate removes the need to provision and manage servers. Fargate will automatically scale the application based on demand. This removes a significant operational burden.
Using ECS along with Fargate provides a managed orchestration layer to easily run and scale the containerized application.
The Application Load Balancer handles distribution of traffic without additional effort.
No code changes are required to move the application to Fargate. The containers can run as-is.

upvoted 3 times

Guru4Cloud 2 years, 10 months ago

Selected Answer: A

A is the correct answer.

AWS Fargate removes the need to provision and manage servers, allowing you to focus on deploying and running applications. Fargate will scale compute capacity up and down automatically based on application load. This removes the operational overhead of managing servers.

upvoted 1 times

james2033 2 years, 10 months ago

Selected Answer: A

Existing: "containerized web-app", "minimum code changes + minimum development effort" --> AWS Fargate + Amazon Elastic Container Services (ECS). Easy question.

upvoted 1 times

MNotABot 2 years, 11 months ago

A
Fargate, ECS, ASG, ALB....What else one will need for a nice sleep?

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: A

Option A (AWS Fargate on Amazon ECS with Service Auto Scaling) is the best choice as it provides a serverless and managed environment for your containerized web application. It requires minimal code changes, offers automatic scaling, and utilizes an Application Load Balancer for request distribution.

Option B (Amazon EC2 instances with an Application Load Balancer) requires manual management of EC2 instances, resulting in more operational overhead compared to option A.

Option C (AWS Lambda with API Gateway) may require significant code changes and restructuring, introducing complexity and potentially increasing development effort.

Option D (AWS ParallelCluster) is not suitable for a containerized web application and involves significant setup and configuration overhead.

upvoted 3 times

Jeeva28 3 years ago

Selected Answer: A

AWS Fargate is a technology that you can use with Amazon ECS to run containers without having to manage servers or clusters of Amazon EC2 instances. With Fargate, you no longer have to provision, configure, or scale clusters of virtual machines to run containers. This removes the need to choose server types, decide when to scale your clusters, or optimize cluster packing.

upvoted 1 times

studynoplay 3 years, 1 month ago

Selected Answer: A

Least Operational Overhead = Serverless

upvoted 1 times

airraid2010 3 years, 2 months ago

Selected Answer: A

AWS Fargate is a technology that you can use with Amazon ECS to run containers without having to manage servers on clusters of Amazon EC2 instances. With Fargate, you no longer have to provision, configure, or scale of virtual machines to run containers.

https://docs.aws.amazon.com/AmazonECS/latest/userguide/what-is-fargate.html

upvoted 1 times

Chalamalli 3 years, 4 months ago

A is correct

upvoted 1 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: A

The best solution to meet the requirements with the least operational overhead is Option A: Use AWS Fargate on Amazon Elastic Container Service (Amazon ECS) to run the containerized web application with Service Auto Scaling. Use an Application Load Balancer to distribute the incoming requests.

upvoted 3 times

Question #113 Topic 1

A company uses 50 TB of data for reporting. The company wants to move this data from on premises to AWS. A custom application in the company's data center runs a weekly data transformation job. The company plans to pause the application until the data transfer is complete and needs to begin the transfer process as soon as possible.
The data center does not have any available network bandwidth for additional workloads. A solutions architect must transfer the data and must configure the transformation job to continue to run in the AWS Cloud.
Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS DataSync to move the data. Create a custom transformation job by using AWS Glue.

B. Order an AWS Snowcone device to move the data. Deploy the transformation application to the device.

C. Order an AWS Snowball Edge Storage Optimized device. Copy the data to the device. Create a custom transformation job by using AWS Glue. Most Voted

D. Order an AWS Snowball Edge Storage Optimized device that includes Amazon EC2 compute. Copy the data to the device. Create a new EC2 instance on AWS to run the transformation application.

Correct Answer: C

Community vote distribution

C (64%) D (36%)

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: C

A. Use AWS DataSync to move the data. Create a custom transformation job by using AWS Glue. - No BW available for DataSync, so "asap" will be weeks/months (?)
B. Order an AWS Snowcone device to move the data. Deploy the transformation application to the device. - Snowcone will just store 14TB (SSD configuration).
C. Order an AWS Snowball Edge Storage Optimized device. Copy the data to the device. Create a custom transformation job by using AWS Glue. SnowBall can store 80TB (ok), takes around 1 week to move the device (faster than A), and AWS Glue allows to do ETL jobs. This is the answer.
D. Order an AWS Snowball Edge Storage Optimized device that includes Amazon EC2 compute. Copy the data to the device. Create a new EC2 instance on AWS to run the transformation application. - Same as C, but the ETL job requires the deployment/configuration/maintenance of an EC2 instance, while Glue is serverless. This means D has more operational overhead than C.

upvoted 77 times

remand 3 years, 4 months ago

I disagree on D. transformation job is already in place.so, all you have to do is deploy and run on ec2.
C takes more effort to build Glue process, like reinventing the wheel . this is unnecessary

upvoted 10 times

xxichlas 2 years, 1 month ago

the differene between snowball with ec2 and without ec2 is $200 (for us east 1). fair to assume aws glue will not be more than that

upvoted 2 times

jdr75 3 years, 2 months ago

I agree. When it said "with least Operational overhead" , it does not takes in account "migration activities" neccesary to reach the "final photo/scenario". In "operational overhead" schema, you're situated in a "final scenario" and you've only take into account how do you operate it, and if the operation of that scheme is ALIGHTED (least effort to operate than original scenario), that's the desired state.

upvoted 6 times

goodmail Highly Voted 3 years, 4 months ago

Selected Answer: D

Why C? This answer misses the part between SnowBall and AWS Glue.
D at least provides a full-step solution that copies data in snowball device, and installs the custom application in device's EC2 to do the transformation job.

upvoted 18 times

happpieee 2 years, 3 months ago

AWS Glue is not part of SnowBall Edge AWS services it can run within. Check it out here : https://docs.aws.amazon.com/snowball/latest/developer-guide/whatisedge.html

upvoted 3 times

pentium75 2 years, 5 months ago

Because AWS Glue means less "operational (!) overhead" than running an EC2 instance.

upvoted 6 times

46e1a46 Most Recent 1 month, 1 week ago

Selected Answer: C

A - Requires network bandwidth, which the data center doesn't have
B- Snowcone only holds 8TB
D- Running the original custom application on EC2 keeps all the operational overhead of maintaining custom code, instances, and infrastructure.

C - Glue is fully managed and has lower overhead

upvoted 1 times

0bf2fba 10 months, 3 weeks ago

Selected Answer: D

I'll go with D.
Option C is very close but AWS Glue requires rebuilding the transformation logic, which adds operational effort compared to reusing the existing app.

upvoted 2 times

kaleido 12 months ago

Selected Answer: C

If the data transform runs in the Snowball edge device, does that count as "running on the AWS cloud"? Honest question.

upvoted 1 times

c12ab95 1 year ago

Selected Answer: D

Option Shortcoming
A AWS DataSync requires network bandwidth, which is unavailable.
B Snowcone's maximum capacity (14 TB SSD) is insufficient for 50 TB.
C AWS Glue would require rewriting the custom application, increasing operational overhead.
Option D minimizes operational overhead by leveraging Snowball for offline transfer and EC2 for running the existing application, ensuring a seamless transition to AWS.

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: C

One requirement is to run transformation with the data in aws, that is the job of glue

upvoted 1 times

architect_kags 1 year, 6 months ago

Selected Answer: D

Since the transformation application already there, Snowball Edge devices with compute capability allow to run applications directly on the device. This means the transformation job can start locally on the Snowball Edge device if needed or directly after the data transfer to AWS, minimizing delays. AWS Glue for transformation adds additional refactoring work and operational overhead.

upvoted 3 times

tom_cruise 1 year, 6 months ago

Selected Answer: D

The only way C can work is by using this: Amazon S3 adapter — Use for programmatic data transfer in to and out of AWS using the Amazon S3 API for Snowball Edge, but it is not indicated in the answer. If it includes the step to transfer the data from the Snowball to S3, then C would be a better answer.

upvoted 2 times

0de7d1b 1 year, 6 months ago

Selected Answer: D

Application still need EC2 instance along data transfer with Snowball

upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans D - but I suspect it should be Ans C because, as stated by 123jhlo (1yr 11mth ago),
D is not serverless:

"**C**. Order an AWS Snowball Edge Storage Optimized device. Copy the data to the device. Create a custom transformation job by using AWS Glue. - SnowBall can store 80TB (ok), takes around 1 week to move the device (faster than A), and AWS Glue allows to do ETL jobs."
D. Order an AWS Snowball Edge Storage Optimized device that includes Amazon EC2 compute. Copy the data to the device. Create a new EC2 instance on AWS to run the transformation application. - Same as C, but the ETL job requires the deployment/configuration/maintenance of an EC2 instance, while Glue is serverless. This means D has more operational overhead than C."

upvoted 3 times

bignatov 1 year, 9 months ago

Selected Answer: D

A and D are wrong. D is the correct answer.
Why not C:

AWS Snowball Edge is a physical device designed for transferring large amounts of data to and from AWS. It includes some compute capabilities, such as running AWS Lambda functions, AWS IoT Greengrass, and EC2 instances, but it does not support AWS Glue.

AWS Glue is a fully managed ETL (Extract, Transform, Load) service that runs within the AWS Cloud. It is designed to work with data stored in AWS services like Amazon S3, Amazon RDS, and Amazon Redshift, among others. AWS Glue is not available on edge devices like Snowball Edge.

upvoted 3 times

Lexxo 1 year, 3 months ago

Yes, AWS Glue can connect to AWS Snowball Edge. AWS Snowball Edge is a data transfer and edge computing device that allows you to move large amounts of data into and out of AWS. You can use AWS Glue to create ETL jobs that read data from Snowball Edge and write it to other AWS services like Amazon S3, Amazon Redshift, or Amazon RDS. --- FROM Copilot

upvoted 1 times

bignatov 1 year, 9 months ago

just correction A and B are wrong and D is the correct answer.

upvoted 1 times

lofzee 2 years ago

gotta be C surely..... LEAST operational overhead.
EC2 = operational overhead.
AWS Glue = managed service with transformation capabilities.

upvoted 2 times

f761d0e 2 years, 1 month ago

The q states that the custom job must remain, so glue is out. Seems like D is the only option as DataSync needs bandwidth.

upvoted 4 times

vip2 2 years, 3 months ago

Selected Answer: C

Using an EC2 instance instead of a managed service like AWS Glue will include more operational overhead for the organization.

upvoted 2 times

Femmyte 2 years, 4 months ago

Selected Answer: D

The answer is D because of the following key points
1. A custom application in the company's data center runs a weekly data transformation job. Which means that the company already has an application that runs the transformation.
2. A solutions architect must transfer the data and must configure the transformation job to continue to run in the AWS Cloud. This shows that the only responsibility of the architect is to transfer the data and configure the existing application to run on the EC2 the architect is going to deploy.

upvoted 3 times

awsgeek75 2 years, 4 months ago

Selected Answer: C

A: Cannot be done because no bandwidth
B: Snowcone is probably to small
D: Doable by EC2 is overhead for transformation when Glue is an option

C: Is correct as Snowball Edge Storage Optimised device is good for storage and Glue can transform once the data is available

upvoted 2 times

Question #114 Topic 1

A company has created an image analysis application in which users can upload photos and add photo frames to their images. The users upload images and metadata to indicate which photo frames they want to add to their images. The application uses a single Amazon EC2 instance and Amazon DynamoDB to store the metadata.
The application is becoming more popular, and the number of users is increasing. The company expects the number of concurrent users to vary significantly depending on the time of day and day of week. The company must ensure that the application can scale to meet the needs of the growing user base.
Which solution meats these requirements?

A. Use AWS Lambda to process the photos. Store the photos and metadata in DynamoDB.

B. Use Amazon Kinesis Data Firehose to process the photos and to store the photos and metadata.

C. Use AWS Lambda to process the photos. Store the photos in Amazon S3. Retain DynamoDB to store the metadata.
Most Voted

D. Increase the number of EC2 instances to three. Use Provisioned IOPS SSD (io2) Amazon Elastic Block Store (Amazon EBS) volumes to store the photos and metadata.

Correct Answer: C

Community vote distribution

C (100%)

Comments

MXB05 Highly Voted 3 years, 8 months ago
Selected Answer: C
Do not store images in databases ;)... correct answer should be C
upvoted 49 times

cookieMr Highly Voted 2 years, 11 months ago
Selected Answer: C
Solution C offloads the photo processing to Lambda. Storing the photos in S3 ensures scalability and durability, while keeping the metadata in DynamoDB allows for efficient querying of the associated information.

Option A does not provide an appropriate solution for storing the photos, as DynamoDB is not suitable for storing large binary data like images.

Option B is more focused on real-time streaming data processing and is not the ideal service for processing and storing photos and metadata in this use case.

Option D involves manual scaling and management of EC2 instances, which is less flexible and more labor-intensive compared to the serverless nature of Lambda. It may not efficiently handle the varying number of concurrent users and can introduce higher operational overhead.

In conclusion, option C provides the best solution for scaling the application to meet the needs of the growing user base by leveraging the scalability and durability of Lambda, S3, and DynamoDB.
upvoted 18 times

michaelmorar Most Recent 1 year, 5 months ago
Selected Answer: C
Bad question - it does not specify a requirement for storing the photos nor does it specify where the old application does so. C is the best answer absent further requirements.
upvoted 1 times

PaulGa 1 year, 8 months ago
Selected Answer: C
Ans C - as well explained by cookieMr (1yr 2mth ago):
"Solution C offloads the photo processing to Lambda. Storing the photos in S3 ensures scalability and durability, while keeping the metadata in DynamoDB allows for efficient querying of the associated information.

Option A does not provide an appropriate solution for storing the photos, as DynamoDB is not suitable for storing large binary data like images.

...option C provides the best solution for scaling the application to meet the needs of the growing user base by leveraging the scalability and durability of Lambda, S3, and DynamoDB."

upvoted 2 times

pranavff_examtopics_1993 1 year, 9 months ago

Selected Answer: C

DynamoDB should not be used for storing images and files in general. Answer should be C.

upvoted 2 times

zliang14 1 year, 9 months ago

DynamoDB is not designed for storing large objects like photos. Amazon S3 is the correct storage service for photos.

upvoted 2 times

snk27 1 year, 10 months ago

Selected Answer: C

s3 bucket is good option to store images

upvoted 3 times

Gape4 1 year, 11 months ago

Selected Answer: C

C is correct answer.

upvoted 1 times

f761d0e 2 years, 1 month ago

C. DB is for data, not for photos. Kinesis doesn't store, it processes streaming.

upvoted 2 times

MehulKapadia 2 years, 2 months ago

Selected Answer: C

DynamoDB items max size is 400kb. so A cannot be right answer.

Correct Answer is C

upvoted 2 times

farnamjam 2 years, 5 months ago

Selected Answer: C

Max size for DDB entry is 400KB.

upvoted 2 times

aptx4869 2 years, 7 months ago

Selected Answer: C

Images (Object) should go in S3 and metadata should go in database (DynamoDB)

upvoted 2 times

Ruffyit 2 years, 7 months ago

Solution C offloads the photo processing to Lambda. Storing the photos in S3 ensures scalability and durability, while keeping the metadata in DynamoDB allows for efficient querying of the associated information.

upvoted 2 times

Ferna 2 years, 7 months ago

Selected Answer: C

Solution C

upvoted 1 times

David_Ang 2 years, 8 months ago

Selected Answer: C

i think is only a confusion of the admin, because it has more sense to store the photos in a S3 bucket is logic.

upvoted 1 times

tom_cruise 2 years, 8 months ago

Selected Answer: C

A does not store data.

upvoted 1 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: C

I stopped at option C

upvoted 1 times

Question #115 Topic 1

A medical records company is hosting an application on Amazon EC2 instances. The application processes customer data files that are stored on Amazon S3. The EC2 instances are hosted in public subnets. The EC2 instances access Amazon S3 over the internet, but they do not require any other network access.
A new requirement mandates that the network traffic for file transfers take a private route and not be sent over the internet.
Which change to the network architecture should a solutions architect recommend to meet this requirement?

A. Create a NAT gateway. Configure the route table for the public subnets to send traffic to Amazon S3 through the NAT gateway.

B. Configure the security group for the EC2 instances to restrict outbound traffic so that only traffic to the S3 prefix list is permitted.

C. Move the EC2 instances to private subnets. Create a VPC endpoint for Amazon S3, and link the endpoint to the route table for the private subnets. **Most Voted**

D. Remove the internet gateway from the VPC. Set up an AWS Direct Connect connection, and route traffic to Amazon S3 over the Direct Connect connection.

Correct Answer: C

Community vote distribution

C (95%) 5%

Comments

cookieMr Highly Voted 2 years, 11 months ago

Selected Answer: C

Option A (creating a NAT gateway) would not meet the requirement since it still involves sending traffic to S3 over the internet. NAT gateway is used for outbound internet connectivity from private subnets, but it doesn't provide a private route for accessing S3.

Option B (configuring security groups) focuses on controlling outbound traffic using security groups. While it can restrict outbound traffic, it doesn't provide a private route for accessing S3.

Option D (setting up Direct Connect) involves establishing a dedicated private network connection between the on-premises environment and AWS. While it offers private connectivity, it is more suitable for hybrid scenarios and not necessary for achieving private access to S3 within the VPC.

In summary, option C provides a straightforward solution by moving the EC2 instances to private subnets, creating a VPC endpoint for S3, and linking the endpoint to the route table for private subnets. This ensures that file transfer traffic between the EC2 instances and S3 remains within the private network without going over the internet.

upvoted 16 times

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago

Selected Answer: C

The correct answer is C. Move the EC2 instances to private subnets. Create a VPC endpoint for Amazon S3, and link the endpoint to the route table for the private subnets.

To meet the new requirement of transferring files over a private route, the EC2 instances should be moved to private subnets, which do not have direct access to the internet. This ensures that the traffic for file transfers does not go over the internet.

To enable the EC2 instances to access Amazon S3, a VPC endpoint for Amazon S3 can be created. VPC endpoints allow resources within a VPC to communicate with resources in other services without the traffic being sent over the internet. By linking the VPC endpoint to the route table for the private subnets, the EC2 instances can access Amazon S3 over a private connection within the VPC.

upvoted 6 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A (Create a NAT gateway) would not work, as a NAT gateway is used to allow resources in private subnets to access the internet, while the requirement is to prevent traffic from going over the internet.

Option B (Configure the security group for the EC2 instances to restrict outbound traffic) would not achieve the goal of routing traffic over a private connection, as the traffic would still be sent over the internet.

Option D (Remove the internet gateway from the VPC and set up an AWS Direct Connect connection) would not be necessary, as the requirement can be met by simply creating a VPC endpoint for Amazon S3 and routing traffic through it.

upvoted 1 times

Kayamables 3 years, 5 months ago

How about the question of moving the instances across subnets. Because according to AWS you can't do it.
https://aws.amazon.com/premiumsupport/knowledge-center/move-ec2-instance/#:~:text=It%27s%20not%20possible%20to%20move,%2C%20Availability%20Zone%2C%20or%20VPC.
Kindly clarify. Maybe I miss something.

upvoted 2 times

pentium75 2 years, 5 months ago

You can't just change the subnet in instance settings, but this article mentions how you CAN move the instance manually.

upvoted 2 times

Vandaman Most Recent 1 year, 3 months ago

Selected Answer: C

C is the best Answer. The requirement is for file transfers to take a private route and not be sent over the internet. This eliminates A and B. D is for Hybrid solutions.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans C - I was going for Ans D...
...but as well explained by Buruguduystunstugudunstuy (1 year, 8 mth ago), C is simpler:
"Option C: Move the EC2 instances to private subnets. Create a VPC endpoint for Amazon S3, and link the endpoint to the route table for the private subnets.
To meet the new requirement of transferring files over a private route, the EC2 instances should be moved to private subnets, which do not have direct access to the internet. This ensures that the traffic for file transfers does not go over the internet.
"Option D (Remove the internet gateway from the VPC and set up an AWS Direct Connect connection) would not be necessary, as the requirement can be met by simply creating a VPC endpoint for Amazon S3 and routing traffic through it."

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: C

C is the correct answer.

upvoted 1 times

Ruffyit 2 years, 7 months ago

C. Move the EC2 instances to private subnets. Create a VPC endpoint for Amazon S3, and link the endpoint to the route table for the private subnets.

upvoted 1 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: C

Move the EC2 instances to private subnets. Create a VPC endpoint for Amazon S3, and link the endpoint to the route table for the private subnets.

upvoted 1 times

sand444 2 years, 9 months ago

Selected Answer: C

link VPC endpoint in route tables ---- EC2 instance to communicate S3 with a private connection in VPC

upvoted 2 times

DavidNamy 3 years, 5 months ago

Selected Answer: C

According to the well-designed framework, option C is the safest and most efficient option.

upvoted 3 times

career360guru 3 years, 5 months ago

Selected Answer: C

Option C

upvoted 1 times

ocbn3wby 3 years, 6 months ago

C is correct.
There is no requirement for public access from internet.

Application must be moved in Private subnet. This is a prerequisite in using VPC endpoints with S3
https://aws.amazon.com/blogs/storage/managing-amazon-s3-access-with-vpc-endpoints-and-s3-access-points/

upvoted 5 times

Wpcorgan 3 years, 6 months ago

C is correct

upvoted 1 times

Jtic 3 years, 6 months ago

Selected Answer: C

Use VPC endpoint

upvoted 1 times

Jtic 3 years, 7 months ago

Selected Answer: C

User VPC endpoint and make the EC2 private

upvoted 1 times

Jtic 3 years, 7 months ago

Use VPC endpoint

upvoted 1 times

backbencher2022 3 years, 7 months ago

Selected Answer: C

VPC endpoint is the best choice to route S3 traffic without traversing internet. Option A alone can't be used as NAT Gateway requires an Internet gateway for outbound internet traffic. Option B would still require traversing through internet and option D is also not a suitable solution

upvoted 4 times

Question #116 *Topic 1*

A company uses a popular content management system (CMS) for its corporate website. However, the required patching and maintenance are burdensome. The company is redesigning its website and wants anew solution. The website will be updated four times a year and does not need to have any dynamic content available. The solution must provide high scalability and enhanced security.
Which combination of changes will meet these requirements with the LEAST operational overhead? (Choose two.)

A. Configure Amazon CloudFront in front of the website to use HTTPS functionality. **Most Voted**

B. Deploy an AWS WAF web ACL in front of the website to provide HTTPS functionality.

C. Create and deploy an AWS Lambda function to manage and serve the website content.

D. Create the new website and an Amazon S3 bucket. Deploy the website on the S3 bucket with static website hosting enabled. **Most Voted**

E. Create the new website. Deploy the website by using an Auto Scaling group of Amazon EC2 instances behind an Application Load Balancer.

Correct Answer: AD

Community vote distribution

AD (89%) 5%

Comments

palermo777 Highly Voted 3 years, 7 months ago
A -> We can configure CloudFront to require HTTPS from clients (enhanced security)
https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/using-https-viewers-to-cloudfront.html
D -> storing static website on S3 provides scalability and less operational overhead, then configuration of Application LB and EC2 instances (hence E is out)

B is out since AWS WAF Web ACL does not to provide HTTPS functionality, but to protect HTTPS only.
upvoted 36 times

Six_Fingered_Jose Highly Voted 3 years, 7 months ago
Selected Answer: AD
agree with A and D

static website -> obviously S3, and S3 is super scalable
CDN -> CloudFront obviously as well, and with HTTPS security is enhanced.

B does not make sense because you are not replacing the CDN with anything,
E works too but takes too much effort and compared to S3, S3 still wins in term of scalability. plus why use EC2 when you are only hosting static website
upvoted 10 times

aussiehoa 3 years, 1 month ago
does not need to have any dynamic content available
upvoted 2 times

Lalo 3 years ago
Amazon CloudFront is for Securely deliver content with low latency and high transfer speeds
But what about the SQLinjection XSS attacks? we use WAF and olso use HTTPS
https://www.f5.com/glossary/web-application-firewall-waf#:~:text=A%20WAF%20protects%20your%20web,and%20what%20traffic%20is%20safe.
WAF protects your web apps by filtering, monitoring, and blocking any malicious HTTP/S traffic traveling to the web application, and prevents any unauthorized data from leaving the app.
Answer is WAF Not Cloudfront
upvoted 3 times

jaradat02 Most Recent 1 year, 10 months ago

Selected Answer: AD

A and D is the safest combination.

upvoted 2 times

David_Ang 2 years, 8 months ago

Selected Answer: AD

these answers are the most common use case for real companies, is like the answers that have more sense

upvoted 3 times

tom_cruise 2 years, 8 months ago

Selected Answer: AD

Web Application Firewall creates rules to block attacks, but it does not create HTTPS. It can only allow HTTPS inbound traffic.

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: AD

Scalability, enhanced security and less operational overhead = CloudFront with HTTPS
Scalability and less operational overhead = S3 bucket with static website hosting

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: AD

A. Amazon CloudFront provides scalable content delivery with HTTPS functionality, meeting security and scalability requirements.

D. Deploying the website on an Amazon S3 bucket with static website hosting reduces operational overhead by eliminating server maintenance and patching.

Why other options are incorrect:

B. AWS WAF does not provide HTTPS functionality or address patching and maintenance.

C. Using AWS Lambda introduces complexity and does not directly address patching and maintenance.

E. Managing EC2 instances and an Application Load Balancer increases operational overhead and does not minimize patching and maintenance tasks.

In summary, configuring Amazon CloudFront for HTTPS and deploying on Amazon S3 with static website hosting provide security, scalability, and reduced operational overhead.

upvoted 2 times

beginnercloud 3 years ago

Selected Answer: AD

AD

A for enhanced security D for static content

upvoted 2 times

studynoplay 3 years, 1 month ago

Selected Answer: AD

LEAST operational overhead = Serverless
https://aws.amazon.com/serverless/

upvoted 4 times

angolateoria 3 years, 1 month ago

AD misses the operational part, how can the app work without a lambda function, an EC2 instance or something?

upvoted 1 times

darn 3 years, 1 month ago

Selected Answer: AD

people do not seem to get the LEAST OPERATIONAL OVERHEAD statement, many people keep voting for options that bring far too Op work

upvoted 2 times

channn 3 years, 2 months ago

Selected Answer: AD

A for enhanced security
D for static content

upvoted 3 times

Erbug 3 years, 2 months ago

Since Amazon S3 is unlimited and you pay as you go so it means there will be no limit to scale as long as your data is going to grow, so D is one of the correct answers and another correct answer is A, because of this: https://docs.aws.amazon.com/AmazonS3/latest/userguide/WebsiteHosting.htm

so my answer is AD.

upvoted 2 times

ManOnTheMoon 3 years, 3 months ago

I vote A & C for the reason being least operational overhead.

upvoted 1 times

Yelizaveta 3 years, 3 months ago

Selected Answer: AD

Here a perfect explanation:
https://aws.amazon.com/premiumsupport/knowledge-center/cloudfront-serve-static-website/

upvoted 4 times

Abdel42 3 years, 4 months ago

Selected Answer: AD

Simple and secure

upvoted 1 times

remand 3 years, 4 months ago

Selected Answer: AD

D. Create the new website and an Amazon S3 bucket. Deploy the website on the S3 bucket with static website hosting enabled.
A. Configure Amazon CloudFront in front of the website to use HTTPS functionality.

By deploying the website on an S3 bucket with static website hosting enabled, the company can take advantage of the high scalability and cost-efficiency of S3 while also reducing the operational overhead of managing and patching a CMS.
By configuring Amazon CloudFront in front of the website, it will automatically handle the HTTPS functionality, this way the company can have a secure website with very low operational overhead.

upvoted 2 times

Question #117 *Topic 1*

A company stores its application logs in an Amazon CloudWatch Logs log group. A new policy requires the company to store all application logs in Amazon OpenSearch Service (Amazon Elasticsearch Service) in near-real time.
Which solution will meet this requirement with the LEAST operational overhead?

A. Configure a CloudWatch Logs subscription to stream the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service). **Most Voted**

B. Create an AWS Lambda function. Use the log group to invoke the function to write the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service).

C. Create an Amazon Kinesis Data Firehose delivery stream. Configure the log group as the delivery streams sources. Configure Amazon OpenSearch Service (Amazon Elasticsearch Service) as the delivery stream's destination.

D. Install and configure Amazon Kinesis Agent on each application server to deliver the logs to Amazon Kinesis Data Streams. Configure Kinesis Data Streams to deliver the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service).

Correct Answer: A

Community vote distribution

A (67%) C (31%) 1

Comments

Six_Fingered_Jose Highly Voted 3 years, 7 months ago
Selected Answer: A
answer is A
https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/CWL_OpenSearch_Stream.html

> You can configure a CloudWatch Logs log group to stream data it receives to your Amazon OpenSearch Service cluster in NEAR REAL-TIME through a CloudWatch Logs subscription

least overhead compared to kinesis
upvoted 103 times

HayLLlHuK 3 years, 5 months ago
Zerotn3 is right! There should be a Lambda for writing into ES
upvoted 1 times

UWSFish 3 years, 7 months ago
Great link. Convinced me
upvoted 5 times

Vandaman 1 year, 3 months ago
Thank you for the link - clear answer
upvoted 2 times

Zerotn3 3 years, 5 months ago
Option A (Configure a CloudWatch Logs subscription to stream the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service)) is not a suitable option, as a CloudWatch Logs subscription is designed to send log events to a destination such as an Amazon Simple Notification Service (Amazon SNS) topic or an AWS Lambda function. It is not designed to write logs directly to Amazon Elasticsearch Service (Amazon ES).
upvoted 4 times

kucyk 3 years, 3 months ago
that is not true, you can stream logs from CloudWatch Logs directly to OpenSearch
upvoted 7 times

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago
Selected Answer: C

The correct answer is C: Create an Amazon Kinesis Data Firehose delivery stream. Configure the log group as the delivery stream source. Configure Amazon OpenSearch Service (Amazon Elasticsearch Service) as the delivery stream's destination.

This solution uses Amazon Kinesis Data Firehose, which is a fully managed service for streaming data to Amazon OpenSearch Service (Amazon Elasticsearch Service) and other destinations. You can configure the log group as the source of the delivery stream and Amazon OpenSearch Service as the destination. This solution requires minimal operational overhead, as Kinesis Data Firehose automatically scales and handles data delivery, transformation, and indexing.

upvoted 19 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A: Configure a CloudWatch Logs subscription to stream the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service) would also work, but it may require more operational overhead as you would need to set up and manage the subscription and ensure that the logs are delivered in near-real time.

Option B: Create an AWS Lambda function. Use the log group to invoke the function to write the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service) would also work, but it may require more operational overhead as you would need to set up and manage the Lambda function and ensure that it scales to handle the incoming logs.

Option D: Install and configure Amazon Kinesis Agent on each application server to deliver the logs to Amazon Kinesis Data Streams. Configure Kinesis Data Streams to deliver the logs to Amazon OpenSearch Service (Amazon Elasticsearch Service) would also work, but it may require more operational overhead as you would need to install and configure the Kinesis Agent on each application server and set up and manage the Kinesis Data Streams.

upvoted 3 times

ocbn3wby 3 years, 4 months ago

https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/CWL_OpenSearch_Stream.html

upvoted 2 times

Lalo 3 years ago

ANSWER A
https://docs.aws.amazon.com/opensearch-service/latest/developerguide/integrations.html
You can use CloudWatch or Kinesis, but in the Kinesis description it never says real time, however in the Cloudwatch description it does say Real time ""You can load streaming data from CloudWatch Logs to your OpenSearch Service domain by using a CloudWatch Logs subscription . For information about Amazon CloudWatch subscriptions, see Real-time processing of log data with subscriptions.""

upvoted 4 times

ln2718 Most Recent 9 months, 4 weeks ago

Selected Answer: C

Chat says answer C -
A isn't supported (CloudWatch Logs can't stream directly to OpenSearch).
B (Lambda) adds code/ops you don't need.
D requires installing agents on servers and using Data Streams—unnecessary overhead.

upvoted 2 times

Mischi 1 year, 5 months ago

Selected Answer: D

In summary, the CloudWatch Logs → Kinesis Data Firehose → Amazon OpenSearch Service (option C) integration is the path recommended by AWS for this type of case. It allows for near real-time transmission, automatic scaling and relatively simple configuration, with the lowest operational overhead .

upvoted 1 times

0de7d1b 1 year, 6 months ago

Selected Answer: A

CloudWatch Logs subscription filter: This is the most straightforward way to stream logs from a CloudWatch Logs group to Amazon OpenSearch Service (Amazon Elasticsearch Service) in near real-time. It eliminates the need for additional components or complex configurations, reducing operational overhead.
Direct integration: CloudWatch Logs can directly stream logs to OpenSearch Service without requiring intermediate services, making it a simple and efficient solution.
Low operational overhead: Once set up, the subscription filter automatically forwards logs to OpenSearch Service with minimal maintenance.

upvoted 2 times

ChymKuBoy 1 year, 7 months ago

Selected Answer: C

C for sure
Simplicity: Kinesis Data Firehose is a managed service that handles the task of capturing, transforming, and loading data into destinations like Amazon OpenSearch Service. This eliminates the need for complex configuration and management.
Scalability: Kinesis Data Firehose can automatically scale to handle varying data volumes, ensuring that logs are ingested in near-real time.
Cost-effectiveness: Kinesis Data Firehose is a pay-as-you-go service, making it a cost-effective option for log ingestion and analysis.

upvoted 2 times

tonybuivannghia 1 year, 8 months ago

Selected Answer: C

I think C is correct because the cloud watch subscription can't stream directly to OpenSearch, it is via Lambda, SNS, FireHouse,....

upvoted 3 times

Tieri 1 year, 8 months ago

You can configure a log group in Amazon CloudWatch Logs, so you can stream data to your Amazon OpenSearch Service cluster in near real-time.

upvoted 1 times

Johnoppong101 1 year, 9 months ago

Selected Answer: A

https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/CWL_OpenSearch_Stream.html

upvoted 2 times

KTEgghead 1 year, 10 months ago

Selected Answer: A

Configure a CloudWatch Logs log group to stream data directly to the Amazon OpenSearch Service cluster. This can be done through a CloudWatch Logs subscription, which allows for real-time processing of log data.

upvoted 1 times

jaradat02 1 year, 10 months ago

Selected Answer: A

A is the correct answer, CloudWatch offers a subscription where you can stream data to other AWS services

upvoted 1 times

Seb888 1 year, 11 months ago

Selected Answer: C

Correct Answer:
C. Create an Amazon Kinesis Data Firehose delivery stream. Configure the log group as the delivery stream's source. Configure Amazon OpenSearch Service (Amazon Elasticsearch Service) as the delivery stream's destination.

Explanation:
Amazon Kinesis Data Firehose is a fully managed service for delivering real-time streaming data to destinations such as Amazon OpenSearch Service
It requires minimal setup and management, making it a low-overhead solution.
By configuring the log group as the source for the Kinesis Data Firehose delivery stream and Amazon OpenSearch Service as the destination, logs ca
be delivered in near-real time with built-in reliability and scalability.

upvoted 2 times

jatric 1 year, 11 months ago

Selected Answer: A

easy enough to figure out. Option A

upvoted 1 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: A

A for sure

upvoted 1 times

824c449 2 years, 1 month ago

Selected Answer: C

It can natively connect to CloudWatch Logs as a source and OpenSearch Service as a destination, handling the delivery of logs efficiently and with minimal setup. This approach offers the least operational overhead by simplifying the data transfer pipeline with automatic scaling and error handling.

upvoted 1 times

zinabu 2 years, 1 month ago

Selected Answer: A

You can configure a CloudWatch Logs log group to stream data it receives to your Amazon OpenSearch Service cluster in near real-time through a CloudWatch Logs subscription.

here is the link/; https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/CWL_OpenSearch_Stream.html

upvoted 1 times

OctavioBatera 2 years, 2 months ago

Selected Answer: A

Answer A.
This doc clarifies the subject: https://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/CWL_OpenSearch_Stream.html

upvoted 1 times

Question #118 *Topic 1*

A company is building a web-based application running on Amazon EC2 instances in multiple Availability Zones. The web application will provide access to a repository of text documents totaling about 900 TB in size. The company anticipates that the web application will experience periods of high demand. A solutions architect must ensure that the storage component for the text documents can scale to meet the demand of the application at all times. The company is concerned about the overall cost of the solution.

Which storage solution meets these requirements MOST cost-effectively?

A. Amazon Elastic Block Store (Amazon EBS)

B. Amazon Elastic File System (Amazon EFS)

C. Amazon OpenSearch Service (Amazon Elasticsearch Service)

D. Amazon S3 **Most Voted**

Correct Answer: D

Community vote distribution

D (98%) 2%

Comments

Azure55 Highly Voted 2 years, 7 months ago

Selected Answer: D

the cost of S3<EFS<EBS

upvoted 17 times

cookieMr Highly Voted 2 years, 11 months ago

Selected Answer: D

Amazon S3 (Simple Storage Service) is a highly scalable and cost-effective storage service. It is well-suited for storing large amounts of data, such as the 900 TB of text documents mentioned in the scenario. S3 provides high durability, availability, and performance.

Option A (Amazon EBS) is block storage designed for individual EC2 instances and may not scale as seamlessly and cost-effectively as S3 for large amounts of data.

Option B (Amazon EFS) is a scalable file storage service, but it may not be the most cost-effective option compared to S3, especially for the anticipated storage size of 900 TB.

Option C (Amazon OpenSearch Service) is a search and analytics service and may not be suitable as the primary storage solution for the text documents.

In summary, Amazon S3 is the recommended choice as it offers high scalability, cost-effectiveness, and durability for storing the large repository of text documents required by the web application.

upvoted 13 times

PaulGa Most Recent 1 year, 8 months ago

Selected Answer: D

Ans D - Amazon S3 is highly scalable, cost-effective storage service, well-suited for large amounts of data. It is highly durable, highly available, and offers good performance.
By comparison, EFS (option B) could do it but is more expensive...

upvoted 3 times

Americanman 1 year, 9 months ago

S3 can be a good option for storing text documents. It allows users to store any file type as objects. (documents, videos, images)

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: D

Using EFS would obviously be the optimal case, we have to use s3 to fulfill the cost efficiency requirement.
upvoted 3 times

theochan 2 years, 4 months ago
Option A : EBS can't be multi-AZ
Option B: EFS is expensive
Option C: ElasticSearch is not for storing
upvoted 4 times

awashenko 2 years, 8 months ago
Selected Answer: D
D is the only real solution here. S3 is the cheapest option for storage and it can scale indefinitely.
upvoted 5 times

Guru4Cloud 2 years, 10 months ago
Selected Answer: D
MOST cost-effective = S3 (unless explicitly stated in the requirements)
upvoted 4 times

Jeeva28 3 years ago
Selected Answer: D
900 in the question to divert our Thinking.When you have keyword least in question S3 will be only thing we should look
upvoted 3 times

Abrar2022 3 years ago
EFS and S3 meet the requirements but S3 is a better option because it is cheaper.
upvoted 2 times

studynoplay 3 years, 1 month ago
Selected Answer: D
MOST cost-effective = S3 (unless explicitly stated in the requirements)
upvoted 3 times

Robrobtutu 3 years, 1 month ago
Selected Answer: D
S3 is the cheapest and most scalable.
upvoted 2 times

jdr75 3 years, 2 months ago
Selected Answer: C
Now in OpenSearch you can reach at 3 PB so option C is better.
With S3 in an intensive scenario the costs of retriving the buckets could be high.
Yes OpenSearch is NOT cheap but this has to be analysed carefully.
So, I opt "C" to increase the discussion.

With UltraWarm, you can retain up to 3 PB of data on a single Amazon OpenSearch Service cluster, while reducing your cost per GB by nearly 90% compared to the warm storage tier. You can also easily query and visualize the data in your Kibana interface (version 7.10 and earlier) or OpenSearch Dashboards. Analyze both your recent (weeks) and historical (months or years) log data without spending hours or days restoring archived logs.

https://aws.amazon.com/es/opensearch-service/features/
upvoted 2 times

Dr_Chomp 3 years, 2 months ago
EFS is a good option but expensive alongside S3 and customer concerned about cost - thus: S3 (D)
upvoted 3 times

frenzoid 3 years, 2 months ago
I wonder why people choose S3, yet S3 max capacity is 5TB 😳.
upvoted 3 times

frenzoid 3 years, 2 months ago
My bad, the 5TB limit is for individual files. S3 has virtually unlimited storage capacity.
upvoted 8 times

Help2023 3 years, 3 months ago
Selected Answer: D
A. It is Not a block storage
B. It is Not a file storage

C. Opensearch is useful but can only accommodate up to 600TiB and is mainly for search and anaytics.
D. S3 is more cost effective than all and can handle all objects like Block, File or Text.

upvoted 5 times

remand 3 years, 4 months ago

Selected Answer: D

D. Amazon S3

Amazon S3 is an object storage service that can store and retrieve large amounts of data at any time, from anywhere on the web. It is designed for high durability, scalability, and cost-effectiveness, making it a suitable choice for storing a large repository of text documents. With S3, you can store and retrieve any amount of data, at any time, from anywhere on the web, and you can scale your storage up or down as needed, which will help to meet the demand of the web application. Additionally, S3 allows you to choose between different storage classes, such as standard, infrequent access, and archive, which will enable you to optimize costs based on your specific use case.

upvoted 2 times

Question #119 Topic 1

A global company is using Amazon API Gateway to design REST APIs for its loyalty club users in the us-east-1 Region and the ap-southeast-2 Region. A solutions architect must design a solution to protect these API Gateway managed REST APIs across multiple accounts from SQL injection and cross-site scripting attacks.
Which solution will meet these requirements with the LEAST amount of administrative effort?

A. Set up AWS WAF in both Regions. Associate Regional web ACLs with an API stage.
B. Set up AWS Firewall Manager in both Regions. Centrally configure AWS WAF rules. Most Voted
C. Set up AWS Shield in bath Regions. Associate Regional web ACLs with an API stage.
D. Set up AWS Shield in one of the Regions. Associate Regional web ACLs with an API stage.

Correct Answer: B

Community vote distribution
B (70%) A (29%)

Comments

Gil80 Highly Voted 3 years, 7 months ago
Selected Answer: B
If you want to use AWS WAF across accounts, accelerate WAF configuration, automate the protection of new resources, use Firewall Manager with AWS WAF
upvoted 45 times

slimen 2 years, 7 months ago
they didn't mention multiple accounts! only 2 regoins
upvoted 3 times

lofzee 2 years ago
wtf? the question says
"to protect these API Gateway managed REST APIs across multiple accounts from SQL injection and cross-site scripting attack"
upvoted 10 times

baku98 2 years, 6 months ago
B is wrong: AWS Firewall Manager cannot create security policies across regions.
Q: Can I create security policies across regions? No, AWS Firewall Manager security policies are region specific. Each Firewall Manager policy can only include resources available in that specified AWS Region. You can create a new policy for each region where you operate.
https://aws.amazon.com/firewall-manager/faqs/#:~:text=No%2C%20AWS%20Firewall%20Manager%20security,in%20that%20specified%20AWS%20Region.
upvoted 9 times

mauroicardi 2 years, 2 months ago
AWS Firewall Manager is integrated with AWS Organizations so you can enable AWS WAF rules, AWS Shield Advanced protections, VPC security groups, AWS Network Firewalls, and Amazon Route 53 Resolver DNS Firewall rules across multiple AWS accounts and resources from a single place.
upvoted 4 times

pentium75 2 years, 5 months ago
That's why B says that you "set up AWS Firewall Manager IN BOTH REGIONS". Still you can "centrally configure" WAF per region, so that you don't have to attach WAF to every individual API.
upvoted 5 times

Nigma Highly Voted 3 years, 7 months ago
B

Using AWS WAF has several benefits. Additional protection against web attacks using criteria that you specify. You can define criteria using characteristics of web requests such as the following:
Presence of SQL code that is likely to be malicious (known as SQL injection).

Presence of a script that is likely to be malicious (known as cross-site scripting).

AWS Firewall Manager simplifies your administration and maintenance tasks across multiple accounts and resources for a variety of protections.

https://docs.aws.amazon.com/waf/latest/developerguide/what-is-aws-waf.html
upvoted 18 times

JayBee65 3 years, 5 months ago
Q: Can I create security policies across regions?

No, AWS Firewall Manager security policies are region specific. Each Firewall Manager policy can only include resources available in that specified AWS Region. You can create a new policy for each region where you operate.

So you could not centrally (i.e. in one place) configure policies, you would need to do this is each region
upvoted 4 times

pentium75 2 years, 5 months ago
"Centrally" on the Firewall Manager per region, as opposed to individually for every single API.
upvoted 3 times

In2718 Most Recent 9 months, 4 weeks ago
Selected Answer: B
Chat -

A. Set up AWS WAF in both Regions. Associate Regional web ACLs with an API stage.
Works, but requires managing rules in each region and account separately. Higher admin overhead.

B. Set up AWS Firewall Manager in both Regions. Centrally configure AWS WAF rules.
Correct. Firewall Manager lets you manage AWS WAF rules centrally across accounts and regions. This gives the least administrative effort.
upvoted 1 times

AwsAbhiKumar 1 year, 3 months ago
Selected Answer: A
AWF has Web ACL which has a rule regarding HTTP header, HHTP body or URL string protects from common attack - SQL Injection and cross site scripting.
upvoted 1 times

Dharmarajan 1 year, 4 months ago
Selected Answer: A
C&D are out of question as AWS Shield is only for DDoS attack prevention.
B is not supported - one single policy cannot be used for multiple regions.
That leaves out A.
upvoted 1 times

Dharmarajan 1 year, 4 months ago
OK I stand corrected: the choice does say "Centrally manager rules" and not "use a sigle policy". In that view B is indeed the most appropriate.
upvoted 1 times

Dharmarajan 1 year, 4 months ago
This is a tricky one. Again. AWS WAF is a product specifically for preventing SQL Injection attacks. So maybe A is indeed the right choice. I believe A is indeed the right answer now, after referring to the doc.
https://docs.aws.amazon.com/waf/latest/developerguide/waf-rule-statement-type-sqli-match.html
upvoted 1 times

FlyingHawk 1 year, 4 months ago
Selected Answer: B
Centralized control with Firewall Manager means you can create and manage WAF rules once and apply them across multiple accounts and Regions, ensuring consistency and compliance.
Firewall Manager automatically applies policies to new and existing resources across accounts, reducing the effort of manually associating web ACLs in each Region and account.
upvoted 2 times

FlyingHawk 1 year, 4 months ago
Selected Answer: B
If only a few accounts, A is more straight. but the question mentions "multiple accounts", may be B is better choice.
upvoted 1 times

tom_cruise 1 year, 6 months ago
Selected Answer: C
The keywords here is "across multiple accounts", not "across multiple regions".

upvoted 1 times

0de7d1b 1 year, 6 months ago

Selected Answer: B

AWS Firewall Manager:

Centralized Management: Allows you to centrally manage security policies across multiple accounts and regions.
WAF Rule Configuration: Enables you to create and manage WAF rules in a single location, simplifying the configuration process.
Automatic Deployment: Automatically deploys WAF rules to protected resources, reducing manual effort.
Policy-Based Control: Provides granular control over security policies, allowing you to tailor them to specific needs.

upvoted 2 times

leoo55 1 year, 7 months ago

upvoted 2 times

ChymKuBoy 1 year, 7 months ago

Selected Answer: B

B for sure
Centralized management: AWS Firewall Manager allows you to centrally manage AWS WAF rules across multiple accounts and regions. This simplifies the configuration and management process.

Consistent security policies: You can enforce consistent security policies across all your API Gateway APIs, ensuring that they are protected from the same threats.

Scalability: AWS Firewall Manager can handle a large number of accounts and resources, making it suitable for global companies with many API Gateway APIs.

upvoted 3 times

PaulGa 1 year, 8 months ago

Selected Answer: A

Ans A - "With AWS WAF, you can create security rules that control bot traffic and block common attack patterns such as SQL injection or cross-site scripting (XSS). Use cases. Filter web traffic."
https://aws.amazon.com › waf

None of the other options can do it.

upvoted 1 times

Americanman 1 year, 9 months ago

AWS WAF helps you to protect your application against common web exploits and bots that can affect availability, compromise security or consume excessive resources.
You can create security rules that will control bot traffic and common attacks like SQL injection or Cross-site scripting (XSS)

upvoted 1 times

jaradat02 1 year, 10 months ago

Selected Answer: B

A is valid, but B achieves the least operational overhead.

upvoted 1 times

TilTil 2 years, 2 months ago

Selected Answer: A

WAF deals well with the types of attacks mentioned. XSS and SQL Injection are both app level attacks hence needs a WAF.

upvoted 2 times

sirasdf 2 years, 3 months ago

B

Option A involves setting up AWS WAF in both regions and associating regional web ACLs with an API stage. While this can provide the necessary protection, it requires more manual configuration in each region, potentially leading to more administrative effort, especially if there are updates or changes needed to be made across multiple regions.

Therefore, Option B is likely to require the least amount of administrative effort.

upvoted 2 times

killbots 2 years, 4 months ago

Selected Answer: A

Original architecture does not have WAFs. B assumes there are WAFs already in place and why would you want to deploy a Firewall Manager to manage 1 Firewall? it adds unnecessary administrative tasks and costs for a tool that is not needed. You would want that if you were managing 10+ Firewalls not just one. A makes the most sense.

upvoted 4 times

Question #120 Topic 1

A company has implemented a self-managed DNS solution on three Amazon EC2 instances behind a Network Load Balancer (NLB) in the us-west-2 Region. Most of the company's users are located in the United States and Europe. The company wants to improve the performance and availability of the solution. The company launches and configures three EC2 instances in the eu-west-1 Region and adds the EC2 instances as targets for a new NLB.
Which solution can the company use to route traffic to all the EC2 instances?

A. Create an Amazon Route 53 geolocation routing policy to route requests to one of the two NLBs. Create an Amazon CloudFront distribution. Use the Route 53 record as the distribution's origin.

B. Create a standard accelerator in AWS Global Accelerator. Create endpoint groups in us-west-2 and eu-west-1. Add the two NLBs as endpoints for the endpoint groups. Most Voted

C. Attach Elastic IP addresses to the six EC2 instances. Create an Amazon Route 53 geolocation routing policy to route requests to one of the six EC2 instances. Create an Amazon CloudFront distribution. Use the Route 53 record as the distribution's origin.

D. Replace the two NLBs with two Application Load Balancers (ALBs). Create an Amazon Route 53 latency routing policy to route requests to one of the two ALBs. Create an Amazon CloudFront distribution. Use the Route 53 record as the distribution's origin.

Correct Answer: B

Community vote distribution

B (79%) A (18%) 3%

Comments

dokaedu Highly Voted 3 years, 7 months ago
B is the correct one for seld manage DNS
If need to use Route53, ALB (layar 7) needs to be used as end points for 2 reginal x 3 EC2s, if it the case answer would be the option 4
upvoted 21 times

RNess 2 years, 7 months ago
Why I need replace NLB to ALB?
upvoted 2 times

pentium75 2 years, 5 months ago
Who said that?
upvoted 3 times

EllenLiu 1 year, 5 months ago
A is not correct as should use NLB or ALB as the distribution origin?
upvoted 1 times

MutiverseAgent 2 years, 11 months ago
After reading the discussion I think the right answer is B, as the service they use is DNS it does not make sense using a cloudfront distribution for this. The scenario would be different if the service were HTTP/HTTPS.
upvoted 6 times

MutiverseAgent 2 years, 11 months ago
Just to complete my previous comment. If the scenario were that the company uses HTTP/HTTPS service, then the correct answer (as the original dokaedu message mentions) would be option D)
upvoted 4 times

LeGloupier Highly Voted 3 years, 7 months ago
Selected Answer: B
for me it is B
upvoted 13 times

90c3163 Most Recent 10 months, 1 week ago

Selected Answer: A

I hate the wording in this. Like it is a DNS service as in DNS servers, or DNS service as in providing DNS selling services. Because the answer would be different if they provided just a bit more context.

upvoted 1 times

satyaammm 1 year, 5 months ago

Selected Answer: B

AWS Global Accelerator is the most suitable here as it provides low latency and makes the AWS Network closer.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: B

Ans B - "AWS Global Accelerator improves the availability and performance of your applications for global users by routing traffic to the optimal endpoint based on performance and policies."

https://aws.amazon.com › global-accelerator › faqs

upvoted 2 times

rityoui 2 years, 3 months ago

Selected Answer: B

i choose a previous until i checked google that tells me "DNS is an Application-layer protocol"

upvoted 3 times

thewalker 2 years, 4 months ago

Selected Answer: A

A seems the right answer.

upvoted 1 times

pentium75 2 years, 5 months ago

Selected Answer: B

Not A: CloudFront is not for DNS
Not C: Involves CloudFront which is not needed, otherwise would work but ignore the NLBs
Not D: ALB can't handle DNS
Leaves B

upvoted 5 times

SaurabhTiwari1 2 years, 5 months ago

Selected Answer: B

Keyword-
AWS global accelerator = Super cop (who direct the traffic and give you the best way to reach your destination)

Geolocation is use for showing web content as you want to show your web content to particular country or continent.
Geolocation has nothing to do with traffic.

upvoted 3 times

SaurabhTiwari1 2 years, 5 months ago

Route 53 geolocation has nothing to do with traffic in the sense that it does not affect the amount or speed of traffic that reaches your resources. It only affects how Route 53 responds to DNS queries based on the location of your users.

upvoted 2 times

Masakichen 2 years, 6 months ago

Option B. Create a standard accelerator in AWS Global Accelerator. Establish endpoint groups in us-west-2 and eu-west-1. Add two NLBs as endpoints of the endpoint group.

AWS Global Accelerator is a network service that can provide a global traffic management solution. By creating a standard accelerator in AWS Global Accelerator, you can guide user traffic to the endpoint closest to them, thereby improving the performance and availability of the application. In this case, you can establish endpoint groups in the us-west-2 and eu-west-1 regions, and add two NLBs as endpoints. In this way, no matter where the user is located, their requests will be routed to the EC2 instance closest to them, thereby improving the performance and availability of DNS resolution. In addition, this design can also provide flexibility and scalability to handle a large amount of traffic. Therefore, this solution can meet you needs.

upvoted 8 times

Ruffyit 2 years, 7 months ago

Global Accelerator: AWS Global Accelerator is designed to improve the availability and performance of applications by using static IP addresses (Anycast IPs) and routing traffic over the AWS global network infrastructure.

Endpoint Groups: By creating endpoint groups in both the us-west-2 and eu-west-1 Regions, the company can effectively distribute traffic to the NLBs in both Regions. This improves availability and allows traffic to be directed to the closest Region based on latency.

upvoted 2 times

tom_cruise 2 years, 8 months ago

Selected Answer: B

Key: route traffic to all the EC2 instances

upvoted 2 times

Hassaoo 2 years, 9 months ago

B. Create a standard accelerator in AWS Global Accelerator. Create endpoint groups in us-west-2 and eu-west-1. Add the two NLBs as endpoints for the endpoint groups.

Here's why this option is the most suitable:

Global Accelerator: AWS Global Accelerator is designed to improve the availability and performance of applications by using static IP addresses (Anycast IPs) and routing traffic over the AWS global network infrastructure.

Endpoint Groups: By creating endpoint groups in both the us-west-2 and eu-west-1 Regions, the company can effectively distribute traffic to the NLBs in both Regions. This improves availability and allows traffic to be directed to the closest Region based on latency.

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

B is the best solution to route traffic to all the EC2 instances across regions.

The key reasons are:

AWS Global Accelerator allows routing traffic to endpoints in multiple AWS Regions. It uses the AWS global network to optimize availability and performance.
Creating an accelerator with endpoint groups in us-west-2 and eu-west-1 allows traffic to be distributed across both regions.
Adding the NLBs in each region as endpoints allows the traffic to be routed to the EC2 instances behind them.
This provides improved performance and availability compared to just using Route 53 geolocation routing.

upvoted 6 times

MNotABot 2 years, 11 months ago

B
route requests to one of the two NLBs --> hence AD out / Attach Elastic IP addresses --> who will pay for it?

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: B

Option B offers a global solution by utilizing Global Accelerator. By creating a standard accelerator and configuring endpoint groups in both Regions the company can route traffic to all the EC2 across multiple regions. Adding the two NLBs as endpoints ensures that traffic is distributed effectively.

Option A does not directly address the requirement of routing traffic to all EC2 instances. It focuses on routing based on geolocation and using CloudFront as a distribution, which may not achieve the desired outcome.

Option C involves managing Elastic IP addresses and routing based on geolocation. However, it may not provide the same level of performance and availability as AWS Global Accelerator.

Option D focuses on ALBs and latency-based routing. While it can be a valid solution, it does not utilize AWS Global Accelerator and may require more configuration and management compared to option B.

upvoted 4 times

beginnercloud 3 years ago

Selected Answer: B

Correctly is B.

if it is self-managed DNS, you cannot use Route 53. There can be only 1 DNS service for the domain.

upvoted 5 times

Question #121 *Topic 1*

A company is running an online transaction processing (OLTP) workload on AWS. This workload uses an unencrypted Amazon RDS DB instance in a Multi-AZ deployment. Daily database snapshots are taken from this instance.
What should a solutions architect do to ensure the database and snapshots are always encrypted moving forward?

A. Encrypt a copy of the latest DB snapshot. Replace existing DB instance by restoring the encrypted snapshot. **Most Voted**

B. Create a new encrypted Amazon Elastic Block Store (Amazon EBS) volume and copy the snapshots to it. Enable encryption on the DB instance.

C. Copy the snapshots and enable encryption using AWS Key Management Service (AWS KMS) Restore encrypted snapshot to an existing DB instance.

D. Copy the snapshots to an Amazon S3 bucket that is encrypted using server-side encryption with AWS Key Management Service (AWS KMS) managed keys (SSE-KMS).

Correct Answer: A

Community vote distribution

A (85%) Other

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: A

"You can enable encryption for an Amazon RDS DB instance when you create it, but not after it's created. However, you can add encryption to an unencrypted DB instance by creating a snapshot of your DB instance, and then creating an encrypted copy of that snapshot. You can then restore a DB instance from the encrypted snapshot to get an encrypted copy of your original DB instance."
https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/encrypt-an-existing-amazon-rds-for-postgresql-db-instance.html

upvoted 67 times

JoeGuan 2 years, 10 months ago

I agree, there is no reason to copy all of the snapshots and ecnrypt them all. You just need one encrypted snapshot, moving forward they will all be encrypted. C is close but I think there is no reason to copy all the snapshots plural. There is a wizard to go through and select the snapshot to encrypt. "In the Amazon RDS console navigation pane, choose Snapshots, and select the DB snapshot you created. For Actions, choose Copy Snapshot. Provide the destination AWS Region and the name of the DB snapshot copy in the corresponding fields. Select the Enable Encryption checkbox. For Master Key, specify the KMS key identifier to use to encrypt the DB snapshot copy. Choose Copy Snapshot. For more information, see Copying a snapshot in the Amazon RDS documentation". What if you had 30 snapshotS? You just need to do it once.

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

In simple terms, you double it the affort of your work and spending money by creating unnessary snapshots... so A is the best choice

upvoted 2 times

Futurebones 3 years ago

How can A gurantee future encryption?

upvoted 4 times

Smart 2 years, 10 months ago

Once DB is encrypted, newer snapshots and read replicas will also be encrypted.

upvoted 8 times

1e22522 1 year, 10 months ago

thats crazy cuh the more you know ig

upvoted 2 times

kruasan Highly Voted 3 years, 1 month ago

Selected Answer: A

You can't restore from a DB snapshot to an existing DB instance; a new DB instance is created when you restore.

upvoted 5 times

rmanuraj Most Recent 1 year, 5 months ago

Selected Answer: A

A seems to be the correct option compared to C. In the question it clearly says the current DB instance is an unencrypted Amazon RDS DB instance. And you can't restore an encrypted snapshot to an unencrypted DB instance.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: A

Ans A - I must admit almsot put Ans C, but re-reading question and seeing comments its clear that encryption is needed "moving forward" so C is overkill...

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: A

Replacing a snapshots creates a new one instead of restoring the old one.

upvoted 2 times

Saadiii 2 years, 1 month ago

Selected Answer: A

I feel this is a bit tricky in the way the question is asked, but C implies that you are encrypting the snapshot. You are not. It is the DB that receives a KMS key upon restoring, but the snapshot is still unencrypted

upvoted 2 times

SinghJagdeep 2 years, 5 months ago

Selected Answer: A

Correct. Please visit for more details.
https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/encrypt-an-existing-amazon-rds-for-postgresql-db-instance.html

upvoted 2 times

ansagr 2 years, 6 months ago

Selected Answer: A

AWS RDS does not support direct restoration of an encrypted snapshot to an existing unencrypted DB instance. When you restore a snapshot, it creates a new DB instance with the same configuration as the original instance.

upvoted 2 times

tom_cruise 2 years, 7 months ago

Selected Answer: A

What's wrong with C is: "Copy the snapshots and enable encryption"

upvoted 3 times

tom_cruise 2 years, 8 months ago

Selected Answer: A

key: snapshots

upvoted 2 times

AntonioMinolfi 2 years, 8 months ago

Selected Answer: A

I was undecided if to choose A or C.
But since you can't restore a snapshot to an existing instance C is out. You can only create a new one.
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_RestoreFromSnapshot.html#:~:text=You%20can%27t%20restore%20from%20a%20DB%20snapshot%20to%20an%20existing%20DB%20instance%3B%20a%20new%20DB%20instance%20is%20created%20when%20you%20restore.

upvoted 2 times

TMabs 2 years, 8 months ago

A makes sence

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: C

A. Replacing the existing DB instance with an encrypted snapshot can result in downtime and potential data loss during migration.

B. Creating a new encrypted EBS volume for snapshots does not address the encryption of the DB instance itself.

D. Copying snapshots to an encrypted S3 bucket only encrypts the snapshots, but does not address the encryption of the DB instance.

Option C is the most suitable as it involves copying and encrypting the snapshots using AWS KMS, ensuring encryption for both the database and snapshots.

upvoted 2 times

BartoszGolebiowski24 2 years, 7 months ago

From the question:
"...What should a solutions architect do to ensure the database and snapshots are always encrypted moving forward?"
I think the question is about encrypting current and future snapshots instead of the old snapshots.

upvoted 2 times

Abrar2022 3 years ago

If daily snapshots are taken from the daily DB instance. Why create another copy? You just need to encrypt the latest daily DB snapshot and the restore from the existing encrypted snapshot.

upvoted 3 times

C_M_M 3 years, 1 month ago

A and C are almost similar except that A is latest snapshot, while C is snapshots (all the snapshots).
I don't see any other difference btw those two options.
Option A is clearly the correct on as all you need is the latest snapshot.

upvoted 3 times

JoeGuan 2 years, 10 months ago

I agree, in the wizard you would select ONE SNAPSHOT (singular in A), not all of the SNAPSHOTS (Plural in C)

upvoted 1 times

rushlav 3 years, 1 month ago

A
You can only encrypt an Amazon RDS DB instance when you create it, not after the DB instance is created.
However, because you can encrypt a copy of an unencrypted snapshot, you can effectively add encryption to an unencrypted DB instance. That is, you can create a snapshot of your DB instance, and then create an encrypted copy of that snapshot. You can then restore a DB instance from the encrypted snapshot, and thus you have an encrypted copy of your original DB instance.
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Overview.Encryption.html

upvoted 2 times

Abhineet9148232 3 years, 2 months ago

Selected Answer: C

Encryption is enabled during the Copy process itself.
https://repost.aws/knowledge-center/encrypt-rds-snapshots

upvoted 1 times

Question #122 Topic 1

A company wants to build a scalable key management infrastructure to support developers who need to encrypt data in their applications.
What should a solutions architect do to reduce the operational burden?

A. Use multi-factor authentication (MFA) to protect the encryption keys.

B. Use AWS Key Management Service (AWS KMS) to protect the encryption keys. Most Voted

C. Use AWS Certificate Manager (ACM) to create, store, and assign the encryption keys.

D. Use an IAM policy to limit the scope of users who have access permissions to protect the encryption keys.

Correct Answer: B

Community vote distribution

B (100%)

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: B

If you are a developer who needs to digitally sign or verify data using asymmetric keys, you should use the service to create and manage the private keys you'll need. If you're looking for a scalable key management infrastructure to support your developers and their growing number of applications, you should use it to reduce your licensing costs and operational burden...
https://aws.amazon.com/kms/faqs/#:~:text=If%20you%20are%20a%20developer%20who%20needs%20to%20digitally,a%20broad%20set%20of%20ndustry%20and%20regional%20compliance%20regimes.

upvoted 25 times

ocbn3wby 3 years, 6 months ago

Most documented answers. Thank you, 123jhl0.

upvoted 6 times

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago

Selected Answer: B

The correct answer is Option B. To reduce the operational burden, the solutions architect should use AWS Key Management Service (AWS KMS) to protect the encryption keys.

AWS KMS is a fully managed service that makes it easy to create and manage encryption keys. It allows developers to easily encrypt and decrypt data in their applications, and it automatically handles the underlying key management tasks, such as key generation, key rotation, and key deletion. This can help to reduce the operational burden associated with key management.

upvoted 9 times

PaulGa Most Recent 1 year, 8 months ago

Selected Answer: B

Ans B - AWS KMS does it all as a managed service...

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: B

to reduce the operational burden, option B is the best choice.

upvoted 2 times

Dani29 2 years, 1 month ago

Selected Answer: B

The correct answer is Option B. To reduce the operational burden, the solutions architect should use AWS Key Management Service (AWS KMS) to protect the encryption keys

upvoted 2 times

Ruffyit 2 years, 7 months ago

AWS KMS handles the encryption key management, rotation, and auditing. This removes the undifferentiated heavy lifting for developers.
KMS integrates natively with many AWS services like S3, EBS, RDS for encryption. This makes it easy to encrypt data.
KMS scales automatically as key usage increases. Developers don't have to worry about provisioning key infrastructure.
Fine-grained access controls are available via IAM policies and grants. KMS is secure by default.
Features like envelope encryption make compliance easier for regulated workloads.
AWS handles the hardware security modules (HSMs) for cryptographic key storage

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

The main reasons are:

AWS KMS handles the encryption key management, rotation, and auditing. This removes the undifferentiated heavy lifting for developers.
KMS integrates natively with many AWS services like S3, EBS, RDS for encryption. This makes it easy to encrypt data.
KMS scales automatically as key usage increases. Developers don't have to worry about provisioning key infrastructure.
Fine-grained access controls are available via IAM policies and grants. KMS is secure by default.
Features like envelope encryption make compliance easier for regulated workloads.
AWS handles the hardware security modules (HSMs) for cryptographic key storage

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: B

By utilizing AWS KMS, the company can offload the operational responsibilities of key management, including key generation, rotation, and protection. AWS KMS provides a scalable and secure infrastructure for managing encryption keys, allowing developers to easily integrate encryption into their applications without the need to manage the underlying key infrastructure.

Option A (MFA), option C (ACM), and option D (IAM policy) are not directly related to reducing the operational burden of key management. While these options may provide additional security measures or access controls, they do not specifically address the scalability and management aspects of a key management infrastructure. AWS KMS is designed to simplify the key management process and is the most suitable option for reducing the operational burden in this scenario.

upvoted 3 times

cheese929 3 years, 1 month ago

Selected Answer: B

B is correct.

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: B

Option B

upvoted 1 times

Wpcorgan 3 years, 6 months ago

B is correct

upvoted 1 times

Wpcorgan 3 years, 6 months ago

B is correct

upvoted 1 times

Jtic 3 years, 7 months ago

Selected Answer: B

If you are responsible for securing your data across AWS services, you should use it to centrally manage the encryption keys that control access to your data. If you are a developer who needs to encrypt data in your applications, you should use the AWS Encryption SDK with AWS KMS to easily generate, use and protect symmetric encryption keys in your code.

upvoted 2 times

Question #123 *Topic 1*

A company has a dynamic web application hosted on two Amazon EC2 instances. The company has its own SSL certificate, which is on each instance to perform SSL termination.
There has been an increase in traffic recently, and the operations team determined that SSL encryption and decryption is causing the compute capacity of the web servers to reach their maximum limit.
What should a solutions architect do to increase the application's performance?

A. Create a new SSL certificate using AWS Certificate Manager (ACM). Install the ACM certificate on each instance.

B. Create an Amazon S3 bucket Migrate the SSL certificate to the S3 bucket. Configure the EC2 instances to reference the bucket for SSL termination.

C. Create another EC2 instance as a proxy server. Migrate the SSL certificate to the new instance and configure it to direct connections to the existing EC2 instances.

D. Import the SSL certificate into AWS Certificate Manager (ACM). Create an Application Load Balancer with an HTTPS listener that uses the SSL certificate from ACM. **Most Voted**

Correct Answer: D

Community vote distribution

D (96%) 4%

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: D

This issue is solved by SSL offloading, i.e. by moving the SSL termination task to the ALB.
https://aws.amazon.com/blogs/aws/elastic-load-balancer-support-for-ssl-termination/

upvoted 23 times

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago

Selected Answer: D

The correct answer is D. To increase the application's performance, the solutions architect should import the SSL certificate into AWS Certificate Manager (ACM) and create an Application Load Balancer with an HTTPS listener that uses the SSL certificate from ACM.

An Application Load Balancer (ALB) can offload the SSL termination process from the EC2 instances, which can help to increase the compute capacity available for the web application. By creating an ALB with an HTTPS listener and using the SSL certificate from ACM, the ALB can handle the SSL termination process, leaving the EC2 instances free to focus on running the web application.

upvoted 14 times

0bf2fba Most Recent 10 months, 3 weeks ago

Selected Answer: D

option D is correct

upvoted 1 times

satyaammm 1 year, 5 months ago

Selected Answer: D

Since the company has its own SSL certificate so its more suitable to import the company's SSL certificate and using HTTPS.

upvoted 1 times

PaulGa 1 year, 8 months ago

Ans D - well explained by Buruguduystunstugudunstuy (1yr, 8mth):
"To increase the application's performance, the solutions architect should import the SSL certificate into AWS Certificate Manager (ACM) and create an Application Load Balancer with an HTTPS listener that uses the SSL certificate from ACM.

An Application Load Balancer (ALB) can offload the SSL termination process from the EC2 instances, which can help to increase the compute capacity available for the web application. By creating an ALB with an HTTPS listener and using the SSL certificate from ACM, the ALB can handle the SSL termination process, leaving the EC2 instances free to focus on running the web application."

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: D

D is the correct answer.

upvoted 1 times

Ruffyit 2 years, 7 months ago

This issue is solved by SSL offloading, i.e. by moving the SSL termination task to the ALB.
https://aws.amazon.com/blogs/aws/elastic-load-balancer-support-for-ssl-termination

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

The key reasons are:

Using an Application Load Balancer with an HTTPS listener allows SSL termination to happen at the load balancer layer.
The EC2 instances behind the load balancer receive only unencrypted traffic, reducing load on them.
Importing the custom SSL certificate into ACM allows the ALB to use it for HTTPS listeners.
This removes the need to install and manage SSL certificates on each EC2 instance.
ALB handles the SSL overhead and scales automatically. The EC2 fleet focuses on app logic.
Options A, B, C don't offload SSL overhead from the EC2 instances themselves.

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: D

By using ACM to manage the SSL certificate and configuring an ALB with HTTPS listener, the SSL termination will be handled by the load balancer instead of the web servers. This offloading of SSL processing to the ALB reduces the compute capacity burden on the web servers and improves their performance by allowing them to focus on serving the dynamic web application.

Option A suggests creating a new SSL certificate using ACM, but it does not address the SSL termination offloading and load balancing capabilities provided by an ALB.

Option B suggests migrating the SSL certificate to an S3 bucket, but this approach does not provide the necessary SSL termination and load balancing functionalities.

Option C suggests creating another EC2 instance as a proxy server, but this adds unnecessary complexity and management overhead without leveraging the benefits of ALB's built-in load balancing and SSL termination capabilities.

Therefore, option D is the most suitable choice to increase the application's performance in this scenario.

upvoted 3 times

dejung 3 years, 4 months ago

Selected Answer: A

Why is A wrong?

upvoted 2 times

Yadav_Sanjay 3 years, 1 month ago

Company uses its own SSL certificate. Option A says.. Create a SSL certificate in ACM

upvoted 4 times

xdkonorek2 2 years, 7 months ago

ec2 instances still would be responsible for decrypting traffic and it wouldn't solve load issue

upvoted 2 times

remand 3 years, 4 months ago

Selected Answer: D

SSL termination is the process of ending an SSL/TLS connection. This is typically done by a device, such as a load balancer or a reverse proxy, that is positioned in front of one or more web servers. The device decrypts incoming SSL/TLS traffic and then forwards the unencrypted request to the web server. This allows the web server to process the request without the overhead of decrypting and encrypting the traffic. The device then re-encrypts the response from the web server and sends it back to the client. This allows the device to offload the SSL/TLS processing from the web servers and also allows for features such as SSL offloading, SSL bridging, and SSL acceleration.

upvoted 5 times

career360guru 3 years, 5 months ago

Selected Answer: D

Option D to offload the SSL encryption workload

upvoted 2 times

Aamee 3 years, 6 months ago

Selected Answer: D

Due to this statement particularly: "The company has its own SSL certificate" as it's not created from AWS ACM itself.

upvoted 1 times

Wpcorgan 3 years, 6 months ago

D is correct

upvoted 1 times

Six_Fingered_Jose 3 years, 7 months ago

Selected Answer: D

agree with D

upvoted 1 times

Question #124 *Topic 1*

A company has a highly dynamic batch processing job that uses many Amazon EC2 instances to complete it. The job is stateless in nature, can be started and stopped at any given time with no negative impact, and typically takes upwards of 60 minutes total to complete. The company has asked a solutions architect to design a scalable and cost-effective solution that meets the requirements of the job.
What should the solutions architect recommend?

A. Implement EC2 Spot Instances. **Most Voted**

B. Purchase EC2 Reserved Instances.

C. Implement EC2 On-Demand Instances.

D. Implement the processing on AWS Lambda.

Correct Answer: A

Community vote distribution

A (96%) 2%

Comments

Kapello10 Highly Voted 3 years, 6 months ago

Selected Answer: A

Cant be implemented on Lambda because the timeout for Lambda is 15mins and the Job takes 60minutes to complete

Answer >> A

upvoted 25 times

Evangelia Highly Voted 3 years, 7 months ago

spot instances

upvoted 6 times

Kazmin Most Recent 1 year, 2 months ago

Selected Answer: B

Spot is a bad choice because process takes up to 60 mins and instances can be down at any time

upvoted 1 times

bmmbmm 6 months, 1 week ago

It says it can be started and stopped at any given time with no negative impact, and the company doesn't have specific scheduled time or an amount of usage for using Reserved Instance.

upvoted 2 times

Neil_12345678 1 year, 3 months ago

Selected Answer: D

Ec2 by themselves are not "scalable" unless they are under an autoscaling group.The question also says that the job can be started and stopped with no negative impact.So even if the execution time of 15mins runs out, it can be started again to complete the 60 mins job. Hence my answer is D

upvoted 1 times

jaradat02 1 year, 10 months ago

Selected Answer: A

A is the correct answer, we can't use lambda because it's limit is 15 mins.

upvoted 3 times

thewalker 2 years, 4 months ago

Selected Answer: A

There is a chance of interrupting the jobs, but as they can be started and stopped at any given time, the MOST COST effective is going for Spot.

upvoted 4 times

Ruffyit 2 years, 7 months ago

Spot Instances provide significant cost savings for flexible start and stop batch jobs.
Purchasing Reserved Instances (B) is better for stable workloads, not dynamic ones.
On-Demand Instances (C) are costly and lack potential cost savings like Spot Instances.
AWS Lambda (D) is not suitable for long-running batch jobs.

upvoted 3 times

tom_cruise 2 years, 8 months ago

Selected Answer: A

key: can be started and stopped at any given time with no negative impact

upvoted 4 times

AbhilashDyadav 2 years, 8 months ago

Selected Answer: A

Spot can do that

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

The key reasons are:

Spot can provide significant cost savings (up to 90%) compared to On-Demand.
Since the job is stateless and can be stopped/restarted anytime, the intermittent availability of Spot is not an issue.
Spot supports the same instance types as On-Demand, so optimal instance types can be chosen.
For a 60+ minute batch job, the chance of Spot interruption is low. But if it happens, the job can just be restarted.
Reserved Instances don't offer any advantage for a highly dynamic job like this.
Lambda is not a good fit given the long runtime requirement.

upvoted 4 times

cookieMr 2 years, 11 months ago

Selected Answer: A

Spot Instances provide significant cost savings for flexible start and stop batch jobs.
Purchasing Reserved Instances (B) is better for stable workloads, not dynamic ones.
On-Demand Instances (C) are costly and lack potential cost savings like Spot Instances.
AWS Lambda (D) is not suitable for long-running batch jobs.

upvoted 2 times

beginnercloud 3 years ago

Selected Answer: A

A is correct

upvoted 1 times

alexiscloud 3 years, 2 months ago

Answer A:
typically takes upwards of 60 minutes total to complete.

upvoted 1 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: A

The correct answer is Option A. To design a scalable and cost-effective solution for the batch processing job, the solutions architect should recommend implementing EC2 Spot Instances.

EC2 Spot Instances allow users to bid on spare Amazon EC2 computing capacity and can be a cost-effective solution for stateless, interruptible workloads that can be started and stopped at any time. Since the batch processing job is stateless, can be started and stopped at any time, and typically takes upwards of 60 minutes to complete, EC2 Spot Instances would be a good fit for this workload.

upvoted 2 times

k1kavi1 3 years, 5 months ago

Selected Answer: A

Spot Instances should be good enough and cost effective because the job can be started and stopped at any given time with no negative impact.

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: A

Option A

upvoted 1 times

Wpcorgan 3 years, 6 months ago

A is correct
upvoted 1 times

Question #125 | Topic 1

A company runs its two-tier ecommerce website on AWS. The web tier consists of a load balancer that sends traffic to Amazon EC2 instances. The database tier uses an Amazon RDS DB instance. The EC2 instances and the RDS DB instance should not be exposed to the public internet. The EC2 instances require internet access to complete payment processing of orders through a third-party web service. The application must be highly available.
Which combination of configuration options will meet these requirements? (Choose two.)

A. Use an Auto Scaling group to launch the EC2 instances in private subnets. Deploy an RDS Multi-AZ DB instance in private subnets. **Most Voted**

B. Configure a VPC with two private subnets and two NAT gateways across two Availability Zones. Deploy an Application Load Balancer in the private subnets.

C. Use an Auto Scaling group to launch the EC2 instances in public subnets across two Availability Zones. Deploy an RDS Multi-AZ DB instance in private subnets.

D. Configure a VPC with one public subnet, one private subnet, and two NAT gateways across two Availability Zones. Deploy an Application Load Balancer in the public subnet.
D. Configure a VPC with two public subnets, two private subnets, and two NAT gateways across two Availability Zones. Deploy an Application Load Balancer in the public subnets. **Most Voted**

Correct Answer: AD

Community vote distribution

AD (62%) | AB (18%) | A (17%) | 2%

Comments

HayLLlHuK Highly Voted 3 years, 5 months ago
A and E!
Application has to be highly available while the instance and database should not be exposed to the public internet, but the instances still requires access to the internet. NAT gateway has to be deployed in public subnets in this case while instances and database remain in private subnets in the VPC, therefore answer is (A) and (E).
https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html

If the instances did not require access to the internet, then the answer could have been
(B) to use a private NAT gateway and keep it in the private subnets to communicate only to the VPCs.

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Scenario2.html
upvoted 33 times

darn 3 years, 1 month ago
your link is right but your voting is wrong, should be AD, although that still doesnt explain why 2 NAT gateways
upvoted 3 times

cheroh_tots 2 years, 3 months ago
Because NAT gateways are availability zone specific, if you need HA you will need a NAT gateway in each availability zone.
upvoted 1 times

ale_brd_111 2 years, 6 months ago
cus application has to be HA, if one NAT gateway fails the other could take the traffic
upvoted 4 times

JA2018 1 year, 6 months ago
It looks like there could be a typo error in the list of option. HayLLiHuK is referring to Option 'E' which is missing.
upvoted 1 times

mabotega Highly Voted 3 years, 7 months ago

Selected Answer: AD

Answer A for: The EC2 instances and the RDS DB instance should not be exposed to the public internet. Answer D for: The EC2 instances require internet access to complete payment processing of orders through a third-party web service. Answer A for: The application must be highly available.

upvoted 28 times

oguzbeliren 2 years, 10 months ago

D allows public internet access which is not desired. The answer is not d.
The most accurate answers are AB

upvoted 2 times

pentium75 2 years, 5 months ago

B is wrong because you can't deploy NAT GW in a private subnet. Correct answer is E (mislabelled as a second D). Stem says that the EC2 instances (!) must not be exposed to the Internet, the Load Balancer can be exposed.

upvoted 3 times

AbhiJo 3 years, 6 months ago

We will require 2 private subnets, D does mention 1 subnet

upvoted 4 times

pentium75 2 years, 5 months ago

There's two D options ;) second is correct

upvoted 5 times

OluwaZettai 1 year, 5 months ago

Why do we require 2?

upvoted 1 times

smd_ 3 years, 1 month ago

why not option B.The EC2 instances can be launched in private subnets across two Availability Zones, and the Application Load Balancer can be deployed in the private subnets. NAT gateways can be configured in each private subnet to provide internet access for the EC2 instances to communicate with the third-party web service.

upvoted 2 times

ruqui 3 years ago

B option wrong! NAT gateways must be created in public subnets!!

upvoted 8 times

x33 2 years, 9 months ago

I think you are wrong on this. In fact, NAT gateways are typically created in private subnets.

upvoted 3 times

RNess 2 years, 7 months ago

NAT Gateway can’t be used by EC2 instance in the same subnet (only from other subnets)

upvoted 4 times

bora4motion Most Recent 1 year, 1 month ago

Selected Answer: A

Correct the answers already! Last option is E not D!
AE

upvoted 1 times

CloudExpert01 1 year, 2 months ago

Selected Answer: AD

If D is chosen: An additional private subnet in the second Availability Zone is needed for high availability.

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: AD

Theright one is A and D. The last option (with numbering incorrectly shown as the second "D") is not correct, since a load balancer is not per AZ, rather for the entire VPC.

upvoted 1 times

mzeynalli 1 year, 7 months ago

Selected Answer: AD

AE
The goal is to create a highly available architecture with EC2 instances and RDS instances that are not exposed to the public internet, while still allowing the EC2 instances to access external services.

upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: AD

And A,D - keeps EC2 and RDS private and highly available, but with public front-end

(Not sure why Author's highlighted answer is C?)

upvoted 1 times

Shailesh1717 1 year, 8 months ago

AD are correct ans

upvoted 1 times

scaredSquirrel 1 year, 10 months ago

AE
A because the EC2 can't be exposed to public
E because each subnet must reside entirely within one Availability Zone and cannot span zones.

source: https://docs.aws.amazon.com/vpc/latest/userguide/configure-subnets.html#:~:text=Delete%20a%20subnet-,Subnet%20basics,of%20a%20single%20Availability%20Zone.

upvoted 1 times

jaradat02 1 year, 10 months ago

A and D is the best combination, it achieves all of the desired requirements.

upvoted 1 times

shil_31 2 years ago

Selected Answer: AD

Option A ensures that EC2 instances and RDS DB instance are not exposed to the public internet, as they are launched in private subnets. Auto Scaling group will also ensure high availability of EC2 instances.
Option D configures a VPC with a public subnet for the load balancer, and a private subnet for the EC2 instances and RDS DB instance. NAT gateways in both Availability Zones will allow EC2 instances to access the internet for payment processing, while keeping them private.

upvoted 3 times

lofzee 2 years ago

A and E (the second D option) .

upvoted 2 times

Yash2804 2 years ago

There might be error in question. i modified it now CE answer seems to be correct

A company runs its two-tier ecommerce website on AWS. The web tier consists of a load balancer that sends traffic to Amazon EC2 instances. The database tier uses an Amazon RDS DB instance. The RDS DB instance should not be exposed to the public internet. The The RDS DB instance require internet access to complete payment processing of orders through a third-party web service. The application must be highly available.

Which combination of configuration options will meet these requirements? (Choose two.)

upvoted 1 times

Solomon2001 2 years, 1 month ago

Selected Answer: AB

Use an Auto Scaling group to launch the EC2 instances in private subnets. Deploy an RDS Multi-AZ DB instance in private subnets.
This ensures that the EC2 instances and the RDS DB instance are not exposed to the public internet.
Configure a VPC with two private subnets and two NAT gateways across two Availability Zones. Deploy an Application Load Balancer in the private subnets.
This allows the EC2 instances in private subnets to access the internet for payment processing through the NAT gateways while keeping them private

upvoted 2 times

EMPERBACH 2 years, 1 month ago

App layer & DB Layer does not expose to Internet -> both in private subnet, access through NAT Gateway
It's enough for all requirements!

upvoted 1 times

SaurabhTiwari1 2 years, 5 months ago

Selected Answer: AD

AD is right , last one D

upvoted 1 times

rlamberti 2 years, 7 months ago

Selected Answer: AD

AE
Two public subnets = two addresses for ALB = high availability
two private subnets with NAT gateway to allow eggress traffic to internet - application tier will be able to complete payment
upvoted 5 times

Question #126 *Topic 1*

A solutions architect needs to implement a solution to reduce a company's storage costs. All the company's data is in the Amazon S3 Standard storage class. The company must keep all data for at least 25 years. Data from the most recent 2 years must be highly available and immediately retrievable.
Which solution will meet these requirements?

A. Set up an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive immediately.

B. Set up an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 2 years. **Most Voted**

C. Use S3 Intelligent-Tiering. Activate the archiving option to ensure that data is archived in S3 Glacier Deep Archive.

D. Set up an S3 Lifecycle policy to transition objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) immediately and to S3 Glacier Deep Archive after 2 years.

Correct Answer: B

Community vote distribution

B (83%) Other

Comments

rjam Highly Voted 3 years, 6 months ago

Selected Answer: B

Why Not C? Because Intelligent Tier the objects are automatically moved to different tiers.
The question says "the data from most recent 2 yrs should be highly available and immediately retrievable", which means in intelligent tier , if you activate archiving option(as Option C specifies) , the objects will be moved to Archive tiers(instant to access to deep archive access tiers) in 90 to 730 days. Remember these archive tiers performance will be similar to S3 glacier flexible and s3 deep archive which means files cannot be retrieved immediately within 2 yrs .

We have a hard requirement in question which says it should be retreivable immediately for the 2 yrs. which cannot be acheived in Intelligent tier. So B is the correct option imho.

Because of the above reason Its possible only in S3 standard and then configure lifecycle configuration to move to S3 Glacier Deep Archive after 2 yrs.

upvoted 18 times

Abdou1604 2 years, 9 months ago

but your S3 intelligent-tiering will move the object to S3 infrequent access tier which a is a single AZ tier , and then the HA requirement will not be respected

upvoted 2 times

sandordini 2 years, 1 month ago

S3 Standard IA is NOT single AZ. (One-Zone IA is single Az.)

upvoted 2 times

sandordini 2 years, 1 month ago

The issue is with the missing "flag" for 2 years, and not S3 Intelligent Tiering. It needs to be B.

upvoted 2 times

MutiverseAgent 2 years, 11 months ago

Mmm.. You can enable Intelligent-Tiering and take advantage of of the infrequent Access tier and thus reducing costs. To avoid moving objects to the deep archive tier before the two years it would be enough to enable ONLY the check "Deep Archive Access tier" and set days to 720 (two years, which is curiously the maximum value), and keep disabled the check "Archive Access tier" to avoid the Intelligent-Tiering move objects to the non-instant retrieval tier. That will work, offcourse this specific configuration is not mention in the question which leaves some doubts about which option is the correct.

upvoted 2 times

MutiverseAgent 2 years, 11 months ago

Just to clarify, my previous comment is about how answer B) might be correct and the MOST cheapest option under the correct configuration.

upvoted 2 times

MutiverseAgent 2 years, 11 months ago

Sorry, I meant answer C) might be correct

upvoted 1 times

TelaO Highly Voted 3 years, 6 months ago

Selected Answer: B

B is the only right answer. C does not indicate archiving after 2 years. If it did specify 2 years, then C would also be an option.

upvoted 9 times

satyaammm Most Recent 1 year, 5 months ago

Selected Answer: B

S3 Lifecycle policies works best here as we no longer need frequently access to data that has been over 2 years and hence moving it to S3 Glacier Deep Archive help reduce costs.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: B

Ans B - I did initially think Ans C, but rjam (1 yr, 10 mth ago) quickly quashed that notion:
"Why Not C? Because Intelligent Tier the objects are automatically moved to different tiers. The question says "the data from most recent 2 yrs should be highly available and immediately retrievable", which means in intelligent tier , if you activate archiving option(as Option C specifies) , the objects will be moved to Archive tiers(instant to access to deep archive access tiers) in 90 to 730 days. Remember these archive tiers performance wil be similar to S3 glacier flexible and s3 deep archive which means files cannot be retrieved immediately within 2 yrs."

upvoted 1 times

bignatov 1 year, 9 months ago

Selected Answer: B

B is correct, because it fits to the requirements and it still cheaper than the option C.

upvoted 1 times

JaegEr_2k1 1 year, 10 months ago

Stupid question:
A: No immediately retrievable and cost
B: No immediately retrievable
C: Not unpredicted access
D: Immediately retrievable but not HA
F*ck the guy who made this question

upvoted 4 times

jaradat02 1 year, 10 months ago

Selected Answer: B

B is the correct choice, the requirements are very clear, intelligent tiering is used only when we don't have a clear pattern for the access of the data, when it's unpredictable.

upvoted 2 times

LoXoL 2 years, 5 months ago

Selected Answer: B

A. We can't move to Glacier immediately as data from last 2 yrs need to be immediately retrievable
B. It's the perfect fit: getting HA and instant access (with current solution = S3 std), then moving to Deep Archive after 2 yrs (very cheap)
C: Highly expensive because of Intelligent Tiering
D: it lacks HA with One Zone

upvoted 2 times

SaurabhTiwari1 2 years, 5 months ago

Selected Answer: B

Data remain in S3 standard storage for 2 years then it will be move to s3 glacier deep archive after 2 year.

upvoted 3 times

Ruffyit 2 years, 7 months ago

but your S3 intelligent-tiering will move the object to S3 infrequent access tier which a is a single AZ tier , and then the HA requirement will not be respected

upvoted 1 times

David_Ang 2 years, 7 months ago

Selected Answer: B

i understand why "B" is more correct than "C" and is because "C" is bad formulated, if in the answer would say "life cycle after 2 years of using intelligent tiring" then it would be the correct answer. so "B" is correct

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: B

I would not opt for C simply because S3IT was specifically designed for scenarios where the access patterns are unknown.
This scenario has clearly known access patterns making option B the best.

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: B

Option A is incorrect because immediately transitioning objects to S3 Glacier Deep Archive would not fulfill the requirement of keeping the most recent 2 years of data highly available and immediately retrievable.

Option C is also incorrect because using S3 Intelligent-Tiering with archiving option would not meet the requirement of immediately retrievable data for the most recent 2 years.

Option D is not the best choice because transitioning objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) and then to S3 Glacier Deep Archive would not satisfy the requirement of immediately retrievable data for the most recent 2 years.

Option B is the correct solution. By setting up an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 2 years, the company can keep all data for at least 25 years while ensuring that data from the most recent 2 years remains highly available and immediately retrievable in the Amazon S3 Standard storage class. This solution optimizes storage costs by leveraging the Glacier Deep Archive for long-term storage.

upvoted 3 times

kambarami 2 years, 9 months ago

this makes sense the question is a bit tricky. I now uderstand that all the data is already kept in S3 Standard meaning immediate retrieval of the most recent data is remains highly available.

upvoted 2 times

Yadav_Sanjay 2 years, 11 months ago

Why not D

upvoted 2 times

RNess 2 years, 7 months ago

"Data from the most recent 2 years must be highly available and immediately retrievable."

upvoted 1 times

RNess 2 years, 7 months ago

Additionally,
S3 Standard Availability = 99.99%
S3 One Zone-IA Availability = 99.5%

upvoted 2 times

Robrobtutu 3 years, 1 month ago

Selected Answer: B

B is the only one possible.

upvoted 1 times

rushlav 3 years, 1 month ago

C would not work as the names of these S3 archives are called Archive Access Tier and Deep Archive access tiers, so since they mention glacier in option C , I think its B which is the correct.

upvoted 2 times

CaoMengde09 3 years, 4 months ago

It's pretty straight forward.

S3 Standard answers for High Availaibility/Immediate retrieval for 2 years. S3 Intelligent Tiering would just incur additional cost of analysis while the company insures that it requires immediate retrieval in any moment and without risk to Availability. So a capital B

upvoted 2 times

Question #127 | Topic 1

A media company is evaluating the possibility of moving its systems to the AWS Cloud. The company needs at least 10 TB of storage with the maximum possible I/O performance for video processing, 300 TB of very durable storage for storing media content, and 900 TB of storage to meet requirements for archival media that is not in use anymore.
Which set of services should a solutions architect recommend to meet these requirements?

A. Amazon EBS for maximum performance, Amazon S3 for durable data storage, and Amazon S3 Glacier for archival storage

B. Amazon EBS for maximum performance, Amazon EFS for durable data storage, and Amazon S3 Glacier for archival storage

C. Amazon EC2 instance store for maximum performance, Amazon EFS for durable data storage, and Amazon S3 for archival storage

D. Amazon EC2 instance store for maximum performance, Amazon S3 for durable data storage, and Amazon S3 Glacier for archival storage Most Voted

Correct Answer: D

Community vote distribution

D (57%) | A (43%)

Comments

Sauran Highly Voted 3 years, 7 months ago
Selected Answer: D
Max instance store possible at this time is 30TB for NVMe which has the higher I/O compared to EBS.

is4gen.8xlarge 4 x 7,500 GB (30 TB) NVMe SSD

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/InstanceStorage.html#instance-store-volumes
upvoted 40 times

ishitamodi4 3 years, 5 months ago
instance store volume for the root volume, the size of this volume varies by AMI, but the maximum size is 10 GB
upvoted 1 times

JayBee65 3 years, 5 months ago
This link shows a max capacity of 30TB, so what is the problem?
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/InstanceStorage.html#instance-store-volumes
upvoted 2 times

JayBee65 3 years, 5 months ago
Only the following instance types support an instance store volume as the root device: C3, D2, G2, I2, M3, and R3, and we're using an I3, so an instance store volume is irrelevant.
upvoted 5 times

antropaws 3 years ago
THE CORRECT ANSWER IS A.

The biggest Instance Store Storage Optimized option (is4gen.8xlarge) has a capacity of only 3TB.

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/instance-store-volumes.html#instance-store-vol-so
upvoted 3 times

michellemeloc 3 years, 1 month ago
Update: i3en.metal and i3en.24xlarge = 8 x 7500 GB (60TB)
upvoted 3 times

Tsige 1 year, 7 months ago

Answer is A
While instance store provides fast local storage, it is ephemeral (data is lost when the instance stops or terminates). It's not suitable for workloads that require persistence, like video processing that may span across multiple EC2 instances over time. EBS provides persistent, high-performance storage.
upvoted 8 times

JA2018 1 year, 6 months ago
err, stem did not mention requirement for durable storage for the 10TB used for video processing....
upvoted 2 times

MiniYang Highly Voted 2 years, 6 months ago
Selected Answer: A
The correct Answer is A :
Amazon EC2 instance store (Instance Store) is usually not the best choice because the storage it provides is temporary and tied to the life cycle of the instance. When an instance is stopped or terminated, data on the instance store is lost.

In this scenario, the company's requirements were to have the maximum possible I/O performance and required durable data storage. Therefore, using Amazon EC2 Instance Store does not meet these requirements because it lacks durability.

In contrast, Amazon EBS (Elastic Block Store) provides persistent regional block storage and can meet the needs of high-performance I/O. Therefore, the answer should include Amazon EBS, not Amazon EC2 instance storage.
upvoted 27 times

pentium75 2 years, 5 months ago
"The company's requirements were to have the maximum possible I/O performance and required durable data storage." Yeah, but not for the same data.

10 TB "maximum possible I/O performance" for processing (= temporary)
300 TB "very durable" (= S3)
900 TB "for archival" (= Glacier)
upvoted 17 times

LoXoL 2 years, 5 months ago
pentium75 is right.
upvoted 6 times

Jarranymo Most Recent 9 months, 1 week ago
Selected Answer: A
Option A: Amazon EBS for maximum performance, Amazon S3 for durable data storage, and Amazon S3 Glacier for archival storage

EBS is great for high-performance storage, so this is a good choice for the 10 TB video processing.

S3 offers the required durability for the 300 TB of media content.

S3 Glacier is a suitable option for the archival storage (900 TB) as it's low-cost and durable.
upvoted 1 times

DougZ 12 months ago
Selected Answer: A
At least 10 TB of storage with the maximum possible I/O performance for video processing.

Amazon EBS (Elastic Block Store): Provides persistent block storage volumes that can be attached to EC2 instances. It offers various volume types, including Provisioned IOPS SSD (io1/io2 Block Express) and General Purpose SSD (gp3), which are designed for high-performance workloads like video processing. EBS volumes are highly available within an Availability Zone.
Amazon EC2 Instance Store: Provides temporary block-level storage for your EC2 instance. It offers very high random I/O performance because it's physically attached to the host machine. However, the data persists only for the lifetime of the instance. If the instance stops or terminates, the data is lost. This makes it unsuitable for long-term video processing data unless you have a robust way to move data in and out quickly and handle transient failures.
upvoted 1 times

Bhuvan_D 1 year ago
Selected Answer: A
Explanation:
1. Maximum I/O Performance for Video Processing (10 TB)
Amazon EBS (Elastic Block Store) provides high-performance block storage that can scale to meet IOPS-intensive workloads like video processing. You can use EBS io2 or io2 Block Express volumes for maximum performance and durability.
2. Durable Storage for Media Content (300 TB)
Amazon S3 offers 11 nines (99.999999999%) of durability, making it ideal for storing large volumes of media content that needs to be safely preserved and frequently accessed.
3. Archival Storage (900 TB)
Amazon S3 Glacier is built for long-term archival at very low cost. It's perfect for media content that is rarely accessed but must be retained for long periods.

upvoted 1 times

MortisG 1 year ago

Selected Answer: A

Correct answer is A. EC2 instance store is not well suited for this scenario since data could be lost if the instance stops or terminates

upvoted 1 times

CloudExpert01 1 year, 2 months ago

Selected Answer: A

Amazon EBS Provisioned IOPS SSD (io2) for the 10 TB of storage needed for high I/O performance video processing.

upvoted 1 times

Mimine87 1 year, 2 months ago

Selected Answer: A

At least 10 TB of storage with max I/O performance (for video processing):
Amazon EBS (Elastic Block Store) provides high-performance block storage for EC2.

Provisioned IOPS SSD (io2) volumes offer the maximum performance and consistency required for intensive workloads like video processing.

❌ EC2 instance store is fast but ephemeral — data is lost if the instance stops or fails.

upvoted 1 times

SirDNS 1 year, 2 months ago

Selected Answer: D

Can be a bit confusing due to the presence of option A: EBS Provisioned IOPS SSD, Durable = S3, Archive = Glacier, but maximum performance is always equal to instance store. Here they do not need a permanent storage with maximum performance. It will only be used for processing and then sent to a highly durable storage.
D logically cannot be the answer

upvoted 1 times

sk1974 1 year, 3 months ago

Selected Answer: A

Instance storage is ephimeral data that gets destroyed when EC2 is stopped/restarted. Unless the question says that data is not important , I will assume that every data that app processed is imporant . So , I will go with A

upvoted 1 times

kyd0nix 1 year, 4 months ago

Selected Answer: D

Ans D.

The requirements for every kind of data is specific. So, you need:
10 TB high I/O / performance = Instance Store (the reqs. doesn't mention anything about durability, just performance)
300 TB very durable = S3
900 TB archival = Glacier

upvoted 1 times

satyaammm 1 year, 5 months ago

Selected Answer: D

EC2 Instance Store Volumes provide the highest storage performance in AWS and S3 provides durable storage while S3 Glacier is best for archival storage.

upvoted 2 times

DavidPhyo 1 year, 5 months ago

Selected Answer: A

answer is A

upvoted 1 times

Mischi 1 year, 5 months ago

Selected Answer: A

The combination of Amazon EBS , Amazon S3 and Amazon S3 Glacier (option A) is the most suitable solution for this media company. It offers:

Maximum I/O performance for video processing with Amazon EBS.
Durability and reliability for media content with Amazon S3.
Cost-effective storage for archived data with Amazon S3 Glacier.

upvoted 2 times

salman7540 1 year, 5 months ago

Selected Answer: A

I prefer A over D.

instance storage is not recommended for video processing because it's designed for temporary storage and doesn't provide the durability and persistence needed for video processing. Instead, Amazon EBS is a better choice for video processing because it provides the required performance, durability, and persistence.
Here's some more information about instance storage and Amazon EBS:
Instance storage
A temporary storage volume that acts as a physical hard drive. It's located on disks that are physically attached to the host computer. Instance storage is ideal for temporary storage of information that changes frequently, such as buffers, caches, and scratch data. Data in an instance store persists during the lifetime of its instance, but it's not persistent through instance stops, terminations, or hardware failures.

upvoted 2 times

kimm_10 1 year, 6 months ago

Selected Answer: B

Why not EC2 instance store?
Instance store is ephemeral (data is lost when the instance stops or terminates).
It lacks durability, making it unsuitable for critical video processing.

upvoted 1 times

0de7d1b 1 year, 6 months ago

Selected Answer: A

Explanation:
Amazon EBS (Elastic Block Store) for maximum performance:

Amazon EBS provides high IOPS and low latency storage, which is ideal for video processing workloads that require fast performance.
Amazon S3 for durable data storage:

Amazon S3 is highly durable, scalable, and designed for storing large amounts of data, such as media content (300 TB). It also provides 99.999999999% (11 nines) durability, making it suitable for this requirement.
Amazon S3 Glacier for archival storage:

Amazon S3 Glacier is a cost-effective storage solution for archiving large amounts of data (900 TB) that is infrequently accessed but still needs to be stored securely and durably.

upvoted 3 times

0de7d1b 1 year, 6 months ago

C & D. Amazon EC2 instance store for maximum performance: EC2 instance store provides temporary block storage tied to the lifecycle of an instance, so it is not durable or persistent, making it unsuitable for video processing workflows that require consistent data availability

upvoted 1 times

Question #128 *Topic 1*

A company wants to run applications in containers in the AWS Cloud. These applications are stateless and can tolerate disruptions within the underlying infrastructure. The company needs a solution that minimizes cost and operational overhead.
What should a solutions architect do to meet these requirements?

A. Use Spot Instances in an Amazon EC2 Auto Scaling group to run the application containers.

B. Use Spot Instances in an Amazon Elastic Kubernetes Service (Amazon EKS) managed node group. **Most Voted**

C. Use On-Demand Instances in an Amazon EC2 Auto Scaling group to run the application containers.

D. Use On-Demand Instances in an Amazon Elastic Kubernetes Service (Amazon EKS) managed node group.

Correct Answer: B

Community vote distribution

B (72%) A (26%) 2%

Comments

bgsanata Highly Voted 3 years ago

Selected Answer: A

Requirement is "minimizes cost and operational overhead"
A is better option than B as EKS add additional cost and operational overhead.

upvoted 18 times

EMPERBACH 2 years, 1 month ago

Really ???
You want more configuration efforts for container workload on EC2, instead of using EKS ::))

upvoted 9 times

MutiverseAgent 2 years, 11 months ago

In my opinion option A) seems to be a reasonable at first because setting up AWS EKS might be seem as an operation overhead comparing to the option of running the containers inside the EC2 using docker just as you we do on your own machines. However, consider installing docker on multiple EC2 instances and manually manage docker instances and images will end up in chaos, so, as a conclusion, the operational cost of setting up AWS EKS will worth the effort.

upvoted 6 times

ruqui 3 years ago

option A is the worst option in terms of operational overhead ... you have to install your own kubernetes cluster!!! B is a more suitable option

upvoted 6 times

MutiverseAgent 2 years, 11 months ago

you do not necessary need to install K8S, in terms of plain containers you can run them using docker just as you do on your own machine.

upvoted 1 times

Lalo 3 years ago

USING SPOT INSTANCES WITH EKS
https://ec2spotworkshops.com/using_ec2_spot_instances_with_eks.html

upvoted 11 times

GalileoEC2 Highly Voted 3 years, 3 months ago

Answer is A:
Amazon ECS: ECS itself is free, you pay only for Amazon EC2 resources you use.
Amazon EKS: The EKS management layer incurs an additional cost of $144 per month per cluster.
Advantages of Amazon ECS include: Spot instances: Because containers are immutable, you can run many workloads using Amazon EC2 Spot Instances (which can be shut down with no advance notice) and save 90% on on-demand instance costs.

upvoted 10 times

46e1a46 Most Recent 1 month, 1 week ago

Selected Answer: D

3 Storage needs 3 different solutions

- Max I/O performance -> EC2 Instance Store. This is physically attached NVMe SSD on the host server, fasters possible I/O on AWS

- Durable Storage for media content -> S3. S3 has 11 nines of durability

- Archival storage -> S3 Glacier. Cheapest option for data that's "not in use anymore"

D

upvoted 1 times

Maco 2 months, 2 weeks ago

Selected Answer: A

Spot Instances
Up to 90% cheaper
Perfect for:
Stateless workloads
Interruptible workloads

upvoted 1 times

Bhuvan_D 1 year ago

Selected Answer: A

Why Option A is best:
Amazon EC2 Auto Scaling Group + Spot Instances:
Spot Instances are highly cost-effective (up to 90% cheaper than On-Demand).
Auto Scaling ensures automatic replacement of interrupted instances.
Stateless nature of apps makes Spot a perfect fit.
Operational overhead is low compared to running a full Kubernetes cluster (like EKS).

upvoted 1 times

MortisG 1 year ago

Selected Answer: B

Answer is B since it's less operational overhead to run containers on EKS (managed service) than running on EC2 instances which implies installing dockers to orchestrate containers)

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: A

ai response "Generally, an ECS EC2 Auto Scaling Group is considered cheaper than an EKS Managed Node Group because with ECS, you only pay for the EC2 instances used to run your containers, while EKS adds an additional cost per cluster hour on top of the compute costs for the managed nodes"

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: B

Definitely B. running containers on spot EC2 instances means there has to be additional configuration requirements every time an instance comes up
That means making a custom AMI with prepping the spot instance to launch the container = additional operational overhead in maintaining the AM
- with EKS, that overhead is eliminated.

upvoted 2 times

satyaammm 1 year, 5 months ago

Selected Answer: B

Spot Instances are the cheapest and most suitable here along with EKS for automation and thereby reducing operational overhead.

upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: A

Ans A - hint: "stateless" and 'don't care' infrastructure - so Spot and containerisation

upvoted 1 times

huaze_lei 1 year, 9 months ago

Spot instance is definitely the answer for interruptible process. It's between A and B now.
I would reckon Option B requires lesser operational overheads than to maintain own fleet of EC2 servers with containers.

upvoted 1 times

SaurabhTiwari1 1 year, 9 months ago

Selected Answer: B

B is correct

upvoted 2 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: B

B for sure

upvoted 2 times

Anthony_Rodrigues 2 years ago

IMHO, the option **B** is the right one.
Breaking down the reasons for it:
1 - Spot is much cheaper than on-demand, which already eliminates C&D for cost related.
2 - Even though we can create a bootstrap script to install docker, managing this can be complicated, especially if any of the applications require more than one instance running.

upvoted 4 times

lofzee 2 years ago

Container in ec2 or container on a container platform?
B

upvoted 3 times

EMPERBACH 2 years, 1 month ago

Selected Answer: B

run applications in containers -> Container service not EC2 (no operational overhead to config container workload on EC2)
Spot instance < On-demand cost

upvoted 3 times

vip2 2 years, 3 months ago

Selected Answer: A

Does A implicitly means run spot Instance on ECS?

upvoted 2 times

Question #129 Topic 1

A company is running a multi-tier web application on premises. The web application is containerized and runs on a number of Linux hosts connected to a PostgreSQL database that contains user records. The operational overhead of maintaining the infrastructure and capacity planning is limiting the company's growth. A solutions architect must improve the application's infrastructure.
Which combination of actions should the solutions architect take to accomplish this? (Choose two.)

A. Migrate the PostgreSQL database to Amazon Aurora. Most Voted

B. Migrate the web application to be hosted on Amazon EC2 instances.

C. Set up an Amazon CloudFront distribution for the web application content.

D. Set up Amazon ElastiCache between the web application and the PostgreSQL database.

E. Migrate the web application to be hosted on AWS Fargate with Amazon Elastic Container Service (Amazon ECS). Most Voted

Correct Answer: AE

Community vote distribution

AE (98%) 2

Comments

ArielSchivo Highly Voted 3 years, 7 months ago

Selected Answer: AE

I would say A and E since Aurora and Fargate are serverless (less operational overhead).

upvoted 14 times

baba365 2 years, 9 months ago

There's a difference between Amazon Aurora and Amazon Aurora Serverless

upvoted 2 times

pentium75 2 years, 5 months ago

"Aurora serverless" is still a flavor of Aurora, it's not a different product.

upvoted 5 times

Manimgh Most Recent 1 year, 2 months ago

Selected Answer: AE

Aurora and Fargate are the answers

upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: AE

Ans A,E -
-PostgreSQL is compatible with Aurora
-Fargate for container service
Both services are serverless

upvoted 3 times

jaradat02 1 year, 10 months ago

Selected Answer: AE

A and E since both Aurora and fargate are serverless.

upvoted 3 times

JTruong 2 years, 5 months ago

Selected Answer: AE

PostgreSQL is compatible w/ Aurora
Fargate & ECS are also paired with containers

A&E

upvoted 3 times

Ruffyit 2 years, 7 months ago

I would say A and E since Aurora and Fargate are serverless (less operational overhead)

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: AE

Requirement is to reduce operational overhead,
Amazon Aurora provides built-in security, continuous backups, serverless compute, up to 15 read replicas, automated multi-Region replication.
AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without managing servers.

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: AE

The reasons are:

Migrating the database to Amazon Aurora provides a high performance, scalable PostgreSQL-compatible database with minimal overhead.
Migrating the containerized web app to Fargate removes the need to provision and manage EC2 instances. Fargate auto-scales.
Together, Aurora and Fargate reduce operational overhead and complexity for the data and application tiers.

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: AE

A is the correct answer because migrating the database to Amazon Aurora reduces operational overhead and offers scalability and automated backups.

E is the correct answer because migrating the web application to AWS Fargate with Amazon ECS eliminates the need for infrastructure management, simplifies deployment, and improves resource utilization.

B. Migrating the web application to Amazon EC2 instances would not directly address the operational overhead and capacity planning concerns mentioned in the scenario.

C. Setting up an Amazon CloudFront distribution improves content delivery but does not directly address the operational overhead or capacity planning limitations.

D. Configuring Amazon ElastiCache improves performance but does not directly address the operational overhead or capacity planning challenges mentioned.

Therefore, the correct answers are A and E as they address the requirements, while the incorrect answers (B, C, D) do not provide the desired solutions.

upvoted 3 times

studynoplay 3 years, 1 month ago

Selected Answer: AE

Improve the application's infrastructure = Modernize Infrastructure = Least Operational Overhead = Serverless

upvoted 4 times

Robrobtutu 3 years, 1 month ago

Selected Answer: AE

A and E are the best options.

upvoted 1 times

bgsanata 3 years, 3 months ago

Selected Answer: AE

A and E

upvoted 1 times

rapatajones 3 years, 4 months ago

Selected Answer: AE

a e..............

upvoted 1 times

goodmail 3 years, 4 months ago

One should that Aurora is not serverless. Aurora serverless and Aurora are 2 Amazon services. I prefer C, however the question does not mention any frontend requirements.

upvoted 1 times

aba2s 3 years, 5 months ago

Selected Answer: AE

Yes, go for A and E since thes two ressources are serverless.

upvoted 3 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: AE

The correct answers are A and E. To improve the application's infrastructure, the solutions architect should migrate the PostgreSQL database to Amazon Aurora and migrate the web application to be hosted on AWS Fargate with Amazon Elastic Container Service (Amazon ECS).

Amazon Aurora is a fully managed, scalable, and highly available relational database service that is compatible with PostgreSQL. Migrating the database to Amazon Aurora would reduce the operational overhead of maintaining the database infrastructure and allow the company to focus on building and scaling the application.

AWS Fargate is a fully managed container orchestration service that enables users to run containers without the need to manage the underlying EC2 instances. By using AWS Fargate with Amazon Elastic Container Service (Amazon ECS), the solutions architect can improve the scalability and efficiency of the web application and reduce the operational overhead of maintaining the underlying infrastructure.

upvoted 2 times

techhb 3 years, 5 months ago

A and E are obvious choices.

upvoted 1 times

Question #130 *Topic 1*

An application runs on Amazon EC2 instances across multiple Availability Zonas. The instances run in an Amazon EC2 Auto Scaling group behind an Application Load Balancer. The application performs best when the CPU utilization of the EC2 instances is at or near 40%.
What should a solutions architect do to maintain the desired performance across all instances in the group?

A. Use a simple scaling policy to dynamically scale the Auto Scaling group.

B. Use a target tracking policy to dynamically scale the Auto Scaling group. **Most Voted**

C. Use an AWS Lambda function ta update the desired Auto Scaling group capacity.

D. Use scheduled scaling actions to scale up and scale down the Auto Scaling group.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago

Selected Answer: B

The correct answer is B. To maintain the desired performance across all instances in the Amazon EC2 Auto Scaling group, the solutions architect should use a target tracking policy to dynamically scale the Auto Scaling group.

A target tracking policy allows the Auto Scaling group to automatically adjust the number of EC2 instances in the group based on a target value for a metric. In this case, the target value for the CPU utilization metric could be set to 40% to maintain the desired performance of the application. The Auto Scaling group would then automatically scale the number of instances up or down as needed to maintain the target value for the metric.

https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-scaling-simple-step.html

upvoted 23 times

tom_cruise Highly Voted 2 years, 8 months ago

Selected Answer: B

target tracking policy = maintain

upvoted 6 times

Rcosmos Most Recent 1 year, 4 months ago

Selected Answer: B

Justificativa:
O que é uma política de rastreamento de destino?
Uma política de rastreio de destino (Target Tracking Scaling Policy) ajusta automaticamente o tamanho do grupo de Auto Scaling para manter uma métrica-alvo específica, como a utilização da CPU, próxima a um valor desejado.
Nesse caso, a métrica-alvo seria 40% de utilização da CPU.
Por que a opção B é ideal?
Automação baseada em métricas:
O rastreamento de destino monitora continuamente a utilização da CPU das instâncias e ajusta a capacidade do grupo para manter o desempenho próximo ao alvo configurado (40%).

upvoted 1 times

satyaammm 1 year, 5 months ago

Selected Answer: B

Since it is mentioned that it works best when CPU usage is around 40% hence using target tracking within ASG's is the most suitable option here.

upvoted 1 times

jaradat02 1 year, 10 months ago

Selected Answer: B

B is the correct answer, since target scaling monitors cloudwatch metrics, while simple/step scaling monitors cloudwatch alarms.

upvoted 3 times

1e22522 1 year, 10 months ago
Useful, jaradat02. Thanks.
upvoted 1 times

awsgeek75 2 years, 4 months ago
Selected Answer: B
40% CPU for best performance is a "target tracking" policy for scaling so B is correct.
A: Wrong policy
CD: Won't achieve 40% CPU
upvoted 3 times

Ruffyit 2 years, 7 months ago
https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-scaling-simple-step.html
upvoted 3 times

youdelin 2 years, 8 months ago
Selected Answer: B
I really don't get what kind of software running like a car with the most economical fuel speed range, but well, the answer is B
upvoted 2 times

TariqKipkemei 2 years, 9 months ago
Selected Answer: B
The application performs best when the CPU utilization of the EC2 instances is at or near 40%.
Target tracking will maintain CPU utilization at 40%. When CloudWatch detects that the average CPU utilization is beyond 40%, it will trigger the target tracking policy to scale out the auto scaling group to meet this target utilization. Once everything is settled and the average CPU utilization has gone below 40%, another scale in action will kick in and reduce the number of auto scaling instances in the auto scaling group.
upvoted 2 times

Guru4Cloud 2 years, 9 months ago
Selected Answer: B
The key reasons are:

A target tracking policy allows defining a specific target metric value to maintain, in this case 40% CPU utilization.
Auto Scaling will automatically add or remove instances to keep utilization at the target level, without manual intervention.
This will dynamically scale the group to maintain performance as load changes.
A simple scaling policy only responds to breaching thresholds, not maintaining a target.
Scheduled actions and Lambda would require manual calculation and updates to track utilization.
Target tracking policies are the native Auto Scaling feature designed to maintain a metric at a target value.
upvoted 4 times

cookieMr 2 years, 11 months ago
Selected Answer: B
Target tracking policy is the most appropriate choice. This policy allows ASG to automatically adjust the desired capacity based on a target metric, such as CPU utilization. By setting the target metric to 40%, ASG will scale the number of instances up or down as needed to maintain the desired CPU utilization level. This ensures that the application's performance remains optimal.

A suggests using a simple scaling policy, which allows for scaling based on a fixed metric or threshold. However, it may not be as effective as a target tracking policy in dynamically adjusting the capacity to maintain a specific CPU utilization level.

C suggests using an Lambda to update the desired capacity. While this can be done programmatically, it would require custom scripting and may no provide the same level of automation and responsiveness as a target tracking policy.

D suggests using scheduled scaling actions to scale up and down ASG at predefined times. This approach is not suitable for maintaining the desired performance in real-time based on actual CPU utilization.
upvoted 3 times

Robrobtutu 3 years, 1 month ago
Selected Answer: B
B of course.
upvoted 1 times

aba2s 3 years, 5 months ago
Selected Answer: B
B seem to the correct response.

With a target tracking scaling policy, you can increase or decrease the current capacity of the group based on a target value for a specific metric. This policy will help resolve the over-provisioning of your resources. The scaling policy adds or removes capacity as required to keep the metric at, or

close to, the specified target value. In addition to keeping the metric close to the target value, a target tracking scaling policy also adjusts to changes in the metric due to a changing load pattern.
upvoted 4 times

orionizzie 3 years, 5 months ago

Selected Answer: B

target tracking - CPU at 40%
upvoted 2 times

career360guru 3 years, 5 months ago

Selected Answer: B

Option B
upvoted 1 times

Wpcorgan 3 years, 6 months ago

B is correct
upvoted 1 times

ArielSchivo 3 years, 7 months ago

Selected Answer: B

Option B. Target tracking policy.

https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-scaling-target-tracking.html
upvoted 5 times

Question #131 *Topic 1*

A company is developing a file-sharing application that will use an Amazon S3 bucket for storage. The company wants to serve all the files through an Amazon CloudFront distribution. The company does not want the files to be accessible through direct navigation to the S3 URL.
What should a solutions architect do to meet these requirements?

A. Write individual policies for each S3 bucket to grant read permission for only CloudFront access.

B. Create an IAM user. Grant the user read permission to objects in the S3 bucket. Assign the user to CloudFront.

C. Write an S3 bucket policy that assigns the CloudFront distribution ID as the Principal and assigns the target S3 bucket as the Amazon Resource Name (ARN).

D. Create an origin access identity (OAI). Assign the OAI to the CloudFront distribution. Configure the S3 bucket permissions so that only the OAI has read permission. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

123jhl0 Highly Voted 2 years, 7 months ago

Selected Answer: D
I want to restrict access to my Amazon Simple Storage Service (Amazon S3) bucket so that objects can be accessed only through my Amazon CloudFront distribution. How can I do that?
Create a CloudFront origin access identity (OAI)
https://aws.amazon.com/premiumsupport/knowledge-center/cloudfront-access-to-amazon-s3/
upvoted 40 times

SimonPark 2 years, 7 months ago
Thanks it convinces me
upvoted 1 times

Guru4Cloud Highly Voted 1 year, 9 months ago

Selected Answer: D
The key reasons are:

An OAI provides secure access between CloudFront and S3 without exposing the S3 bucket publicly.
The OAI is associated with the CloudFront distribution.
The S3 bucket policy limits access only to that OAI.
This ensures only CloudFront can access the objects, not direct S3 access.
Option A is complex to manage individual bucket policies.
Option B exposes credentials that aren't needed.
Option C works but OAI is the preferred method.
So using an origin access identity provides the most secure way to serve private S3 content through CloudFront. The OAI prevents direct public access to the S3 bucket.
upvoted 10 times

46e1a46 Most Recent 1 month, 1 week ago

Selected Answer: D
Origin Access Identity is the aws mechanism for restricting an S3 bucket so it can only be accessed through CloudFront, not through direct S3 URLs. You attach the OAI to the CloudFront distribution, then write a bucket policy granting read access only to that OAI
upvoted 1 times

satyaammm 1 year, 5 months ago

Selected Answer: D
OAI's are designed specifically for S3 origins. It helps provide security for CloudFront.

upvoted 1 times

xdkonorek2 1 year, 7 months ago

Selected Answer: D

C would also work but missing important details in the answer
D is legacy and architect should not recommend it

upvoted 4 times

tom_cruise 1 year, 8 months ago

Selected Answer: D

"If your users try to access objects using Amazon S3 URLs, they're denied access. The origin access identity has permission to access objects in your Amazon S3 bucket, but users don't."
https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/private-content-restricting-access-to-s3.html

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: D

To meet the requirements of serving files through CloudFront while restricting direct access to the S3 bucket URL, the recommended approach is to use an origin access identity (OAI). By creating an OAI and assigning it to the CloudFront distribution, you can control access to the S3 bucket. This setup ensures that the files stored in the S3 bucket are only accessible through CloudFront and not directly through the S3 bucket URL. Requests made directly to the S3 URL will be blocked.

Option A suggests writing individual policies for each S3 bucket, which can be cumbersome and difficult to manage, especially if there are multiple buckets involved.

Option B suggests creating an IAM user and assigning it to CloudFront, but this does not address restricting direct access to the S3 bucket URL.

Option C suggests writing an S3 bucket policy with CloudFront distribution ID as the Principal, but this alone does not provide the necessary restrictions to prevent direct access to the S3 bucket URL.

upvoted 4 times

antropaws 2 years ago

DECEMBER 2022 UPDATE:

Restricting access to an Amazon S3 origin:

CloudFront provides two ways to send authenticated requests to an Amazon S3 origin: origin access control (OAC) and origin access identity (OAI). We recommend using OAC because it supports:

All Amazon S3 buckets in all AWS Regions, including opt-in Regions launched after December 2022
Amazon S3 server-side encryption with AWS KMS (SSE-KMS)
Dynamic requests (PUT and DELETE) to Amazon S3

OAI doesn't work for the scenarios in the preceding list, or it requires extra workarounds in those scenarios.

https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/private-content-restricting-access-to-s3.html

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: D

The correct answer is D. To meet the requirements, the solutions architect should create an origin access identity (OAI) and assign it to the CloudFront distribution. The S3 bucket permissions should be configured so that only the OAI has read permission.

An OAI is a special CloudFront user that is associated with a CloudFront distribution and is used to give CloudFront access to the files in an S3 bucket. By using an OAI, the company can serve the files through the CloudFront distribution while preventing direct access to the S3 bucket.

https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/private-content-restricting-access-to-s3.html

upvoted 6 times

career360guru 2 years, 5 months ago

Selected Answer: D

D is the right answer

upvoted 1 times

gloritown 2 years, 6 months ago

Selected Answer: D

D is correct but instead of OAI using OAC would be better since OAI is legacy

upvoted 3 times

Robrobtutu 2 years, 1 month ago

Thanks, I didn't know about OAC.

upvoted 1 times

Wpcorgan 2 years, 6 months ago

D is correct

upvoted 1 times

Question #132 *Topic 1*

A company's website provides users with downloadable historical performance reports. The website needs a solution that will scale to meet the company's website demands globally. The solution should be cost-effective, limit the provisioning of infrastructure resources, and provide the fastest possible response time.
Which combination should a solutions architect recommend to meet these requirements?

A. Amazon CloudFront and Amazon S3 **Most Voted**

B. AWS Lambda and Amazon DynamoDB

C. Application Load Balancer with Amazon EC2 Auto Scaling

D. Amazon Route 53 with internal Application Load Balancers

Correct Answer: A

Community vote distribution

A (96%) 2%

Comments

G3 Highly Voted 3 years, 4 months ago

Selected Answer: A

Historical reports = Static content = S3

upvoted 22 times

dokaedu Highly Voted 3 years, 7 months ago

A is the correct answer

The solution should be cost-effective, limit the provisioning of infrastructure resources, and provide the fastest possible response time.

upvoted 11 times

Manimgh Most Recent 1 year, 2 months ago

Selected Answer: A

Question mentioned cost effective and S3 is the most cost effective answer

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: A

Ans A - S3 is designed to optimise costs, highly scalable and can store static content such as website. CloudFront is designed to securely deliver content with low latency and high transfer rate.

upvoted 2 times

huaze_lei 1 year, 9 months ago

Selected Answer: A

Historical data will not change, and hence they are static content. So the answer is S3 with distributed content (Cloudfront)

upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: A

A is the most suitable choice because the content is static and downloadable

upvoted 2 times

KTEgghead 1 year, 11 months ago

Selected Answer: A

A. S3 is designed to optimize storage costs, is highly scalable and can hold static content e.g. a website. CloudFront is designed to securely deliver content with low latency and high transfer speeds.

upvoted 2 times

MrPCarrot 2 years, 4 months ago
Bringing content closer to users, Answer is A
upvoted 3 times

TariqKipkemei 2 years, 9 months ago
Selected Answer: A
Global, cost-effective, serverless, low latency = CloudFront with S3
Static content = S3
upvoted 6 times

Guru4Cloud 2 years, 9 months ago
Selected Answer: A
Historical reports = Static content = S3
upvoted 3 times

cookieMr 2 years, 11 months ago
By using CloudFront, the website can leverage the global network of edge locations to cache and deliver the performance reports to users from the nearest edge location, reducing latency and providing fast response times. Amazon S3 serves as the origin for the files, where the reports are stored.

Option B is incorrect because AWS Lambda and Amazon DynamoDB are not the most suitable services for serving downloadable files and meeting the website demands globally.

Option C is incorrect because using an Application Load Balancer with Amazon EC2 Auto Scaling may require more infrastructure provisioning and management compared to the CloudFront and S3 combination. Additionally, it may not provide the same level of global scalability and fast response times as CloudFront.

Option D is incorrect because while Amazon Route 53 is a global DNS service, it alone does not provide the caching and content delivery capabilities required for serving the downloadable reports. Internal Application Load Balancers do not address the global scalability and caching requirements specified in the scenario.
upvoted 5 times

Bmarodi 2 years, 11 months ago
Very good explanations!
upvoted 1 times

Buruguduystunstugudunstuy 3 years, 5 months ago
Selected Answer: A
The correct answer is Option A. To meet the requirements, the solutions architect should recommend using Amazon CloudFront and Amazon S3.

By combining Amazon CloudFront and Amazon S3, the solutions architect can provide a scalable and cost-effective solution that limits the provisioning of infrastructure resources and provides the fastest possible response time.

https://aws.amazon.com/cloudfront/

https://aws.amazon.com/s3/
upvoted 4 times

techhb 3 years, 5 months ago
A is correct
upvoted 1 times

career360guru 3 years, 5 months ago
Selected Answer: A
A is the best and most cost effective option if only download of the static pre-created report(no data processing before downloading) is a requirement.
upvoted 2 times

Wpcorgan 3 years, 6 months ago
A is correct
upvoted 1 times

sdasdawa 3 years, 7 months ago
Selected Answer: A
https://www.examtopics.com/discussions/amazon/view/27935-exam-aws-certified-solutions-architect-associate-saa-c02/
upvoted 2 times

Nirmal3331 3 years, 7 months ago
Selected Answer: A
https://www.examtopics.com/discussions/amazon/view/27935-exam-aws-certified-solutions-architect-associate-saa-c02/

upvoted 2 times

Question #133 *Topic 1*

A company runs an Oracle database on premises. As part of the company's migration to AWS, the company wants to upgrade the database to the most recent available version. The company also wants to set up disaster recovery (DR) for the database. The company needs to minimize the operational overhead for normal operations and DR setup. The company also needs to maintain access to the database's underlying operating system.
Which solution will meet these requirements?

A. Migrate the Oracle database to an Amazon EC2 instance. Set up database replication to a different AWS Region.

B. Migrate the Oracle database to Amazon RDS for Oracle. Activate Cross-Region automated backups to replicate the snapshots to another AWS Region.

C. Migrate the Oracle database to Amazon RDS Custom for Oracle. Create a read replica for the database in another AWS Region. **Most Voted**

D. Migrate the Oracle database to Amazon RDS for Oracle. Create a standby database in another Availability Zone.

Correct Answer: C

Community vote distribution

C (50%) A (40%) 9%

Comments

ArielSchivo Highly Voted 3 years, 7 months ago

Option C since RDS Custom has access to the underlying OS and it provides less operational overhead. Also, a read replica in another Region can be used for DR activities.

https://aws.amazon.com/blogs/database/implementing-a-disaster-recovery-strategy-with-amazon-rds/

upvoted 45 times

KalarAzar 2 years, 12 months ago

You can't create cross-Region replicas in RDS Custom for Oracle: https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/custom-rr.html#custom-rr.limitations

upvoted 23 times

brushek Highly Voted 3 years, 8 months ago

Selected Answer: C

It should be C:

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/rds-custom.html
and
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/working-with-custom-oracle.html

upvoted 21 times

bhgt 2 years, 8 months ago

how it is C when the read replica is not meant for DR

upvoted 3 times

wsdasdasdqwdaw 2 years, 7 months ago

If the source DB instance fails, you can promote your Read Replica to a standalone source server.

upvoted 14 times

CloudExpert01 Most Recent 1 year, 2 months ago

Selected Answer: C

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/working-with-custom-oracle.html

upvoted 1 times

AshishDhole 1 year, 4 months ago

Selected Answer: C

The company also needs to maintain access to the database's underlying operating system --> only RDS Custom can provide this facility

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: C

you can create a read replica in another region which is less operational cost for DR

upvoted 1 times

0de7d1b 1 year, 6 months ago

Selected Answer: C

Amazon RDS Custom for Oracle:

Amazon RDS Custom for Oracle provides a managed database service while also allowing access to the underlying operating system, which is required in this scenario.
This option strikes a balance between minimizing operational overhead (as AWS manages backups, patching, and scaling) and retaining flexibility for customizations or specific access requirements.

upvoted 2 times

TheTeaBoy 1 year, 7 months ago

A: Incorrect - Question states to minimise operational overhead, building an EC2 instance and managing that doesn't reduce the operational overhead.
B: Incorrect - Yes that ticks the DR aspect, but Amazon RDS for Oracle doesn't provide access to the underlying operating system of the database server.
C: Correct - Amazon RDS Custom provides a limited amount of operating system access of the database server(s), so that's good. Creating a read replica in another AWS Region covers DR. Its not asking for High Availability (HA) in this case.
D: Incorrect - Amazon RDS for Oracle doesn't provide access to the underlying operating system of the database server. A standby database in another availability, could be considered as covering DR.

upvoted 3 times

mzeynalli 1 year, 7 months ago

Selected Answer: C

Amazon RDS Custom for Oracle is the best choice because it allows access to the underlying OS, provides an easier path for upgrades, minimizes operational overhead, and supports setting up cross-region read replicas for a DR solution.

upvoted 2 times

19d92c7 1 year, 7 months ago

It should be standby:
. When the primary database fails,
Amazon RDS promotes the secondary database to primary. Because it assumes the primary databases endpoint,
the EC2 instances can resume traffic with the new primary database. Meanwhile, a new standby database is
created in the other Availability Zone.

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans D - use regular RDS Oracle; no need for RDS Custom because the question doesn't state and special customisations... the only bit unanswered is "...maintain access to company's underlying o/s"

upvoted 2 times

LuongTo 1 year, 7 months ago

D is out since DR is more about "different region", not just "different AZ"

upvoted 2 times

bignatov 1 year, 9 months ago

Selected Answer: C

Option C provides the right balance of managed service convenience, access to the underlying OS, and effective disaster recovery with cross-region read replicas

upvoted 2 times

MatAlves 1 year, 9 months ago

Selected Answer: A

"You can't create cross-Region RDS Custom for Oracle replicas." So it can't be C.
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/custom-rr.html

B and D don't provide access to underlying OS. So the only option left is A, which won't help minimize the operational overhead though.

upvoted 5 times

scaredSquirrel 1 year, 10 months ago

Selected Answer: C

B, D - do not have access to underlying OS.
Both A and C could work, but C is less Overhead.

C - Amazon RDS Custom for Oracle actually supports creating read replicas. "Creating an RDS Custom for Oracle replica is similar to creating an RDS for Oracle replica, but with important differences." (https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/custom-rr.html). The read replica is not meant for disaster recovery, but it could work as such when no better options are available.

upvoted 2 times

MatAlves 1 year, 9 months ago

"You can't create cross-Region RDS Custom for Oracle replicas."

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/custom-rr.html

upvoted 1 times

Abbas_Abi_AWS 1 year, 10 months ago

Selected Answer: A

You can't create cross-Region RDS Custom for Oracle replicas.

upvoted 5 times

jaradat02 1 year, 10 months ago

Selected Answer: C

Since we need to have access to the underlying infrastructure, C makes sense.

upvoted 1 times

ChinthaGurumurthi 1 year, 10 months ago

Selected Answer: C

Its C
Key : Access to underlying O.S - RDS custom can give you this feature
General RDS for Oracle(or any) - you can't access underlying O.S
So, definitely C considering this point in the question

upvoted 1 times

freedafeng 1 year, 11 months ago

Defintely not C. You cannot create a read replica of RDS custom for Oracle in a different region: https://aws.amazon.com/blogs/database/build-high-availability-for-amazon-rds-custom-for-oracle-using-read-replicas/

upvoted 1 times

Question #134 *Topic 1*

A company wants to move its application to a serverless solution. The serverless solution needs to analyze existing and new data by using SL. The company stores the data in an Amazon S3 bucket. The data requires encryption and must be replicated to a different AWS Region.
Which solution will meet these requirements with the LEAST operational overhead?

A. Create a new S3 bucket. Load the data into the new S3 bucket. Use S3 Cross-Region Replication (CRR) to replicate encrypted objects to an S3 bucket in another Region. Use server-side encryption with AWS KMS multi-Region kays (SSE-KMS). Use Amazon Athena to query the data.

B. Create a new S3 bucket. Load the data into the new S3 bucket. Use S3 Cross-Region Replication (CRR) to replicate encrypted objects to an S3 bucket in another Region. Use server-side encryption with AWS KMS multi-Region keys (SSE-KMS). Use Amazon RDS to query the data.

C. Load the data into the existing S3 bucket. Use S3 Cross-Region Replication (CRR) to replicate encrypted objects to an S3 bucket in another Region. Use server-side encryption with Amazon S3 managed encryption keys (SSE-S3). Use Amazon Athena to query the data. **Most Voted**

D. Load the data into the existing S3 bucket. Use S3 Cross-Region Replication (CRR) to replicate encrypted objects to an S3 bucket in another Region. Use server-side encryption with Amazon S3 managed encryption keys (SSE-S3). Use Amazon RDS to query the data.

Correct Answer: C

Community vote distribution

C (53%) A (47%)

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: C

SSE-KMS vs SSE-S3 - The last seems to have less overhead (as the keys are automatically generated by S3 and applied on data at upload, and don't require further actions. KMS provides more flexibility, but in turn involves a different service, which finally is more "complex" than just managing one (S3). So A and B are excluded. If you are in doubt, you are having 2 buckets in A and B, while just keeping one in C and D.
https://s3browser.com/server-side-encryption-types.aspx
Decide between C and D is deciding on Athena or RDS. RDS is a relational db, and we have documents on S3, which is the use case for Athena. Athena is also serverless, which eliminates the need of controlling the underlying infrastructure and capacity. So C is the answer.
https://aws.amazon.com/athena/

upvoted 67 times

markw92 2 years, 12 months ago

See comment from Nicknameinvalid below. You get your answer.

upvoted 2 times

MutiverseAgent 2 years, 11 months ago

It'a since replication works for new objects but not for the existing ones, untless you use batch replication which is not the case.

upvoted 1 times

Chiznitz 2 years, 7 months ago

Answer A has you move the data before you enable replication, therefore there is no difference between A and C when it comes to the point in time you enable replication. I agree A would be a better choice if the order of operations said, create a bucket->Enable encryption->move files...but it doesn't. It has you create the bucket and move the files.

upvoted 3 times

dokaedu Highly Voted 3 years, 7 months ago

Answer is A:
Amazon S3 Bucket Keys reduce the cost of Amazon S3 server-side encryption using AWS Key Management Service (SSE-KMS). This new bucket-level key for SSE can reduce AWS KMS request costs by up to 99 percent by decreasing the request traffic from Amazon S3 to AWS KMS. With a few clicks

in the AWS Management Console, and without any changes to your client applications, you can configure your bucket to use an S3 Bucket Key for AWS KMS-based encryption on new objects.
The Existing S3 bucket might have uncrypted data - encryption will apply new data received after the applying of encryption on the new bucket.
upvoted 32 times

s50600822 3 years, 1 month ago
Don't know what "kays" are, could they be a trap?
upvoted 2 times

Bmarodi 3 years ago
Kays = keys, mistype i think.
upvoted 1 times

RBSK 3 years, 6 months ago
Cost reduction is in comparison bet Bucket level KMS key and object level KMS key. Not between SSE-KMS and SSE-S3. Hence its a wrong comparison
upvoted 2 times

RODROSKAR 3 years, 7 months ago
Reducing cost was never the target, it's LEAST operational. In that regard SSE-S3 AWS fully managed.
upvoted 6 times

LuckyAro 3 years, 4 months ago
I didn't read anywhere in the question where cost was an issue of consideration, so how you made it a main issue here is beyond me.
upvoted 6 times

46e1a46 Most Recent 1 month, 1 week ago
Selected Answer: A
WHY YOU NEED A NEW BUCKET!!!

S3 Cross Region Replication only replicates objects uploaded AFTER CRR is enabled. It does not retroactively replicate objects in the bucket
upvoted 1 times

607d657 10 months, 2 weeks ago
Selected Answer: A
- Serverless Analysis: Amazon Athena is a serverless query service that lets you analyze data directly in S3 using SQL — no infrastructure to manage.
- Encryption: SSE-KMS with multi-Region keys provides strong encryption and seamless replication support across Regions.
- Replication: S3 CRR handles replication automatically once configured, minimizing manual effort.
- Operational Simplicity: Creating a new bucket ensures clean configuration and avoids legacy settings or permissions issues.
upvoted 1 times

3386cca 10 months, 2 weeks ago
Selected Answer: A
It's A. or C because of athena. But can't be C because ❌ SSE-S3 encryption is not supported with CRR by default. You need SSE-KMS or unencryptec data for CRR.
upvoted 1 times

Clouddon 11 months, 1 week ago
Selected Answer: A
A. Create a new S3 bucket. Load the data into the new S3 bucket. Use S3 Cross-Region Replication (CRR) to replicate encrypted objects to an S3 bucket in another Region. Use server-side encryption with AWS KMS multi-Region keys (SSE-KMS). Use Amazon Athena to query the data.
upvoted 1 times

MortisG 1 year ago
Selected Answer: A
Correct answer is A. SSE-S3 is not supported for CRR.
upvoted 1 times

CloudExpert01 1 year, 2 months ago
Selected Answer: C
since the requirement is "The serverless solution needs to analyze existing and new data", the answer C seems to be more appropriate since all the data will be in 1 bucket
upvoted 1 times

Mimine87 1 year, 2 months ago
Selected Answer: A
Amazon S3 + Athena is a fully serverless data analytics solution (no infrastructure to manage).

S3 Cross-Region Replication (CRR) enables automatic, asynchronous replication of objects across AWS Regions, satisfying the replication requirement.

SSE-KMS with multi-Region keys ensures that the data is encrypted and can be decrypted in the destination Region, which is crucial for cross-region analytics and compliance.

Athena is used for analyzing data directly in S3 using SQL, which meets the requirement to analyze existing and new data using SQL (SL was a typo).

This option delivers everything with the least operational overhead — no server provisioning, no DB management, and built-in encryption + replication.

upvoted 1 times

Faraz999 1 year, 2 months ago

Selected Answer: A

you can apply cross-region replication to an existing Amazon S3 bucket, but it will only replicate new objects after the replication rule is configured. To replicate existing objects, you'll need to use S3 Batch Replication.

upvoted 1 times

MPG1970 1 year, 2 months ago

Selected Answer: A

I went for A because, if you start to encrypt only new data then
1. You will have an inconsistently solution for security. I think it fair to assume that if they want all new data encrypted, then all data should be encrypted
2. the serverless function will become more complex as it will have to establish whether the data is encrypted or not every time it retrieves an object.

Therefore you need to create a new S3 bucket so that all data will be encrypted

upvoted 1 times

SirDNS 1 year, 2 months ago

Selected Answer: C

fewer services = less operational overhead
managed services/serverless = less operational overhead

upvoted 1 times

tch 1 year, 3 months ago

Selected Answer: A

If you want a straightforward approach to encrypting your S3 data without managing your own encryption keys, SSE-S3 is a good option.

you need AWS KMS for this complex solution

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: A

A new bucket is needed to encrypt objects. C uses existing bucket.

upvoted 1 times

LeonSauveterre 1 year, 4 months ago

Selected Answer: A

About typos: "kays" should be "keys", and "SL" should be "SQL". So you gotta choose between A and C (We need Athena!).

About option C: SSE-S3 is a valid encryption method, just not as suitable as SSE-KMS with multi-Region keys for CRR. SSE-KMS with multi-Region keys simplifies key management in the destination Region.

upvoted 2 times

Rcosmos 1 year, 4 months ago

Selected Answer: U

A melhor opção para atender aos requisitos com a menor sobrecarga operacional é a opção A:

A. Crie um novo bucket do S3. Carregue os dados no novo bucket do S3. Use a replicação entre regiões (CRR) do S3 para replicar objetos criptografados para um bucket do S3 em outra região. Use a criptografia no lado do servidor com chaves multirregionais do AWS KMS (SSE-KMS). Use o Amazon Athena para consultar os dados.Conclusão:
A opção A é a escolha ideal, pois combina os recursos sem servidor do Amazon Athena, a segurança avançada do SSE-KMS com chaves multirregionais, e a replicação automática entre regiões com o S3 CRR, tudo com a menor sobrecarga operacional.

upvoted 1 times

skylerwhite 1 year, 5 months ago

Selected Answer: A

what is the different between create a new S3 bucket and load data into the existing S3 bucket? I don't get it this. Please..

upvoted 1 times

Question #135 Topic 1

A company runs workloads on AWS. The company needs to connect to a service from an external provider. The service is hosted in the provider's VPC. According to the company's security team, the connectivity must be private and must be restricted to the target service. The connection must be initiated only from the company's VPC.
Which solution will mast these requirements?

A. Create a VPC peering connection between the company's VPC and the provider's VPC. Update the route table to connect to the target service.

B. Ask the provider to create a virtual private gateway in its VPC. Use AWS PrivateLink to connect to the target service.

C. Create a NAT gateway in a public subnet of the company's VPUpdate the route table to connect to the target service.

D. Ask the provider to create a VPC endpoint for the target service. Use AWS PrivateLink to connect to the target service.
Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: D

AWS PrivateLink provides private connectivity between VPCs, AWS services, and your on-premises networks, without exposing your traffic to the public internet. AWS PrivateLink makes it easy to connect services across different accounts and VPCs to significantly simplify your network architecture.
Interface **VPC endpoints**, powered by AWS PrivateLink, connect you to services hosted by AWS Partners and supported solutions available in AWS Marketplace.
https://aws.amazon.com/privatelink/

upvoted 33 times

remand Highly Voted 3 years, 4 months ago

Selected Answer: D

The solution that meets these requirements best is option D.

By asking the provider to create a VPC endpoint for the target service, the company can use AWS PrivateLink to connect to the target service. This enables the company to access the service privately and securely over an Amazon VPC endpoint, without requiring a NAT gateway, VPN, or AWS Direct Connect. Additionally, this will restrict the connectivity only to the target service, as required by the company's security team.

Option A VPC peering connection may not meet security requirement as it can allow communication between all resources in both VPCs.
Option B, asking the provider to create a virtual private gateway in its VPC and use AWS PrivateLink to connect to the target service is not the optimal solution because it may require the provider to make changes and also you may face security issues.
Option C, creating a NAT gateway in a public subnet of the company's VPC can expose the target service to the internet, which would not meet the security requirements.

upvoted 11 times

PaulGa Most Recent 1 year, 8 months ago

Selected Answer: D

Ans D - create a unique, private only link: "Ask the provider to create a VPC endpoint for the target service. Use AWS PrivateLink to connect to the target service"

upvoted 2 times

lofzee 2 years ago

Selected Answer: D

no split decisions on this answer eh? not like the last one. lol

upvoted 3 times

RNess 2 years, 7 months ago

Selected Answer: D

AWS PrivateLink / VPC Endpoint Services:
• Connect services privately from your service VPC to customers VPC
• Doesn't need VPC Peering, public Internet, NAT Gateway, Route Tables
• Must be used with Network Load Balancer & ENI

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: D

option D is correct

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

The best solution to meet the requirements is option D:

Ask the provider to create a VPC endpoint for the target service
Use AWS PrivateLink to connect to the target service
The reasons are:

PrivateLink provides private connectivity between VPCs without using public internet.
The provider creates a VPC endpoint in their VPC for the target service.
The company uses PrivateLink to securely access the endpoint from their VPC.
Connectivity is restricted only to the target service.
The connection is initiated only from the company's VPC.
Options A, B, C would expose the connection to the public internet or require infrastructure changes in the provider's VPC.

PrivateLink enables private, restricted connectivity to the target service without VPC peering or public exposure.

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: D

Option C meets the requirements of establishing a private and restricted connection to the service hosted in the provider's VPC. By asking the provider to create a VPC endpoint for the target service, you can establish a direct and private connection from your company's VPC to the target service. AWS PrivateLink ensures that the connectivity remains within the AWS network and does not require internet access. This ensures both privacy and restriction to the target service, as the connection can only be initiated from your company's VPC.

A. VPC peering does not restrict access only to the target service.
B. PrivateLink is typically used for accessing AWS services, not external services in a provider's VPC.
C. NAT gateway does not provide a private and restricted connection to the target service.

Option D is the correct choice as it uses AWS PrivateLink and VPC endpoint to establish a private and restricted connection from the company's VPC to the target service in the provider's VPC.

upvoted 4 times

Abrar2022 3 years ago

VPC Endpoint (Target Service) - for specific services (not accessing whole vpc)
VPC Peering - (accessing whole VPC)

upvoted 4 times

Abrar2022 3 years ago

VPC Peering Connection:
All resources in a VPC, such as ECSs and load balancers, can be accessed.

VPC Endpoint:
Allows access to a specific service or application. Only the ECSs and load balancers in the VPC for which VPC endpoint services are created can be accessed.

upvoted 2 times

eugene_stalker 3 years ago

Selected Answer: D

Option D, but seems that it is vise versa. Customer needs to create Privatelink and and you VPC endpoint to connect to Privatelink

upvoted 2 times

studynoplay 3 years, 1 month ago

AWS PrivateLink / VPC Endpoint Services:
• Connect services privately from your service VPC to customers VPC
• Doesn't need VPC Peering, public Internet, NAT Gateway, Route Tables
• Must be used with Network Load Balancer & ENI

upvoted 3 times

Help2023 3 years, 3 months ago

Selected Answer: D

D. Here you are the one initiating the connection

upvoted 2 times

devonwho 3 years, 4 months ago

Selected Answer: D

PrivateLink is a more generalized technology for linking VPCs to other services. This can include multiple potential endpoints: AWS services, such as Lambda or EC2; Services hosted in other VPCs; Application endpoints hosted on-premises.

https://www.tinystacks.com/blog-post/aws-vpc-peering-vs-privatelink-which-to-use-and-when/

upvoted 2 times

devonwho 3 years, 4 months ago

Selected Answer: D

While VPC peering enables you to privately connect VPCs, AWS PrivateLink enables you to configure applications or services in VPCs as endpoints that your VPC peering connections can connect to.

upvoted 3 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: D

The solution that meets these requirements is Option D:

* Ask the provider to create a VPC endpoint for the target service.
* Use AWS PrivateLink to connect to the target service.

Option D involves asking the provider to create a VPC endpoint for the target service, which is a private connection to the service that is hosted in the provider's VPC. This ensures that the connection is private and restricted to the target service, as required by the company's security team. The company can then use AWS PrivateLink to connect to the target service over the VPC endpoint. AWS PrivateLink is a fully managed service that enables you to privately access services hosted on AWS, on-premises, or in other VPCs. It provides secure and private connectivity to services by using private IP addresses, which ensures that traffic stays within the Amazon network and does not traverse the public internet.

Therefore, Option D is the solution that meets the requirements.

upvoted 5 times

Buruguduystunstugudunstuy 3 years, 5 months ago

AWS PrivateLink documentation: https://docs.aws.amazon.com/privatelink/latest/userguide/what-is-privatelink.html

upvoted 2 times

techhb 3 years, 5 months ago

D is right,if requirement was to be ok with public internet then option C was ok.

upvoted 1 times

Question #136 *Topic 1*

A company is migrating its on-premises PostgreSQL database to Amazon Aurora PostgreSQL. The on-premises database must remain online and accessible during the migration. The Aurora database must remain synchronized with the on-premises database.
Which combination of actions must a solutions architect take to meet these requirements? (Choose two.)

A. Create an ongoing replication task. **Most Voted**

B. Create a database backup of the on-premises database.

C. Create an AWS Database Migration Service (AWS DMS) replication server. **Most Voted**

D. Convert the database schema by using the AWS Schema Conversion Tool (AWS SCT).

E. Create an Amazon EventBridge (Amazon CloudWatch Events) rule to monitor the database synchronization.

Correct Answer: AC

Community vote distribution

AC (92%) 7%

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: AC

AWS Database Migration Service (AWS DMS) helps you migrate databases to AWS quickly and securely. The source database remains fully operational during the migration, minimizing downtime to applications that rely on the database.
... With AWS Database Migration Service, you can also continuously replicate data with low latency from any supported source to any supported target.
https://aws.amazon.com/dms/

upvoted 33 times

pentium75 2 years, 5 months ago

PostgreSQL -> Aurora PostgreSQL requires schema conversion per https://aws.amazon.com/dms/schema-conversion-tool/

upvoted 3 times

LoXoL 2 years, 5 months ago

SCT is compatible with PostgreSQL as source and Aurora PostgreSQL as destination, but not required.

upvoted 3 times

gustavtd Highly Voted 3 years, 5 months ago

Selected Answer: AC

AC, here it is clearly shown https://docs.aws.amazon.com/zh_cn/dms/latest/sbs/chap-manageddatabases.postgresql-rds-postgresql.html

upvoted 11 times

LuckyAro 3 years, 4 months ago

You nailed it !

upvoted 1 times

0de7d1b Most Recent 1 year, 6 months ago

Selected Answer: AC

DMS helps to migrate the data from onpremise to aws and require replication task

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: CE

Ans C, E -
C: AWS Database Migration Service to migrate databases to AWS, source database remains fully operational during the migration, avoiding application downtime.
https://aws.amazon.com/dms/

E: monitor with CloudWatch
As for C - not convinced: its PostgreSQL to PostgreSQL migration... no SCT is needed?
upvoted 1 times

MatAlves 1 year, 8 months ago
No, since you're using the same schema for Postgresql.
upvoted 3 times

jaradat02 1 year, 10 months ago
Selected Answer: AC
A and c obviously
upvoted 1 times

jatric 1 year, 11 months ago
Selected Answer: AC
AC is more accurate as DMS will help to migrate the on-premises data base to cloud with ease and for ongoing replication to synchronized the datbase "ongiong replication task" will be helpfull.

And yes its PostgreSQL to PostgreSQL migration so no SCT is needed ehre
upvoted 2 times

BombArat 1 year, 11 months ago
Selected Answer: AC
WeLL CHATGPT says SCT is not required so , AC makes sense
upvoted 2 times

CCCat 1 year, 12 months ago
Selected Answer: AC
Keywords:
- migrating its on-premises PostgreSQL database to Amazon Aurora PostgreSQL
- The Aurora database must remain synchronized with the on-premises database
Analysis:
Option A satisfy the requirement of "synchronized with the on-premises database"
Option C suits for the homogeneous database migration.
Option D is not needed in this scenario, it suits for the heterogeneous database.

Homogeneous database migration tools: https://docs.aws.amazon.com/prescriptive-guidance/latest/migration-oracle-database/homogeneous-migration-tools.html

Heterogeneous database migration tools: https://aws.amazon.com/dms/schema-conversion-tool/
upvoted 2 times

Hopeyemi 2 years, 1 month ago
To meet the requirements of migrating an on-premises PostgreSQL database to Amazon Aurora PostgreSQL while keeping the on-premises database online and ensuring synchronization with the Aurora database, the following actions need to be taken:

Create an ongoing replication task (Option A): This action involves setting up continuous replication between the on-premises PostgreSQL database and the Aurora PostgreSQL database. This ensures that changes made to the on-premises database are replicated to the Aurora database in real-time, keeping them synchronized.
Create an AWS Database Migration Service (AWS DMS) replication server (Option C): AWS DMS provides a reliable and efficient way to migrate databases to AWS while minimizing downtime. By creating an AWS DMS replication server, you can configure and manage the replication tasks between the on-premises database and the Aurora database.
upvoted 4 times

Alphateccc 2 years, 3 months ago
answer is CD : postgresql and aurora postgresql have different schemes, you need sct for conversion and dms for the migration (replication)
upvoted 1 times

vip2 2 years, 3 months ago
Selected Answer: AC
AC
perform ongoing replication using AWS DMS to keep the source and target databases in sync
upvoted 2 times

farnamjam 2 years, 5 months ago
Selected Answer: AC
DMS has Continuous Data Replication using CDC
upvoted 3 times

Michael_Li 2 years, 6 months ago

CD
A is out because it does not specify what is the service to perform the replication task, clearly what needed here is DMS
B is out because backup is solution to keep 2 DB in sync, backup and restore takes long time
C is correct as DMS takes care both full load and ongoing replication, see this youtube video https://www.youtube.com/watch?v=VhXDa9SPDLw
D is right as from to PostgreSQL to Amazon Aurora PostgreSQL you need AWS Schema Conversion Tool, see https://aws.amazon.com/dms/schema-conversion-tool/
E is out monitor itself doen't perform the replication work, if we have to choose 3 options then we can have E selected

upvoted 2 times

pentium75 2 years, 5 months ago

https://docs.aws.amazon.com/zh_cn/dms/latest/sbs/chap-manageddatabases.postgresql-rds-postgresql-ongoing-replication.html literally says that you must "configure the ongoing replication task"

upvoted 2 times

Mikado211 2 years, 6 months ago

Well technically when you operate such task, you must create a database on the cloud, then operate a migration using DMS and none of the propositions give you those two tasks separately. Sometimes those questions can be really frustrating.

upvoted 1 times

Amitabha09 2 years, 8 months ago

C. Create an AWS Database Migration Service (AWS DMS) replication server.
E. Create an Amazon EventBridge (Amazon CloudWatch Events) rule to monitor the database synchronization.

AWS DMS can replicate data from on-premises databases to Aurora PostgreSQL in real time, so the on-premises database will remain online and accessible during the migration. AWS DMS can also automatically convert the database schema, so there is no need to use AWS SCT.

An Amazon EventBridge rule can be used to monitor the database synchronization and send notifications if any errors occur. This is important because it allows the solutions architect to quickly identify and resolve any issues that may arise during the migration.

A database backup of the on-premises database is not necessary because AWS DMS will replicate the data in real time. Creating an ongoing replication task is not necessary because AWS DMS will automatically create an ongoing replication task when the replication server is created.

upvoted 2 times

David_Ang 2 years, 7 months ago

Mate you can monitor everything you want but it is not going to make sure the synchronization is working, an alert is not going to help.

upvoted 4 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: AC

Create an AWS Database Migration Service (AWS DMS) replication server then create an ongoing replication task

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: AC

A) Create an ongoing replication task

C) Create an AWS Database Migration Service (AWS DMS) replication server

The key reasons are:

An ongoing DMS replication task keeps the source and target databases synchronized during the migration.
The DMS replication server manages and executes the replication tasks.
Together, these will continuously replicate changes from on-prem to Aurora to keep them in sync.
A database backup alone wouldn't maintain synchronization.

upvoted 2 times

Question #137 *Topic 1*

A company uses AWS Organizations to create dedicated AWS accounts for each business unit to manage each business unit's account independently upon request. The root email recipient missed a notification that was sent to the root user email address of one account. The company wants to ensure that all future notifications are not missed. Future notifications must be limited to account administrators.
Which solution will meet these requirements?

A. Configure the company's email server to forward notification email messages that are sent to the AWS account root user email address to all users in the organization.

B. Configure all AWS account root user email addresses as distribution lists that go to a few administrators who can respond to alerts. Configure AWS account alternate contacts in the AWS Organizations console or programmatically. Most Voted

C. Configure all AWS account root user email messages to be sent to one administrator who is responsible for monitoring alerts and forwarding those alerts to the appropriate groups.

D. Configure all existing AWS accounts and all newly created accounts to use the same root user email address. Configure AWS account alternate contacts in the AWS Organizations console or programmatically.

Correct Answer: B

Community vote distribution

B (86%) 14%

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: B

Use a group email address for the management account's root user
https://docs.aws.amazon.com/organizations/latest/userguide/orgs_best-practices_mgmt-acct.html#best-practices_mgmt-acct_email-address

upvoted 28 times

cookieMr Highly Voted 2 years, 11 months ago

Selected Answer: B

Option B ensures that all future notifications are not missed by configuring the AWS account root user email addresses as distribution lists that are monitored by a few administrators. By setting up alternate contacts in the AWS Organizations console or programmatically, the notifications can be sent to the appropriate administrators responsible for monitoring and responding to alerts. This solution allows for centralized management of notifications and ensures they are limited to account administrators.

A. Floods all users with notifications, lacks granularity.
C. Manual forwarding introduces delays, centralizes responsibility.
D. No flexibility for specific account administrators, limits customization.

upvoted 14 times

15df3d0 Most Recent 1 year, 3 months ago

Selected Answer: D

//docs.aws.amazon.com/organizations/latest/userguide/orgs_best-practices_mgmt-acct.html#best-practices_mgmt-acct_email-address

upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: B

Ans B - as opposed to option D, because the organisation account structure implies there is more than one root account: "The root email recipient missed a notification that was sent to the root user email address of one account."

upvoted 2 times

MatAlves 1 year, 9 months ago

Selected Answer: B

"Use a group email address for root user credentials:

Use an email address that is managed by your business and forwards received messages directly to a group of users. If AWS must contact the owner of the account, this approach reduces the risk of delays in responding, even if individuals are on vacation, out sick, or have left the business. The email address used for the root user should not be used for other purposes."
https://docs.aws.amazon.com/IAM/latest/UserGuide/root-user-best-practices.html#ru-bp-group

upvoted 2 times

MandAsh 2 years ago

Lol.. How is this AWS related question. Isnt it general knowledge.

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: B

No idea why "D" would be correct answer unless there is some missing context in the question or the answer. "B" is best practice as pointed out in other links.

upvoted 1 times

David_Ang 2 years, 7 months ago

Selected Answer: B

the only answer with sense is "B", because "A" is not exclusive, "C" is exactly the case the want to avoid, and "D" just don't make sense

upvoted 2 times

tom_cruise 2 years, 8 months ago

Selected Answer: B

distribution list is the way to go

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

The reasons are:

Alternate contacts allow defining other users to receive root emails.
Distribution lists ensure multiple admins get notified.
Limits notifications to account admins rather than all users.
Using the same root email address for all accounts (Option D) is not recommended.
Relying on one admin or external forwarding (Options A, C) introduces delays or single points of failure.

upvoted 2 times

Itsume 2 years, 11 months ago

all admins need access or else some wont get the right mails and cant do their job,
sending it only to a few would disrupt the workflowso it is D

upvoted 1 times

fishy_resolver 3 years ago

Selected Answer: D

From the links provided below there are no mention of having a distribution list capability within AWS:
https://docs.aws.amazon.com/organizations/latest/userguide/orgs_best-practices_mgmt-acct.html#best-practices_mgmt-acct_email-address

As per link for best practices:
Use a group email address for the management account's root user!

upvoted 2 times

Abrar2022 3 years ago

The clue is in the pudding!!

Question: account "administrators"
Answer: Configure all AWS account root user email addresses as distribution lists that go to a few "administrators"

upvoted 3 times

Rainchild 3 years, 1 month ago

Selected Answer: B

Option A: wrong - sends email to everybody
Option B: correct (but sub-optimal because distribution lists aren't all that secure)
Option C: wrong - single point of failure on the new administrator
Option D: wrong - each root email address must be unique, you can't change them all to the same one

upvoted 3 times

jdr75 3 years, 2 months ago

Selected Answer: B

The more aligned answer to this article:

https://docs.aws.amazon.com/organizations/latest/userguide/orgs_best-practices_mgmt-acct.html#best-practices_mgmt-acct_email-address

is B.

D would be best if it'd said that the email you configure as "root user email address" will be a distribution list.
The phrase "all future notifications are not missed" points to D, cos' it said:
".. and all newly created accounts to use the same root user email address"
so the future account that will be created will be covered with the business policy.

It's not 100% clear, but I'll choose B.

upvoted 3 times

TheAbsoluteTruth 3 years, 2 months ago

Una pregunta si la gente va votando las preguntas por que los administradores no cambian la respuesta correcta. Es a interpretación y ya?

upvoted 1 times

jdr75 3 years, 2 months ago

El administrador de "examtopics" pasa olímpicamente de marcar la respuesta correcta y es evidente que muchas respuestas que indica como "correctas" no lo son. Dice muy poco del servicio que dan.

upvoted 1 times

jaswantn 3 years, 2 months ago

Using the method of crossing out the option that does not fit....
Option A: address to all users of organization (wrong)
Option B: go to a few administration who can respond to alerts (question says to send notification to administrators not a selected few)
Option C: send to one administrator and giving him responsibility (wrong)
Option D: correct (as this is the one option left after checking all others).

upvoted 1 times

Question #138 Topic 1

A company runs its ecommerce application on AWS. Every new order is published as a massage in a RabbitMQ queue that runs on an Amazon EC2 instance in a single Availability Zone. These messages are processed by a different application that runs on a separate EC2 instance. This application stores the details in a PostgreSQL database on another EC2 instance. All the EC2 instances are in the same Availability Zone.
The company needs to redesign its architecture to provide the highest availability with the least operational overhead.
What should a solutions architect do to meet these requirements?

A. Migrate the queue to a redundant pair (active/standby) of RabbitMQ instances on Amazon MQ. Create a Multi-AZ Auto Scaling group for EC2 instances that host the application. Create another Multi-AZ Auto Scaling group for EC2 instances that host the PostgreSQL database.

B. Migrate the queue to a redundant pair (active/standby) of RabbitMQ instances on Amazon MQ. Create a Multi-AZ Auto Scaling group for EC2 instances that host the application. Migrate the database to run on a Multi-AZ deployment of Amazon RDS for PostgreSQL. **Most Voted**

C. Create a Multi-AZ Auto Scaling group for EC2 instances that host the RabbitMQ queue. Create another Multi-AZ Auto Scaling group for EC2 instances that host the application. Migrate the database to run on a Multi-AZ deployment of Amazon RDS for PostgreSQL.

D. Create a Multi-AZ Auto Scaling group for EC2 instances that host the RabbitMQ queue. Create another Multi-AZ Auto Scaling group for EC2 instances that host the application. Create a third Multi-AZ Auto Scaling group for EC2 instances that host the PostgreSQL database

Correct Answer: B

Community vote distribution

B (100%)

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: B

Migrating to Amazon MQ reduces the overhead on the queue management. C and D are dismissed.
Deciding between A and B means deciding to go for an AutoScaling group for EC2 or an RDS for Postgress (both multi- AZ). The RDS option has less operational impact, as provide as a service the tools and software required. Consider for instance, the effort to add an additional node like a read replica, to the DB.
https://docs.aws.amazon.com/amazon-mq/latest/developer-guide/active-standby-broker-deployment.html
https://aws.amazon.com/rds/postgresql/

upvoted 29 times

UWSFish 3 years, 7 months ago

Yes but active/standby is fault tolerance, not HA. I would concede after thinking about it that B is probably the answer that will be marked correct but its not a great question.

upvoted 4 times

EKA_CloudGod 3 years, 6 months ago

This also helps anyone in doubt; https://docs.aws.amazon.com/amazon-mq/latest/developer-guide/active-standby-broker-deployment.html

upvoted 2 times

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago

Selected Answer: B

To meet the requirements of providing the highest availability with the least operational overhead, the solutions architect should take the following actions:

* By migrating the queue to Amazon MQ, the architect can take advantage of the built-in high availability and failover capabilities of the service, which will help ensure that messages are delivered reliably and without interruption.

* By creating a Multi-AZ Auto Scaling group for the EC2 instances that host the application, the architect can ensure that the application is highly available and able to handle increased traffic without the need for manual intervention.

* By migrating the database to a Multi-AZ deployment of Amazon RDS for PostgreSQL, the architect can take advantage of the built-in high availability and failover capabilities of the service, which will help ensure that the database is always available and able to handle increased traffic.

Therefore, the correct answer is Option B.

upvoted 8 times

zx9r Most Recent 1 year, 6 months ago

Selected Answer: B

B makes the most sense

upvoted 1 times

jaradat02 1 year, 10 months ago

Selected Answer: B

B is the correct answer

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

CD, you cannot have EC2 scaling work with RabbitMQ as only once instance can be active
A: Is good but B is better
B: Correct due to usage of RDS for PG so less overhead

upvoted 3 times

Gizmo2022 1 year, 6 months ago

Thank you for explaining the answer

upvoted 1 times

chandu7024 2 years, 8 months ago

Agree with B

upvoted 1 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: B

B offers high availability and low operational overheads.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

Option B is the best solution to meet the high availability and low overhead requirements:

Migrate the queue to redundant Amazon MQ
Use Auto Scaling groups across AZs for the application
Migrate the database to Multi-AZ RDS PostgreSQL
The reasons are:

Amazon MQ provides a managed, highly available RabbitMQ cluster
Multi-AZ Auto Scaling distributes the application across AZs
RDS PostgreSQL is managed, multi-AZ capable database
Together this architecture removes single points of failure
RDS and MQ reduce operational overhead over self-managed

upvoted 6 times

MNotABot 2 years, 11 months ago

B
least operational overhead (Amazon RDS for PostgreSQL --> hence AD out / C says EC2 so out --> Hence B)

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: B

Option B provides the highest availability with the least operational overhead. By migrating the queue to a redundant pair of RabbitMQ instances on Amazon MQ, the messaging system becomes highly available. Creating a Multi-AZ Auto Scaling group for EC2 instances hosting the application ensures that it can automatically scale and maintain availability across multiple Availability Zones. Migrating the database to a Multi-AZ deployment of Amazon RDS for PostgreSQL provides automatic failover and data replication across multiple Availability Zones, enhancing availability and reducing operational overhead.

A. Incorrect because it does not address the high availability requirement for the RabbitMQ queue and the PostgreSQL database.

C. Incorrect because it does not provide redundancy for the RabbitMQ queue and does not address the high availability requirement for the PostgreSQL database.

D. Incorrect because it does not address the high availability requirement for the RabbitMQ queue and does not provide redundancy for the application instances.

upvoted 4 times

Gary_Phillips_2007 3 years, 3 months ago

Selected Answer: B

B for me.

upvoted 1 times

techhb 3 years, 5 months ago

Selected Answer: B

B is right all explanations below are correct

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: B

Option B is right answer

upvoted 1 times

Wpcorgan 3 years, 6 months ago

B for me

upvoted 1 times

Question #139 Topic 1

A reporting team receives files each day in an Amazon S3 bucket. The reporting team manually reviews and copies the files from this initial S3 bucket to an analysis S3 bucket each day at the same time to use with Amazon QuickSight. Additional teams are starting to send more files in larger sizes to the initial S3 bucket.
The reporting team wants to move the files automatically analysis S3 bucket as the files enter the initial S3 bucket. The reporting team also wants to use AWS Lambda functions to run pattern-matching code on the copied data. In addition, the reporting team wants to send the data files to a pipeline in Amazon SageMaker Pipelines.
What should a solutions architect do to meet these requirements with the LEAST operational overhead?

A. Create a Lambda function to copy the files to the analysis S3 bucket. Create an S3 event notification for the analysis S3 bucket. Configure Lambda and SageMaker Pipelines as destinations of the event notification. Configure s3:ObjectCreated:Put as the event type.

B. Create a Lambda function to copy the files to the analysis S3 bucket. Configure the analysis S3 bucket to send event notifications to Amazon EventBridge (Amazon CloudWatch Events). Configure an ObjectCreated rule in EventBridge (CloudWatch Events). Configure Lambda and SageMaker Pipelines as targets for the rule.

C. Configure S3 replication between the S3 buckets. Create an S3 event notification for the analysis S3 bucket. Configure Lambda and SageMaker Pipelines as destinations of the event notification. Configure s3:ObjectCreated:Put as the event type.

D. Configure S3 replication between the S3 buckets. Configure the analysis S3 bucket to send event notifications to Amazon EventBridge (Amazon CloudWatch Events). Configure an ObjectCreated rule in EventBridge (CloudWatch Events). Configure Lambda and SageMaker Pipelines as targets for the rule. Most Voted

Correct Answer: D

Community vote distribution

D (76%) 13% 9%

Comments

Six_Fingered_Jose Highly Voted 3 years, 7 months ago

Selected Answer: D

i go for D here
A and B says you are copying the file to another bucket using lambda,
C an D just uses S3 replication to copy the files,

They are doing exactly the same thing while C and D do not require setting up of lambda, which should be more efficient

The question says the team is manually copying the files, automatically replicating the files should be the most efficient method vs manually copying or copying with lambda.

upvoted 31 times

Abdou1604 2 years, 8 months ago

but the reporting team also wants to use AWS Lambda functions to run pattern-matching code on the copied , S3 replication cons is copying everything

upvoted 2 times

pentium75 2 years, 5 months ago

The Lambda functions should run "on the copied data", so first copy, THEN run Lambda function, which is achieved by D.

upvoted 5 times

vipyodha 2 years, 11 months ago

yes d because of least operational overhead and also s3 event notification can only send to sns.sqs.and lambda , not to sagemaker.eventbridge can send to sagemaker

upvoted 22 times

Tsige 1 year, 7 months ago

S3 Replication: Configuring S3 replication between the initial and analysis S3 buckets automates the process of moving files between the buckets without the need to manually copy files or run a Lambda function for this purpose. This reduces operational overhead.
S3 Event Notifications: Once files are replicated to the analysis bucket, you can configure S3 event notifications for the s3:ObjectCreated event. This event triggers actions (such as invoking Lambda functions and sending data to SageMaker Pipelines) when new files are placed in the analysis bucket.
The answer is C

upvoted 4 times

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: B

C and D aren't answers as replicating the S3 bucket isn't efficient, as other teams are starting to use it to store larger docs not related to the reporting, making replication not useful.
As Amazon SageMaker Pipelines, ..., is now supported as a target for routing events in Amazon EventBridge, means the answer is B
https://aws.amazon.com/about-aws/whats-new/2021/04/new-options-trigger-amazon-sagemaker-pipeline-executions/

upvoted 18 times

vipyodha 2 years, 11 months ago

but B is not least operational overhead , D is least operational overhead

upvoted 2 times

jdr75 3 years, 2 months ago

You misinterpret it ... the reporting team is overload, cos' more teams request their services uploading more data to the bucket. That's the reason reporting team need to automate the process. So ALL the bucket objects need to be copied to other bucket, and the replication is better an cheape than use Lambda. So the answer is D.

upvoted 3 times

LuckyAro 3 years, 4 months ago

Nowhere in the question did they mention that other files were unrelated to reporting ....
"The reporting team wants to move the files automatically to analysis S3 bucket as the files enter the initial S3 bucket" where did it say they were unrelated files ? except for conjecture.

upvoted 7 times

KADSM 3 years, 7 months ago

Not sure how far lambda will cope up with larger files with the timelimit in place.

upvoted 4 times

byteb 2 years, 5 months ago

"The reporting team wants to move the files automatically to analysis S3 bucket as the files enter the initial S3 bucket." Replication is asynchronous, with lambda the data will be available faster. So I think A is the answer.

upvoted 1 times

In2718 Most Recent 9 months, 2 weeks ago

Selected Answer: C

Native S3 replication automatically copies objects with no custom code. After replication, S3 event notification on the analysis bucket triggers Lambda + SageMaker. Very little overhead.

upvoted 2 times

gianola 1 year ago

Selected Answer: D

No answer is valid as the question asked how to "move" the files and they are getting duplicated. Otherwise I'd go for D.

upvoted 1 times

Soliner_Bilgi_Teknolojileri 1 year ago

Selected Answer: C

✔ S3 Replication: Replication job 100% automated
✔ S3 event notification: Triggered when copied files arrive
✔ Both Lambda and SageMaker Pipelines (via Step Function or Lambda proxy) can be given as destinations
✅ Minimum operational burden + full automation
✅ s3:ObjectCreated:Put → appropriate trigger type

upvoted 1 times

ChhatwaniB 1 year, 1 month ago

Selected Answer: D

Amazon S3 can send event notification messages to the following destinations.

Amazon Simple Notification Service (Amazon SNS) topics
Amazon Simple Queue Service (Amazon SQS) queues
AWS Lambda
Amazon EventBridge

However, only one destination type can be specified for each event notification.
https://docs.aws.amazon.com/AmazonS3/latest/userguide/notification-how-to-event-types-and-destinations.html
upvoted 2 times

SirDNS 1 year, 2 months ago

Selected Answer: C

Why do we need to add EventBridge (additional layer/operational overhead) when S3 Event Notification does the job?
upvoted 1 times

AwsAbhiKumar 1 year, 3 months ago

Selected Answer: C

The question is little confusing between C and D.
Option C suggest to use S3 events notification in combination with AWS lambda and SageMaker. But S3 events notification dont't have native integration to directly trigger Amazon SageMaker Pipelines. But it can work around this limitation by having the S3 event trigger a Lambda function, and then that Lambda function can call the SageMaker Pipelines API to start a pipeline execution.(Option doesn't directly suggests this combo)

Option D is fine but it uses S3 replication combined with EventBridge for notifications, which introduces an extra layer (EventBridge rules) that is unnecessary since S3 event notifications can directly trigger Lambda and SageMaker Pipelines. as suggested above.
upvoted 2 times

hpirnaj 1 year, 5 months ago

Selected Answer: C

c is the answer , S3 event notification can directly invoke a Lambda function, which can then in turn trigger a SageMaker Pipeline execution, effectively allowing an S3 event to initiate a SageMaker pipeline through a Lambda intermediary. we dont need to invoke EventBridge
upvoted 3 times

salman7540 1 year, 5 months ago

Selected Answer: D

S3 replication can use to copy files in different buckets so we don't need lambda.
S3 events can't be sent directly to sagemaker so we have to utilise eventbridge who supports many targets including sagemaker.
upvoted 5 times

rmanuraj 1 year, 5 months ago

Selected Answer: D

In the case of S3 event notification only one destination type can be specified for each event notification.
https://docs.aws.amazon.com/AmazonS3/latest/userguide/notification-how-to-event-types-and-destinations.html#supported-notification-destinations
upvoted 3 times

PSH123 1 year, 6 months ago

Selected Answer: C

gpt said 'C' is solution
upvoted 1 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans D - least operational overhead using replication; I was initially going for Ans C until I spotted S3 event notification can only send to SQS, SNS, Lambda - not directly to Sagemaker; but Eventbridge can send to Sagemaker. Not sure why author prefers A...?
upvoted 4 times

jatric 1 year, 11 months ago

Selected Answer: D

S3 event can't be use to notify sagemaker, So C can't be right option. AB required lambda which is not unnecessary
upvoted 4 times

lofzee 2 years ago

Answer is D because it requires least operational overhead and S3 replication does the copying for you.
Also read this https://docs.aws.amazon.com/AmazonS3/latest/userguide/EventNotifications.html
Lambda and Sagemaker are not supported destinations for S3 Event Notifications
upvoted 4 times

lofzee 2 years ago

Sorry i meant Sagemaker is not supported as an S3 Event Notification. Lambda is though. Still doesn't change what the answer is.... D
upvoted 4 times

andyngkh86 2 years, 4 months ago

I go for C, because option C no need to configure event notifications, but D need to extra work to configure the event notification, for the least operation, option C is best choice

upvoted 2 times

Marco_St 2 years, 6 months ago

Selected Answer: D

B is the first option I denied. Since it makes the event happens inside the analysis bucket to trigger the lambda function. so if the lambda function is running code to copy files from initial bucket to analysis bucket. Then this lambda function should be triggered by the event in initial bucket like once the data reaches in the initial bucket then lambda is triggered. D is the answer.

upvoted 2 times

Question #140 Topic 1

A solutions architect needs to help a company optimize the cost of running an application on AWS. The application will use Amazon EC2 instances, AWS Fargate, and AWS Lambda for compute within the architecture.
The EC2 instances will run the data ingestion layer of the application. EC2 usage will be sporadic and unpredictable. Workloads that run on EC2 instances can be interrupted at any time. The application front end will run on Fargate, and Lambda will serve the API layer. The front-end utilization and API layer utilization will be predictable over the course of the next year.
Which combination of purchasing options will provide the MOST cost-effective solution for hosting this application? (Choose two.)

A. Use Spot Instances for the data ingestion layer Most Voted

B. Use On-Demand Instances for the data ingestion layer

C. Purchase a 1-year Compute Savings Plan for the front end and API layer. Most Voted

D. Purchase 1-year All Upfront Reserved instances for the data ingestion layer.

E. Purchase a 1-year EC2 instance Savings Plan for the front end and API layer.

Correct Answer: AC

Community vote distribution

AC (90%) 10%

Comments

SimonPark Highly Voted 3 years, 7 months ago

Selected Answer: AC

EC2 instance Savings Plan saves 72% while Compute Savings Plans saves 66%. But according to link, it says "Compute Savings Plans provide the mos flexibility and help to reduce your costs by up to 66%. These plans automatically apply to EC2 instance usage regardless of instance family, size, AZ, region, OS or tenancy, and also apply to Fargate and Lambda usage." EC2 instance Savings Plans are not applied to Fargate or Lambda

upvoted 21 times

aba2s Highly Voted 3 years, 5 months ago

Selected Answer: AC

Compute Savings Plans can be used for EC2 instances and Fargate. Whereas EC2 Savings Plans support EC2 only.

upvoted 10 times

ln2718 Most Recent 11 months, 2 weeks ago

Selected Answer: AD

One other follow-up item on splitting hairs for whether C or D is the best answer... you can pull of a combination of reserved instances that would be more ideal for answering the question as D, but again in the real world, C would be more correct.

"Yes, you can leverage Reserved Instances with Amazon ECS (Elastic Container Service) to reduce costs for your container workloads. While ECS itself doesn't have specific "ECS Reserved Instances", you can achieve cost savings by using Reserved Instances with the underlying EC2 instances that support your ECS cluster, especially when using EC2 launch types. "

upvoted 1 times

ln2718 11 months, 2 weeks ago

Selected Answer: AD

Ah, I see, the highest responses are sent to the top. Got it, well that may be the best possible, though not always accurate.

upvoted 1 times

ln2718 11 months, 2 weeks ago

Selected Answer: AD

I'm guessing the responses to my answers are running through an AI generated lookup and reply as they are instantaneous and not completely addressing my comments... interesting. Well, so much for posting discussions))

upvoted 1 times

ln2718 11 months, 2 weeks ago

Selected Answer: AD

Yes, that is correct, for the real world

upvoted 1 times

ln2718 11 months, 2 weeks ago

Selected Answer: AD

Follow-up, the better answer is still D. but in the real world, the answer is C, not D. In the real world there is no such thing as a predictable, I would not rely on a reserved instance for such but have the 6% cost difference vs the risk as a factor for using a compute savings plan. For the test, A, D; for the real world A, C.

upvoted 1 times

ln2718 11 months, 2 weeks ago

Selected Answer: AD

A is obvious. D takes some real understanding of the cost of a known reserved instance overhead vs the flexibility of compute savings plan. Since the performance of the next year is known, a reserved savings plan is actually cheaper than a compute savings plan.

"Reserved Instances (RIs), especially Standard RIs, traditionally offered the highest potential discounts (up to 72% for EC2 instances) but came with rigid terms. You had to commit to specific instance types, regions, operating systems, and tenancies for a 1 or 3-year term."

You have to assume that the question is stating correctly the performance of the next year is known, "The front-end utilization and API layer utilization will be predictable over the course of the next year." C is a good answer, but D is actually better.

upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: AC

Ans A, C -
A: Spot obvious for unpredictable, 'don't care' usage
C: Not so obvious... but its more than just EC2 - its about Compute power using Fargate, Lambda, API call processing so it has to be C (as opposed to E)

upvoted 2 times

huaze_lei 1 year, 9 months ago

Selected Answer: AC

Be mindful that the question is asking about API. So it should be Compute Savings Plans.

If it is for EC2, the Reserved Instance will be correct.

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: AC

Compute Savings Plans can also apply to Fargate and Lambda Usage.

upvoted 5 times

AKBM7829 2 years, 9 months ago

BC is the answer
data ingestion = Spot Instance but
Keyword "Usage Unpredictable" : On-Demand

and for APi its Compute Savings Plan

upvoted 1 times

awashenko 2 years, 8 months ago

Spot instances can auto scale so Spot instance is correct.

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: AC

The two most cost-effective purchasing options for this architecture are:

A) Use Spot Instances for the data ingestion layer

C) Purchase a 1-year Compute Savings Plan for the front end and API layer

The reasons are:

Spot Instances provide the greatest savings for flexible, interruptible EC2 workloads like data ingestion.
Savings Plans offer significant discounts for predictable usage like the front end and API layer.
All Upfront and partial/no Upfront RI's don't align well with the sporadic EC2 usage.
On-Demand is more expensive than Spot for flexible EC2 workloads.

By matching purchasing options to the workload patterns, Spot for unpredictable EC2 and Savings Plans for steady-state usage, the solutions architect optimizes cost efficiency.

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: AC

Using Spot Instances for the data ingestion layer will provide the most cost-effective option for sporadic and unpredictable workloads, as Spot Instances offer significant cost savings compared to On-Demand Instances (Option A).

Purchasing a 1-year Compute Savings Plan for the front end and API layer will provide cost savings for predictable utilization over the course of a year (Option C).

Option B is less cost-effective as it suggests using On-Demand Instances for the data ingestion layer, which does not take advantage of cost-saving opportunities.

Option D suggests purchasing 1-year All Upfront Reserved instances for the data ingestion layer, which may not be optimal for sporadic and unpredictable workloads.

Option E suggests purchasing a 1-year EC2 instance Savings Plan for the front end and API layer, but Compute Savings Plans are typically more suitable for predictable workloads.

upvoted 4 times

Abrar2022 3 years ago

Spot instances for data injection because the task can be terminated at anytime and tolerate disruption. Compute Saving Plan is cheaper than EC2 instance Savings plan.

upvoted 2 times

Abrar2022 3 years ago

EC2 instance Savings Plans are not applied to Fargate or Lambda

upvoted 1 times

Noviiiice 3 years, 2 months ago

Why not B?

upvoted 1 times

SkyZeroZx 3 years, 2 months ago

because onDemand is more expensive than spot additionally that the workload has no problem with being interrupted at any time

upvoted 2 times

Question #141 *Topic 1*

A company runs a web-based portal that provides users with global breaking news, local alerts, and weather updates. The portal delivers each user a personalized view by using mixture of static and dynamic content. Content is served over HTTPS through an API server running on an Amazon EC2 instance behind an Application Load Balancer (ALB). The company wants the portal to provide this content to its users across the world as quickly as possible.
How should a solutions architect design the application to ensure the LEAST amount of latency for all users?

A. Deploy the application stack in a single AWS Region. Use Amazon CloudFront to serve all static and dynamic content by specifying the ALB as an origin. **Most Voted**

B. Deploy the application stack in two AWS Regions. Use an Amazon Route 53 latency routing policy to serve all content from the ALB in the closest Region.

C. Deploy the application stack in a single AWS Region. Use Amazon CloudFront to serve the static content. Serve the dynamic content directly from the ALB.

D. Deploy the application stack in two AWS Regions. Use an Amazon Route 53 geolocation routing policy to serve all content from the ALB in the closest Region.

Correct Answer: A

Community vote distribution

A (71%) B (24%) 3%

Comments

huiy Highly Voted 3 years, 8 months ago

Selected Answer: A

Answer is A.
Amazon CloudFront is a web service that speeds up distribution of your static and dynamic web content
https://www.examtopics.com/discussions/amazon/view/81081-exam-aws-certified-solutions-architect-associate-saa-c02/

upvoted 36 times

MutiverseAgent 2 years, 10 months ago

Also, option B does not use CloudFront which means all the traffic will go through the internet; So, despite deploying resources in two regions and using the lowest latency point, that public internet connection might probably be slower than a connection through a private aws network as Cloudfront can use.

upvoted 5 times

Six_Fingered_Jose Highly Voted 3 years, 7 months ago

Selected Answer: B

Answer should be B,

CloudFront reduces latency if its only static content, which is not the case here.
For Dynamic content, CF cant cache the content so it sends the traffic through the AWS Network which does reduces latency, but it still has to travel through another region.

For the case with 2 region and Route 53 latency routing, Route 53 detects the nearest resouce (with lowest latency) and routes the traffic there. Because the traffic does not have to travel to resources far away, it should have the least latency in this case here.

upvoted 17 times

Aamee 3 years, 6 months ago

Can you pls. provide a ref. link from where this info. got extracted?

upvoted 1 times

manuelemg2007 2 years, 10 months ago

this is link https://aws.amazon.com/es/blogs/aws-spanish/cloudfront-para-la-distribucion-de-contenido-estatico-y-dinamico/

upvoted 4 times

Abdou1604 2 years, 8 months ago
What about accross the word :)
upvoted 4 times

Onimole 3 years, 7 months ago
Cf works for both static and dynamic content
upvoted 13 times

awsgeek75 2 years, 4 months ago
two regions won't cover the whole world.
upvoted 1 times

SnaxAttax Most Recent 6 months, 2 weeks ago
Selected Answer: A
Answer is A-- CF serves static and dynamic content globally. Did not select multi-region because it is irrelevant to this question, as resiliency is not mentioned.
upvoted 1 times

network_enthusiast 1 year, 1 month ago
Selected Answer: B
B is the correct answer
upvoted 1 times

tch 1 year, 2 months ago
Selected Answer: C
Dynamic content can also be served via CloudFront by configuring the WordPress website as an origin. Since dynamic content includes personalized content, you need to configure CloudFront to forward certain HTTP cookies and HTTP headers as part of a request to your custom origin server. CloudFront uses the forwarded cookie values as part of the key that identifies a unique object in its cache. To ensure that you maximize the caching efficiency, you should configure CloudFront to only forward those HTTP cookies and HTTP headers that really vary the content (not cookies that are only used on the client side or by third-party applications, for example, for web analytics).
upvoted 1 times

Dharmarajan 1 year, 4 months ago
Selected Answer: A
A is correct, because the dynamic content is also cached - which means first time the data is brought in, then subsequently it does not have to go to the source to get the same content.
There are many ways to set these combinations up, even with Route 53, but of the given options, "A" seems to fit the bill best.
upvoted 1 times

satyaammm 1 year, 4 months ago
Selected Answer: A
CloudFront is suitable for both static and dynamic content.
upvoted 1 times

FlyingHawk 1 year, 4 months ago
Selected Answer: B
The best approach combines multi-region deployment for dynamic content and Amazon CloudFront for static content. However, none of the option: explicitly describe this combination. Among the given options, Option B is the best because:

It uses two regions to reduce latency for dynamic content.

Route 53 latency routing ensures users are directed to the closest region.

For a more optimal solution (not listed in the options):

Deploy the application stack in multiple AWS Regions.

Use Amazon CloudFront to serve static content from edge locations.

Use Route 53 latency routing to direct users to the closest region for dynamic content.
upvoted 1 times

FlyingHawk 1 year, 4 months ago
A : CloudFront is a global CDN that caches static content at edge locations, reducing latency for users worldwide.

However, dynamic content served through CloudFront with a single ALB origin will still incur latency for users far from the single AWS Region.

This solution does not address latency for dynamic content effectively
upvoted 1 times

FlyingHawk 1 year, 4 months ago

B - Deploying in two regions reduces latency for users closer to those regions.

Route 53 latency routing ensures users are directed to the closest region.

However, this solution does not leverage a CDN for static content, which could further reduce latency.

Verdict: Better than Option A but not optimal for static content delivery.

upvoted 1 times

rmanuraj 1 year, 5 months ago

Selected Answer: A

Amazon cloud front is a better choice in terms of delivering both static and dynamic content globally. Also option B says deploy the application stacl to only two regions. but the use case is to access the portal globally. Don't think Amazon Route 53 latency routing policy will have a bigger impact in terms of low latency.

upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: B

Ans B - "If your application is hosted in multiple AWS Regions, you can improve performance for your users by serving their requests from the AWS Region that provides the lowest latency" ...because it needs to be dynamic: "Latency between hosts on the internet can change over time as a result of changes in network connectivity and routing. Latency-based routing is based on latency measurements taken over a period of time, and the measurements reflect these changes. A request that is routed to the Oregon Region this week might be routed to the Singapore Region next week."

https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy-latency.html

upvoted 2 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: B

B for sure

upvoted 1 times

ManikRoy 2 years, 1 month ago

With Amazon CloudFront, your end users connections are terminated at CloudFront locations closer to them, which helps in reducing the overall round trip time required to establish a connection. This is irrespective of static a dynamic content.

upvoted 2 times

ManikRoy 2 years, 1 month ago

Selected Answer: A

You can still have improved performance by distributing the dynamic traffic through CDN instead of ALB. Refer below link.
Also for other 2 options, using just 2 other regions for world wide distribution doesn't make much of a sense.
https://aws.amazon.com/cloudfront/dynamic-content/

upvoted 3 times

Uzbekistan 2 years, 2 months ago

Selected Answer: C

CloudFront for Static Content: By leveraging Amazon CloudFront, static content such as images, stylesheets, and scripts can be cached and distributed globally across a network of edge locations. This ensures that users receive static content from the nearest edge location, reducing latency and improving performance.

Serve Dynamic Content from ALB: Since dynamic content requires real-time processing and cannot be effectively cached at edge locations, serving dynamic content directly from the Application Load Balancer (ALB) is appropriate. The ALB can handle dynamic requests efficiently within the AWS Region where the application is deployed.

upvoted 3 times

eb7be10 2 years, 1 month ago

C was my choice for the reasons stated here. What am I missing?

upvoted 3 times

lofzee 2 years ago

How can you SERVE content from a load balancer?
Amazon Cloudfront is designed for static and dynamic content. Why would you pick any other option that isn't A?

upvoted 3 times

tch 1 year, 2 months ago

https://docs.aws.amazon.com/whitepapers/latest/best-practices-wordpress/dynamic-content.html

upvoted 1 times

Parul25 2 years, 4 months ago

CloudFront improves the performance, availability, and security of your dynamic content but not the latency as compared to Route 53 Latency Routing policy. Hence option B
https://aws.amazon.com/cloudfront/dynamic-content/

upvoted 2 times

Parul25 2 years, 4 months ago

I choose option B.

While CloudFront can accelerate content delivery by caching static content at edge locations, it may not be the most effective solution in this scenario. Since the portal delivers a mixture of static and dynamic content, leveraging Route 53 latency routing for dynamic content delivery ensures that users are directed to the nearest AWS Region hosting the dynamic content.

upvoted 1 times

pentium75 2 years, 5 months ago

Selected Answer: A

"Least amount of latency for all users" "across the world" = CloudFront, thus B and D are out. Also, deploying the stack in "two regions" would benefit those two regions, but not users "across the world".

CloudFront can also cache dynamic content, thus A.

upvoted 9 times

Question #142 Topic 1

A gaming company is designing a highly available architecture. The application runs on a modified Linux kernel and supports only UDP-based traffic. The company needs the front-end tier to provide the best possible user experience. That tier must have low latency, route traffic to the nearest edge location, and provide static IP addresses for entry into the application endpoints.
What should a solutions architect do to meet these requirements?

A. Configure Amazon Route 53 to forward requests to an Application Load Balancer. Use AWS Lambda for the application in AWS Application Auto Scaling.

B. Configure Amazon CloudFront to forward requests to a Network Load Balancer. Use AWS Lambda for the application in an AWS Application Auto Scaling group.

C. Configure AWS Global Accelerator to forward requests to a Network Load Balancer. Use Amazon EC2 instances for the application in an EC2 Auto Scaling group. Most Voted

D. Configure Amazon API Gateway to forward requests to an Application Load Balancer. Use Amazon EC2 instances for the application in an EC2 Auto Scaling group.

Correct Answer: C

Community vote distribution

C (100%)

Comments

dokaedu Highly Voted 3 years, 7 months ago
Correct Answer: C
AWS Global Accelerator and Amazon CloudFront are separate services that use the AWS global network and its edge locations around the world. CloudFront improves performance for both cacheable content (such as images and videos) and dynamic content (such as API acceleration and dynamic site delivery). Global Accelerator improves performance for a wide range of applications over TCP or UDP by proxying packets at the edge t applications running in one or more AWS Regions. Global Accelerator is a good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP, as well as for HTTP use cases that specifically require static IP addresses or deterministic, fast regional failover. Both services integrate with AWS Shield for DDoS protection.
upvoted 91 times

praveenas400 3 years, 5 months ago
Explained very well. ty
upvoted 2 times

iCcma 3 years, 6 months ago
Thank you, your explanation helped me to better understand even the answer of question 29
upvoted 4 times

stepman 3 years, 6 months ago
On top of this, lambda would not be able to run application that is running on a modified Linux kernel. The answer is C .
upvoted 9 times

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago
Selected Answer: C
The correct answer is Option C. To meet the requirements;

* AWS Global Accelerator is a service that routes traffic to the nearest edge location, providing low latency and static IP addresses for the front-end tier. It supports UDP-based traffic, which is required by the application.

* A Network Load Balancer is a layer 4 load balancer that can handle UDP traffic and provide static IP addresses for the application endpoints.

* An EC2 Auto Scaling group ensures that the required number of Amazon EC2 instances is available to meet the demand of the application. This will help the front-end tier to provide the best possible user experience.

Option A is not a valid solution because Amazon Route 53 does not support UDP traffic.

Option B is not a valid solution because Amazon CloudFront does not support UDP traffic.
Option D is not a valid solution because Amazon API Gateway does not support UDP traffic.
upvoted 10 times

Buruguduystunstugudunstuy 3 years, 5 months ago
My mistake, correction on Option A, it is the Application Load Balancers do not support UDP traffic. They are designed to load balance HTTP and HTTPS traffic, and they do not support other protocols such as UDP.
upvoted 5 times

Dharmarajan Most Recent 1 year, 4 months ago
Selected Answer: C
Gobal Accelarator rules the roost in this use case! gives a static IP, and does everything asked here.
upvoted 1 times

PaulGa 1 year, 8 months ago
Selected Answer: C
Ans C - hint: "That tier must have low latency, route traffic to the nearest edge location, and provide static IP addresses for entry into the application endpoints."

"AWS Global Accelerator... provides static IP addresses that provide a fixed entry point to your applications and eliminate the complexity of managin specific IP addresses for different AWS Regions and Availability Zones... [to] always routes user traffic to the optimal endpoint based on performance, reacting instantly to changes in application health, your user's location, and policies that you configure."

https://aws.amazon.com/global-accelerator/faqs/
upvoted 3 times

huaze_lei 1 year, 9 months ago
Selected Answer: C
Non HTTP/S, the answer is always Global Accelerator without doubts. GA serves like Cloudfront on providing low latency through edge locations, with the exception of handling different protocols.
upvoted 3 times

lofzee 2 years ago
Whenever I see this line : "and provide static IP addresses for entry into the application endpoints." - my brain automatically thinks Global Accelerator.
upvoted 1 times

sidharthwader 2 years, 3 months ago
If the situation demands for UDP or some protocols that are not at application level then it would be better to use Global Accelerator and here they need top notch perfromance hence using it with NLB would be the best answer. Cloud Front does not support UDP nor does it support use of NLB
upvoted 2 times

Murtadhaceit 2 years, 6 months ago
Selected Answer: C
UDP: NLB.
Static IP: Global Accelerator.
upvoted 3 times

TariqKipkemei 2 years, 9 months ago
Selected Answer: C
UDP, static IP = Global Accelerator and Network Load Balancer
upvoted 2 times

Guru4Cloud 2 years, 9 months ago
Selected Answer: C
AWS Global Accelerator provides static IP addresses that serve as a fixed entry point to application endpoints. This allows optimal routing to the nearest edge location.
Using a Network Load Balancer (NLB) allows support for UDP traffic, as NLBs can handle TCP and UDP protocols.
The application runs on a modified Linux kernel, so using Amazon EC2 instances directly will provide the needed customization and low latency.
The EC2 instances can be auto scaled based on demand to provide high availability.
API Gateway and Application Load Balancer are more suited for HTTP/HTTPS and REST API type workloads. For a UDP gaming workload, Global Accelerator + NLB + EC2 is a better architectural fit.
upvoted 3 times

cookieMr 2 years, 11 months ago
Selected Answer: C
AWS Global Accelerator is designed to improve the availability and performance of applications by routing traffic through the AWS global network tc the nearest edge locations, reducing latency. By configuring AWS Global Accelerator to forward requests to a Network Load Balancer, UDP-based traffic can be efficiently distributed across multiple EC2 instances in an Auto Scaling group. Using Amazon EC2 instances for the application allows fo

customization of the Linux kernel and support for UDP-based traffic. This solution provides static IP addresses for entry into the application endpoints, ensuring consistent access for users.

Option A suggests using AWS Lambda for the application, but Lambda is not suitable for long-running UDP-based applications and may not provide the required low latency.
Option B suggests using CloudFront, which is primarily designed for HTTP/HTTPS traffic and does not have native support for UDP-based traffic.
Option D suggests using API Gateway, which is primarily used for RESTful APIs and does not support UDP-based traffic.

upvoted 4 times

Abrar2022 3 years ago

aws global accelarator provides static IP addresses.

upvoted 1 times

Bmarodi 3 years ago

Selected Answer: C

My choice is option C, due to the followings: Amazon Global accelator route the traffic to nearest edge locations, it supports UDP-based traffic, and i provides static ip addresses as well, hence C is right answer.

upvoted 2 times

bakamon 3 years, 2 months ago

Answer : C
CloudFront : Doesn't support static IP addresses
ALB : Doesn't support UDP

upvoted 1 times

Devsin2000 3 years, 3 months ago

C - https://aws.amazon.com/global-accelerator/

upvoted 1 times

SilentMilli 3 years, 5 months ago

Selected Answer: C

To meet the requirements of providing low latency, routing traffic to the nearest edge location, and providing static IP addresses for entry into the application endpoints, the best solution would be to use AWS Global Accelerator. This service routes traffic to the nearest edge location and provide: static IP addresses for the application endpoints. The front-end tier should be configured with a Network Load Balancer, which can handle UDP-based traffic and provide high availability. Option C, "Configure AWS Global Accelerator to forward requests to a Network Load Balancer. Use Amazon EC2 instances for the application in an EC2 Auto Scaling group," is the correct answer.

upvoted 1 times

techhb 3 years, 5 months ago

Selected Answer: C

C is obvious choice here.

upvoted 1 times

Question #143 Topic 1

A company wants to migrate its existing on-premises monolithic application to AWS. The company wants to keep as much of the front-end code and the backend code as possible. However, the company wants to break the application into smaller applications. A different team will manage each application. The company needs a highly scalable solution that minimizes operational overhead.
Which solution will meet these requirements?

A. Host the application on AWS Lambda. Integrate the application with Amazon API Gateway.

B. Host the application with AWS Amplify. Connect the application to an Amazon API Gateway API that is integrated with AWS Lambda.

C. Host the application on Amazon EC2 instances. Set up an Application Load Balancer with EC2 instances in an Auto Scaling group as targets.

D. Host the application on Amazon Elastic Container Service (Amazon ECS). Set up an Application Load Balancer with Amazon ECS as the target. **Most Voted**

Correct Answer: D

Community vote distribution

D (78%) B (19%) 2%

Comments

Ken701 Highly Voted 3 years, 7 months ago
I think the answer here is "D" because usually when you see terms like "monolithic" the answer will likely refer to microservices.
upvoted 40 times

allanm 10 months, 1 week ago
Yes but Microservices is a broad term through. Stuff like Lambda, SNS/SQS etc are all considered microservices. Within this question, I think the main differentiator is that "the code needs to be kept the same" which means lambda is out of the question and EC2/ALBs need to come into play hence D
upvoted 1 times

Bevemo Highly Voted 3 years, 7 months ago
Selected Answer: D
D is organic pattern, lift and shift, decompose to containers, first making most use of existing code, whilst new features can be added over time with lambda+api gw later.
A is leapfrog pattern. requiring refactoring all code up front.
upvoted 21 times

Mimine87 Most Recent 1 year, 2 months ago
Selected Answer: D
Monolith to Smaller Apps (Microservices-Ready)
ECS (or EKS) is ideal when breaking a monolithic app into smaller services (aka microservices).

Different teams can manage individual containers running different parts of the app.
upvoted 1 times

Dharmarajan 1 year, 4 months ago
Selected Answer: D
D because the operational overhead is the smallest among the given options. The company may do all that breaking up of functionalities and let teams manage the parts, but operationally for the site, hosting in Containers is the lowest maintenance. No ASG tuning, no ALB limitations and so or
upvoted 1 times

FlyingHawk 1 year, 4 months ago
Selected Answer: D

The company wants to keep as much of the front-end code and the backend code as possible, so containization is less code changes than B which uses lambda.

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: C

Auto scaling group meets highly scalable requirement. D is not right 1. it is unknown if the app can be containerized , 2. and it maintains EC2 as C, sc D has no operational advantage. Microservice is not equivalent to container.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: D

Ans D - hint: "...break the application into smaller applications"

upvoted 3 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans D - hint: "The company wants to keep as much of the front-end code and the backend code as possible. However, the company wants to break the application into smaller applications." Containerisation will help the company achieve a scaleable, more manageable solution.

upvoted 2 times

pedro_vieira 1 year, 10 months ago

For the folks suggesting Amplify: Have any of you actually shipped anything on Amplify? There are tons of adaptations needed to port a monolith to Amplify, specially around the backend that will need severe refactor.
Answer D allows for decomposing the application into different containers, enabling a distributed monolith.

upvoted 4 times

ManikRoy 2 years, 1 month ago

Selected Answer: B

Amazon API Gateway and Amplify both server less. Also you can import your code from GitHub in the amplify.

upvoted 2 times

ManikRoy 2 years, 1 month ago

Option D does not mention AWS Fargate which would cover the 'least operational overhead ' part.

upvoted 1 times

ManikRoy 2 years, 1 month ago

The company wants to keep much of its existing code. So the preferable solution is ECS. However the option D does not mention AWS Fargate which would cover the 'least operational overhead ' part.

upvoted 2 times

jbkrishna 2 years, 2 months ago

Different teams working means " microservices based architecture" so basically decoupling the application ..u can achieve this only by containerizing the app so answer is D

upvoted 4 times

NayeraB 2 years, 3 months ago

Selected Answer: B

B allows for a serverless architecture using AWS Lambda functions, which are highly scalable and require minimal operational overhead. AWS Amplify can help in managing the front-end code, while Amazon API Gateway integrated with AWS Lambda can handle the backend services.

D imo is not the best option in this scenario. While ECS can be a good choice for containerized workloads, it might introduce more operational overhead compared to a serverless solution like AWS Lambda and AWS Amplify.

upvoted 4 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

I have a problem with this question.
"The company wants to keep as much of the front-end code and the backend code as possible"
So containerization is the solution here (D)? ABC don't make much sense so I will go with D but using containers for FE/BE code and configuring ALB for ECS (hopefully for frontend containers) is a pain in practice. Maybe this is worded in a bad way.

upvoted 5 times

vip2 2 years, 5 months ago

Selected Answer: B

Original state: monolithic with FE and BE code
Wanted state: seperate to mutilple components for diff. teams as Microservices

B is correct to decouple monolithic to microservices.

D still keep monolithic application in ECS.

upvoted 3 times

06042022 2 years, 5 months ago

IT is B. AWS amplify.
AWS Amplify will help seperate FE and BE. I agree with MM_Korvinus answer.

upvoted 2 times

JTruong 2 years, 5 months ago

Selected Answer: D

https://aws.amazon.com/tutorials/break-monolith-app-microservices-ecs-docker-ec2/module-three/
This page explained clearly why D is the correct answer

upvoted 3 times

Question #144 Topic 1

A company recently started using Amazon Aurora as the data store for its global ecommerce application. When large reports are run, developers report that the ecommerce application is performing poorly. After reviewing metrics in Amazon CloudWatch, a solutions architect finds that the ReadIOPS and CPUUtilizalion metrics are spiking when monthly reports run.
What is the MOST cost-effective solution?

A. Migrate the monthly reporting to Amazon Redshift.

B. Migrate the monthly reporting to an Aurora Replica. Most Voted

C. Migrate the Aurora database to a larger instance class.

D. Increase the Provisioned IOPS on the Aurora instance.

Correct Answer: B

Community vote distribution

B (98%) 2%

Comments

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago

Selected Answer: B

Option B: Migrating the monthly reporting to an Aurora Replica may be the most cost-effective solution because it involves creating a read-only copy of the database that can be used specifically for running large reports without impacting the performance of the primary database. This solution allows the company to scale the read capacity of the database without incurring additional hardware or I/O costs.

upvoted 17 times

Buruguduystunstugudunstuy 3 years, 5 months ago

The incorrect solutions are:

Option A: Migrating the monthly reporting to Amazon Redshift may not be cost-effective because it involves creating a new data store and potentially significant data migration and ETL costs.

Option C: Migrating the Aurora database to a larger instance class may not be cost-effective because it involves changing the underlying hardware of the database and potentially incurring additional costs for the larger instance.

Option D: Increasing the Provisioned IOPS on the Aurora instance may not be cost-effective because it involves paying for additional I/O capacity that may not be necessary for other workloads on the database.

upvoted 11 times

Mikado211 Highly Voted 2 years, 6 months ago

Selected Answer: B

Report = Aurora replica

upvoted 7 times

Dharmarajan Most Recent 1 year, 4 months ago

Selected Answer: B

B. Doesnt say "Read Replica" but I guess it is implied - what other kind of replica is there? only an active-active standby, still that would solve the issue.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: B

Ans B - Using Aurora Replica means you can work off a snapshot without impacting the master DB

upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: B

Ans B - making reports is temporary feature every month, so creating Aurora Replica is sufficient and efficient. Redshift is unnecessary (and means further overheads); C, D permanently upgrade capability which is not needed

upvoted 2 times

1166ae3 1 year, 11 months ago

Selected Answer: D

This approach focuses directly on mitigating the observed IOPS spikes, which are likely contributing to the performance degradation during heavy report processing, without introducing additional complexities or higher operational costs associated with other options.

upvoted 1 times

lofzee 2 years ago

Aye B.

upvoted 1 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: B

Migrate the monthly reporting to an Aurora Replica

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

Aurora Replicas utilize the same storage as the primary instance so there is no additional storage cost.
Replicas can be created and destroyed easily to match reporting needs.
The primary Aurora instance size does not need to be changed, avoiding additional cost.
Workload is offloaded from the primary instance, improving its performance.
No major software/configuration changes needed compared to options like Redshift.

upvoted 4 times

cd93 2 years, 10 months ago

I don't understand why doubling everything (instances, network cost, maintenance effort, and especially storage) can be considered "cost-saving" for a simple monthly report...
An instance upgrade can very well be much cheaper. This question is very vague and does not provide enough information.

upvoted 2 times

cd93 2 years, 9 months ago

Silly me, I thought upgrading instance type includes storage upgrade (increase read iops) lol. The question pointed out that hard drive is also a limiting factor, so correct answer is B.

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: B

B is correct because migrating the monthly reporting to an Aurora Replica can offload the reporting workload from the primary Aurora instance, reducing the impact on its performance during large reports. Using an Aurora Replica provides scalability and allows the replica to handle the read-intensive reporting queries, improving the overall performance of the ecommerce application.

A is wrong because migrating to Amazon Redshift introduces additional costs and complexity, and it may not be necessary to switch to a separate data warehousing service for this specific issue.

C is wrong because simply increasing the instance class of the Aurora database may not be the most cost-effective solution if the performance issue can be resolved by offloading the reporting workload to an Aurora Replica.

D is wrong because increasing the Provisioned IOPS alone may not address the issue of spikes in CPUUtilization during large reports, as it primarily focuses on storage performance rather than overall database performance.

upvoted 5 times

Abrar2022 3 years ago

By using an Aurora Replica for running large reports, the primary database will be relieved of the additional read load, improving performance for the ecommerce application.

upvoted 2 times

Bmarodi 3 years ago

Selected Answer: B

Option B is right answer.

upvoted 1 times

studynoplay 3 years, 1 month ago

Finally a question where there are no controversies

upvoted 3 times

SilentMilli 3 years, 5 months ago

Selected Answer: B

The most cost-effective solution for addressing high ReadIOPS and CPU utilization when running large reports would be to migrate the monthly reporting to an Aurora Replica. An Aurora Replica is a read-only copy of an Aurora database that is updated in real-time with the primary database. By using an Aurora Replica for running large reports, the primary database will be relieved of the additional read load, improving performance for the ecommerce application. Option B, "Migrate the monthly reporting to an Aurora Replica," is the correct answer.

upvoted 2 times

career360guru 3 years, 5 months ago

Selected Answer: B

B is the best option

upvoted 2 times

sanket1990 3 years, 6 months ago

Selected Answer: B

B is correct

upvoted 1 times

Question #145 Topic 1

A company hosts a website analytics application on a single Amazon EC2 On-Demand Instance. The analytics software is written in PHP and uses a MySQL database. The analytics software, the web server that provides PHP, and the database server are all hosted on the EC2 instance. The application is showing signs of performance degradation during busy times and is presenting 5xx errors. The company needs to make the application scale seamlessly.
Which solution will meet these requirements MOST cost-effectively?

A. Migrate the database to an Amazon RDS for MySQL DB instance. Create an AMI of the web application. Use the AMI to launch a second EC2 On-Demand Instance. Use an Application Load Balancer to distribute the load to each EC2 instance.

B. Migrate the database to an Amazon RDS for MySQL DB instance. Create an AMI of the web application. Use the AMI to launch a second EC2 On-Demand Instance. Use Amazon Route 53 weighted routing to distribute the load across the two EC2 instances.

C. Migrate the database to an Amazon Aurora MySQL DB instance. Create an AWS Lambda function to stop the EC2 instance and change the instance type. Create an Amazon CloudWatch alarm to invoke the Lambda function when CPU utilization surpasses 75%.

D. Migrate the database to an Amazon Aurora MySQL DB instance. Create an AMI of the web application. Apply the AMI to a launch template. Create an Auto Scaling group with the launch template Configure the launch template to use a Spot Fleet. Attach an Application Load Balancer to the Auto Scaling group. Most Voted

Correct Answer: D

Community vote distribution

D (78%) A (20%) 2%

Comments

Konb Highly Voted 3 years, 1 month ago

Selected Answer: D

I was tempted to pick A but then I realized there are two key requirements:
- scale seamlessly
- cost-effectively

None of A-C give seamless scalability. A and B are about adding second instance (which I assume does not match to "scale seamlessly"). C is about changing instance type.

Therefore D is only workable solution to the scalability requirement

upvoted 21 times

NayeraB 2 years, 3 months ago

But wouldn't RDS scale as well? Also Spot instances seems like a bit of a risky decision

upvoted 2 times

pbpally 3 years, 1 month ago

Yup. Got me too. I picked A, saw D, and then reread the "scale seamlessly" part. D is correct.

upvoted 6 times

hpirnaj 1 year, 5 months ago

Aurora is way more expensive than MySQL. but . I would still pick option D because it will scale seamlessly with ASG and Aurora . A is not scale at al

upvoted 1 times

genny Highly Voted 3 years, 2 months ago

Selected Answer: A

I wouldn't run my website on spot instances. Spot instances might be terminated at any time, and since I need to run analytics application it's not an option for me. And using route 53 for load balancing of 2 instances is an overkill. I go with A.

upvoted 12 times

AZ_Master 2 years, 6 months ago

It is spot fleet - not spot instances. They can include On-Demand instances and can also maintain the target capacity automatically.

A Spot Fleet is a set of Spot Instances and optionally On-Demand Instances that is launched based on criteria that you specify. The Spot Fleet selects the Spot capacity pools that meet your needs and launches Spot Instances to meet the target capacity for the fleet. By default, Spot Fleets are set to maintain target capacity by launching replacement instances after Spot Instances in the fleet are terminated.

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/spot-fleet.html

upvoted 15 times

eb7be10 2 years, 1 month ago

While Spot Fleet is temporary, the Auto-Scaling Group builds it back again and again. D is the best answer.

upvoted 2 times

network_enthusiast Most Recent 1 year, 1 month ago

Selected Answer: D

D is the answer

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: D

Both ec2 and db scale better in D.

upvoted 1 times

satyaammm 1 year, 4 months ago

Selected Answer: D

Only D refers to seamlessly scale ... rest are about increasing the number of instances or the size of the instance.

upvoted 1 times

Rcosmos 1 year, 4 months ago

Selected Answer: D

Conclusão:
A opção D é a solução mais econômica e eficaz porque combina o poder de escalabilidade automática do grupo de Auto Scaling, a economia das instâncias Spot, e a alta performance do Amazon Aurora MySQL. Essa abordagem atende aos requisitos de custo, desempenho e escalabilidade de forma abrangente.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: D

Ans D - Like Konb (1 year, 5 months ago) I almost picked A, but for the same reasons of seamless scaling it has to be D

upvoted 2 times

carlossarmient 1 year, 8 months ago

D aswer is forcing you to use the new "feature" of aws. if you have a performance issue with the ddbb, front and back all in on your ec2 intances. the best way to solve that issue is move de ddbb to RDS and create a new ec2 intances. BUT we need to sell the aws unique service.

upvoted 2 times

PaulGa 1 year, 8 months ago

Selected Answer: D

Ans D - I was tempted with AAns D - I was tempted with A but then thought that's too obvious and scaling might be an issue... so looking at Spot Fleets "Fleets provide the following features and benefits, enabling you to maximize cost savings and optimize availability and performance when running applications on multiple EC2 instances" it has to be Ans D.

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/Fleets.html

upvoted 2 times

bignatov 1 year, 9 months ago

Selected Answer: A

Option A is the best balance between cost-effectiveness and scalability. It allows the application to scale horizontally with minimal changes while ensuring the database is managed and can scale independently, reducing the risk of performance degradation during peak times.

upvoted 1 times

bignatov 1 year, 9 months ago

i think it is not D, because amazon aurora is much expensive and also the spot instances can be interrupted, even they are most cost effective in the requirements is not mentioned that the workload is stateless and can be interrupted.

upvoted 2 times

maryam_sh 1 year, 10 months ago

Selected Answer: A

A, not D > Using Spot Instances with an Auto Scaling group adds complexity and risk to the infrastructure

upvoted 1 times

MatAlves 1 year, 9 months ago

It is not "spot instances", but "spot fleet".

A Spot Fleet is a set of Spot Instances and optionally On-Demand Instances that is launched based on criteria that you specify. The Spot Fleet selects the Spot capacity pools that meet your needs and launches Spot Instances to meet the target capacity for the fleet. By default, Spot Fleets are set to maintain target capacity by launching replacement instances after Spot Instances in the fleet are terminated.

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/spot-fleet.html

upvoted 1 times

jaradat02 1 year, 10 months ago

Selected Answer: D

I don't think D is the optimal solution. We certainly can find other solutions that are more cost effective and fulfill the same requirements, but among the provided options, I think D is the most reasonable.

upvoted 2 times

gt520490aws 1 year, 11 months ago

PHP....

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

I'll pick D because it sounds fun :)

upvoted 4 times

pentium75 2 years, 5 months ago

Selected Answer: D

"Scale seamlessly", none of A-C include scaling at all.

upvoted 2 times

xdkonorek2 2 years, 7 months ago

Selected Answer: D

spot instance receives 2 minutes interruption notice, it should be enough for requests to finish, it's quite unusal for app to run longer requests only option D allow for seamless scaling with autoscaling group

upvoted 2 times

BrijMohan08 2 years, 8 months ago

Selected Answer: B

Option B is a cost-effective choice that combines the benefits of database migration to RDS, horizontal scaling with EC2 instances, and control over traffic distribution with Route 53 weighted routing, making it the best solution for the given requirements.

upvoted 2 times

pentium75 2 years, 5 months ago

But there's no scaling at all in B.

upvoted 4 times

Question #146 Topic 1

A company runs a stateless web application in production on a group of Amazon EC2 On-Demand Instances behind an Application Load Balancer. The application experiences heavy usage during an 8-hour period each business day. Application usage is moderate and steady overnight. Application usage is low during weekends.
The company wants to minimize its EC2 costs without affecting the availability of the application.
Which solution will meet these requirements?

A. Use Spot Instances for the entire workload.

B. Use Reserved Instances for the baseline level of usage. Use Spot instances for any additional capacity that the application needs. **Most Voted**

C. Use On-Demand Instances for the baseline level of usage. Use Spot Instances for any additional capacity that the application needs.

D. Use Dedicated Instances for the baseline level of usage. Use On-Demand Instances for any additional capacity that the application needs.

Correct Answer: B

Community vote distribution

B (64%) C (28%) 8%

Comments

rob74 Highly Voted 3 years, 7 months ago

Selected Answer: B

In the Question is mentioned that it has o Demand instances...so I think is more cheapest reserved and spot

upvoted 26 times

Qjb8m9h Highly Voted 3 years, 6 months ago

Answer is B: Reserved is cheaper than on demand the company has. And it's meet the availabilty (HA) requirement as to spot instance that can be disrupted at any time.
PRICING BELOW.
On-Demand: 0% There's no commitment from you. You pay the most with this option.
Reserved : 40%-60%1-year or 3-year commitment from you. You save money from that commitment.
Spot 50%-90% Ridiculously inexpensive because there's no commitment from the AWS side.

upvoted 13 times

3d54049 Most Recent 1 year ago

Selected Answer: B

Reserved and spot instances are cheaper than on-demand ones.

upvoted 1 times

manal001 1 year, 3 months ago

Selected Answer: C

couldn't decide between B and C, but C seems right. The thing with reserved instances is that you have to commit for 1 or 3 years, which is not mentioned in this use case.

upvoted 2 times

zdi561 1 year, 4 months ago

Selected Answer: C

Though ASG is not mentioned but it is implied. Mixing on-demand and spot is the best to save money. Reserved is expensive because you need to pay on the capacity all times.

upvoted 1 times

satyaammm 1 year, 4 months ago

Selected Answer: B

Since Reserved Instances are cheaper than On-Demand and also Spot Instances are best suited here for additional requitements.
upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: B

Ans B - Reserved guarantees baseline level operation; Spot for peaks - its stateless
upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: B

B is the correct answer, using reserved instances is definitely more cost effective than using on-demand instances.
upvoted 2 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: B

B for sure
upvoted 1 times

ManikRoy 2 years, 1 month ago

Selected Answer: B

Agree with others
upvoted 1 times

pentium75 2 years, 5 months ago

Selected Answer: B

This is a bit unclear, but B seems the best option of the ones given.

Usage is either "heavy" (during the 8 hours), "moderate and steady" (overnight) or "low" (during weekends). So there is always SOME usage, which could be covered by a few Reserved Instances (which would be cheaper than On-Demand Instances).

A - "Spot instances for the entire workload", might 'affect the availability of the application'
B - Seems the best answer
C - More expensive than B
D - Dedicated instances aka dedicated hardware -> very expensive
upvoted 6 times

awsgeek75 2 years, 4 months ago
Agree, very little clarity between B and C but B makes more sense.
upvoted 3 times

HackPack 2 years, 5 months ago
I vote for C:
Please explain me if I am wrong:
If application experiences heavy usage during an 8-hour period each business day and all other time we don't need them? it mean than on-demand price will be only 33% from total cost so saving will be near 66%, more than reserved instances all other load we can cover by spot instances.
So why it not C?
upvoted 1 times

dungtrungpham 2 years, 4 months ago
You got it wrong.
You need the application all the time (24/7) because it says: "moderate and steady overnight, low usage at the weekend", not 8 hours a day
upvoted 4 times

VladanO 2 years, 6 months ago
Selected Answer: C
On-Demand Instances are more appropriate than Reserved Instances because "The application is used heavily for a period of 8 hours every weekday requirements.
upvoted 2 times

rcptryk 2 years, 6 months ago

Selected Answer: C

The answer should be C. Because if reserved is chosen, you have to pay for every hour. I calculate from this pages (if I'm wrong please correct me)
https://aws.amazon.com/ec2/pricing/reserved-instances/pricing/#:~:text=Reserved%20Instances%20provide%20you%20with,instances%20when%20you%20need%20them.
Example: for t4g.nano
Reserved instances (0.003X24X365)+(1.90X12)=49.08
On demand instance (0.0042X8X365)=12.264
it will be added spot instances
upvoted 3 times

pentium75 2 years, 5 months ago

"The baseline level of usage" is the minimum usage that is always there (even at night and during weekends), for THAT you can use Reserved Instance.

upvoted 4 times

Marco_St 2 years, 6 months ago

Selected Answer: B

B, since the application needs to be on 24/7 for business days; on weekends, it can be off at any moment. The question mentions something like 8 hour per business day but!!! this is just for heavy usage, the application is also on during overnight.

upvoted 2 times

Juliez 2 years, 7 months ago

Why it's not A ? the application is "stateless" so it can be interrupted at any moment and the spot option is the cheaper one.

upvoted 2 times

pentium75 2 years, 5 months ago

But there might not be any Spot Instances available and the app would go offline.

upvoted 2 times

vi24 2 years, 2 months ago

The statelessness of a web application does not necessarily mean that it's okay to be interrupted. Statelessness refers to how the application handles requests and manages session data, not its ability to handle interruptions.

upvoted 1 times

StudyAllNite 2 years, 7 months ago

Selected Answer: C

If we assume moderate usage of 8 hours on average every day a week, this should be on demand, since it is not a 24/7 server. There is downtime on the weekends and after the initial 8 hours.

upvoted 2 times

SVDK 2 years, 5 months ago

There is no downtime. The application runs all the time (even weekends). Weekends is the base workload which we cover with reserved instances, the higher workloads during the week is covered by spot instances.

upvoted 2 times

Question #147 Topic 1

A company needs to retain application log files for a critical application for 10 years. The application team regularly accesses logs from the past month for troubleshooting, but logs older than 1 month are rarely accessed. The application generates more than 10 TB of logs per month.
Which storage option meets these requirements MOST cost-effectively?

A. Store the logs in Amazon S3. Use AWS Backup to move logs more than 1 month old to S3 Glacier Deep Archive.

B. Store the logs in Amazon S3. Use S3 Lifecycle policies to move logs more than 1 month old to S3 Glacier Deep Archive. Most Voted

C. Store the logs in Amazon CloudWatch Logs. Use AWS Backup to move logs more than 1 month old to S3 Glacier Deep Archive.

D. Store the logs in Amazon CloudWatch Logs. Use Amazon S3 Lifecycle policies to move logs more than 1 month old to S3 Glacier Deep Archive.

Correct Answer: B

Community vote distribution

B (100%)

Comments

rjam Highly Voted 3 years, 6 months ago

Selected Answer: B

Why not AwsBackup? No Glacier Deep is supported by AWS Backup

https://docs.aws.amazon.com/aws-backup/latest/devguide/s3-backups.html
AWS Backup allows you to backup your S3 data stored in the following S3 Storage Classes:
• S3 Standard
• S3 Standard - Infrequently Access (IA)
• S3 One Zone-IA
• S3 Glacier Instant Retrieval
• S3 Intelligent-Tiering (S3 INT)

upvoted 11 times

tdkcumberland 3 years, 6 months ago

AWS BackUp costs something, setting up S3 LCP doesn't.

upvoted 8 times

cookieMr Highly Voted 2 years, 11 months ago

Selected Answer: B

B is the most cost-effective solution. Storing the logs in S3 and using S3 Lifecycle policies to transition logs older than 1 month to S3 Glacier Deep Archive allows for cost optimization based on data access patterns. Since logs older than 1 month are rarely accessed, moving them to S3 Glacier Deep Archive helps minimize storage costs while still retaining the logs for the required 10-year period.

A is incorrect because using AWS Backup to move logs to S3 Glacier Deep Archive can incur additional costs and complexity compared to using S3 Lifecycle policies directly.

C adds unnecessary complexity and costs by involving CloudWatch Logs and AWS Backup when direct management through S3 is sufficient.

D is incorrect because using S3 Lifecycle policies to move logs from CloudWatch Logs to S3 Glacier Deep Archive is not a valid option. CloudWatch Logs and S3 have separate storage mechanisms, and S3 Lifecycle policies cannot be applied directly to CloudWatch Logs.

upvoted 7 times

satyaammm Most Recent 1 year, 4 months ago

Selected Answer: B

S3 and S3 Lifecycle policies are the most suited here.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: B

Ans B - S3 with Glacier Deep plus Lifecycle management

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: B

S3 Lifecycle policies to the rescue

upvoted 3 times

Mamiololo 3 years, 4 months ago

B is correct..

upvoted 1 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: B

Option B (Store the logs in Amazon S3. Use S3 Lifecycle policies to move logs more than 1-month-old to S3 Glacier Deep Archive) would meet these requirements in the most cost-effective manner.

This solution would allow the application team to quickly access the logs from the past month for troubleshooting, while also providing a cost-effective storage solution for the logs that are rarely accessed and need to be retained for 10 years.

upvoted 2 times

career360guru 3 years, 5 months ago

Selected Answer: B

Option B is most cost effective. Moving logs to Cloudwatch logs may incure additional cost.

upvoted 2 times

Wpcorgan 3 years, 6 months ago

B is correct

upvoted 1 times

ArielSchivo 3 years, 7 months ago

Selected Answer: B

S3 + Glacier is the most cost effective.

upvoted 4 times

Bevemo 3 years, 7 months ago

Selected Answer: B

D works, archive cloudwatch logs to S3 .... but is an additional service to pay for over B.

upvoted 2 times

Aamee 3 years, 6 months ago

CloudWatch logs can't store around 10 TB of data per month I believe so both C and D options are ruled out already.

upvoted 1 times

masetromain 3 years, 7 months ago

Selected Answer: B

https://www.examtopics.com/discussions/amazon/view/80772-exam-aws-certified-solutions-architect-associate-saa-c02/

upvoted 2 times

Question #148 Topic 1

A company has a data ingestion workflow that includes the following components:
An Amazon Simple Notification Service (Amazon SNS) topic that receives notifications about new data deliveries
An AWS Lambda function that processes and stores the data
The ingestion workflow occasionally fails because of network connectivity issues. When failure occurs, the corresponding data is not ingested unless the company manually reruns the job.
What should a solutions architect do to ensure that all notifications are eventually processed?

A. Configure the Lambda function for deployment across multiple Availability Zones.

B. Modify the Lambda function's configuration to increase the CPU and memory allocations for the function.

C. Configure the SNS topic's retry strategy to increase both the number of retries and the wait time between retries.

D. Configure an Amazon Simple Queue Service (Amazon SQS) queue as the on-failure destination. Modify the Lambda function to process messages in the queue. Most Voted

Correct Answer: D

Community vote distribution

D (91%) 9%

Comments

bunnychip Highly Voted 3 years, 7 months ago

Selected Answer: D

ensure that all notifications are eventually processed

upvoted 17 times

Help2023 Highly Voted 3 years, 3 months ago

Selected Answer: D

This is why https://docs.aws.amazon.com/sns/latest/dg/sns-message-delivery-retries.html

upvoted 6 times

satyaammm Most Recent 1 year, 4 months ago

Selected Answer: D

SQS queues are suited here for message buffering.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: D

Ans D - using SQS ensure the data is captured and not lost for processing

upvoted 2 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: D

D for sure

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

Configure an Amazon Simple Queue Service (Amazon SQS) queue as the on-failure destination. Modify the Lambda function to process messages in the queue.

upvoted 5 times

CaoMengde09 3 years, 4 months ago

C is not the right answer since after several retries SNS discard the message which doesn't align with the reqirement. D is the right answer

upvoted 4 times

CaoMengde09 3 years, 4 months ago

Best solution to process failed SNS notifications is using sns-dead-letter-queues (SQS Queue for reprocessing)
https://docs.aws.amazon.com/sns/latest/dg/sns-dead-letter-queues.html

upvoted 4 times

SilentMilli 3 years, 5 months ago

Selected Answer: D

To ensure that all notifications are eventually processed, the best solution would be to configure an Amazon Simple Queue Service (SQS) queue as the on-failure destination for the SNS topic. This will allow the notifications to be retried until they are successfully processed. The Lambda function can then be modified to process messages in the queue, ensuring that all notifications are eventually processed. Option D, "Configure an Amazon Simple Queue Service (Amazon SQS) queue as the on-failure destination. Modify the Lambda function to process messages in the queue," is the correct answer.

upvoted 4 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: D

I choose Option D as the correct answer.

To ensure that all notifications are eventually processed, the solutions architect can set up an Amazon SQS queue as the on-failure destination for the Amazon SNS topic. This way, when the Lambda function fails due to network connectivity issues, the notification will be sent to the queue instead of being lost. The Lambda function can then be modified to process messages in the queue, ensuring that all notifications are eventually processed.

upvoted 5 times

techhb 3 years, 5 months ago

Selected Answer: D

Option D to ensure that all notifications are eventually processed you need to use SQS.

upvoted 3 times

career360guru 3 years, 5 months ago

Selected Answer: C

Option C is right option.
SNS does not have any "On Failure" delivery destination. One need to configure dead-letter queue and configure SQS to read from there. So given this option D is incorrect.

upvoted 2 times

JayBee65 3 years, 5 months ago

I don't think that's right
"A dead-letter queue is an Amazon SQS queue that an Amazon SNS subscription can target for messages that can't be delivered to subscribers successfully. Messages that can't be delivered due to client errors or server errors are held in the dead-letter queue for further analysis or reprocessing" from https://docs.aws.amazon.com/sns/latest/dg/sns-dead-letter-queues.html.
This is pretty much what is being described in D.
Plus C will only retry message processing, and network problems could still prevent the message from being processed, but the question states "ensure that all notifications are eventually processed". So C does not meet the requirements but D does look to do this.

upvoted 8 times

Gajendr 2 years, 5 months ago

+ https://docs.aws.amazon.com/sns/latest/dg/sns-message-delivery-retries.html
""To keep the message after the retries specified in the delivery policy are exhausted, configure your subscription to move undeliverables messages to a dead-letter queue (DLQ). For more information"" So D

upvoted 1 times

NikaCZ 3 years, 5 months ago

Selected Answer: D

Is correct.

upvoted 1 times

NikaCZ 3 years, 5 months ago

If you want to ensure that all notifications are eventually processed you need to use SQS.

upvoted 2 times

Wajif 3 years, 6 months ago

Selected Answer: D

C isnt specific. Hence D

upvoted 1 times

LeGloupier 3 years, 6 months ago

Selected Answer: C

"on-failure destination" doesn't exist, only dead letter queue exist.
that's why I am leaning for C
upvoted 1 times

Wajif 3 years, 6 months ago
Dead letter queue doesnt exist in SNS. They are specifically saying a new queue will be configured for failures from SNS. Hence D
upvoted 3 times

Wpcorgan 3 years, 6 months ago
D is correct
upvoted 1 times

Question #149 *Topic 1*

A company has a service that produces event data. The company wants to use AWS to process the event data as it is received. The data is written in a specific order that must be maintained throughout processing. The company wants to implement a solution that minimizes operational overhead.
How should a solutions architect accomplish this?

A. Create an Amazon Simple Queue Service (Amazon SQS) FIFO queue to hold messages. Set up an AWS Lambda function to process messages from the queue. **Most Voted**

B. Create an Amazon Simple Notification Service (Amazon SNS) topic to deliver notifications containing payloads to process. Configure an AWS Lambda function as a subscriber.

C. Create an Amazon Simple Queue Service (Amazon SQS) standard queue to hold messages. Set up an AWS Lambda function to process messages from the queue independently.

D. Create an Amazon Simple Notification Service (Amazon SNS) topic to deliver notifications containing payloads to process. Configure an Amazon Simple Queue Service (Amazon SQS) queue as a subscriber.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Buruguduystunstugudunstuy Highly Voted 3 years, 5 months ago

Selected Answer: A

The correct solution is Option A. Creating an Amazon Simple Queue Service (Amazon SQS) FIFO queue to hold messages and setting up an AWS Lambda function to process messages from the queue will ensure that the event data is processed in the correct order and minimize operational overhead.

Option B is incorrect because using Amazon Simple Notification Service (Amazon SNS) does not guarantee the order in which messages are delivered.

Option C is incorrect because using an Amazon SQS standard queue does not guarantee the order in which messages are processed.

Option D is incorrect because using an Amazon SQS queue as a subscriber to an Amazon SNS topic does not guarantee the order in which message: are processed.

upvoted 6 times

alect096 Highly Voted 3 years, 5 months ago

Selected Answer: A

"The data is written in a specific order that must be maintained throughout processing" --> FIFO

upvoted 5 times

Dharmarajan Most Recent 1 year, 4 months ago

Selected Answer: A

Bst wouldbe to set up a SQS FIFO queue! anytime "Order" is important, FIFO rules. Other options dont have ordering as a priority and they dont guarantee order.

upvoted 1 times

satyaammm 1 year, 4 months ago

Selected Answer: A

SQS FIFO queues maintain the order and hence are suitable here.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: A

Ans A - SQS (FIFO) ensures data is processed in the order it is received
upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: A

"The data is written in a specific order that must be maintained throughout processing."
A without reading other options.
upvoted 2 times

theochan 2 years, 4 months ago

Selected Answer: A

Easiest question ever?
upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: A

"specific order" = must be FIFO queue = only mentioned in A
upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

Create an Amazon Simple Queue Service (Amazon SQS) FIFO queue to hold messages. Set up an AWS Lambda function to process messages from the queue.
upvoted 2 times

cookieMr 2 years, 11 months ago

A is the correct solution. By creating an Amazon Simple Queue Service (Amazon SQS) FIFO queue to hold messages and setting up an AWS Lambda function to process messages from the queue, the company can ensure that the order of the event data is maintained throughout processing. SQS FIFO queues guarantee the order of messages and are suitable for scenarios where strict message ordering is required.

B is incorrect because Amazon Simple Notification Service (Amazon SNS) topics are not designed to preserve message order. SNS is a publish-subscribe messaging service and does not guarantee the order of message delivery.

C is incorrect because using an SQS standard queue does not guarantee the order of message processing. SQS standard queues provide high throughput and scale, but they do not guarantee strict message ordering.

D is incorrect because configuring an SQS queue as a subscriber to an SNS topic does not ensure message ordering. SNS topics distribute messages to subscribers independently, and the order of message processing is not guaranteed.
upvoted 5 times

cheese929 3 years, 1 month ago

Selected Answer: A

A is correct. Use FIFO to process in the specific order required
upvoted 3 times

WherecanIstart 3 years, 3 months ago

Selected Answer: A

Option A is correct...data is processed in the correct order
upvoted 2 times

techhb 3 years, 5 months ago

Only A is right option here.
upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: A

Option A is the best option.
upvoted 2 times

NikaCZ 3 years, 5 months ago

Selected Answer: A

specific order = FIFO
upvoted 2 times

k1kavi1 3 years, 5 months ago

Selected Answer: A

A is correct
upvoted 1 times

david76x 3 years, 6 months ago

Selected Answer: A

Definitely A

upvoted 1 times

Question #150 *Topic 1*

A company is migrating an application from on-premises servers to Amazon EC2 instances. As part of the migration design requirements, a solutions architect must implement infrastructure metric alarms. The company does not need to take action if CPU utilization increases to more than 50% for a short burst of time. However, if the CPU utilization increases to more than 50% and read IOPS on the disk are high at the same time, the company needs to act as soon as possible. The solutions architect also must reduce false alarms.
What should the solutions architect do to meet these requirements?

A. Create Amazon CloudWatch composite alarms where possible. **Most Voted**

B. Create Amazon CloudWatch dashboards to visualize the metrics and react to issues quickly.

C. Create Amazon CloudWatch Synthetics canaries to monitor the application and raise an alarm.

D. Create single Amazon CloudWatch metric alarms with multiple metric thresholds where possible.

Correct Answer: A

Community vote distribution

A (99%)

Comments

123jhl0 Highly Voted 3 years, 7 months ago

Selected Answer: A

Composite alarms determine their states by monitoring the states of other alarms. You can **use composite alarms to reduce alarm noise**. For example, you can create a composite alarm where the underlying metric alarms go into ALARM when they meet specific conditions. You then can set up your composite alarm to go into ALARM and send you notifications when the underlying metric alarms go into ALARM by configuring the underlying metric alarms never to take actions. Currently, composite alarms can take the following actions:
https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/Create_Composite_Alarm.html

upvoted 32 times

cookieMr Highly Voted 2 years, 11 months ago

Selected Answer: A

By creating composite alarms in CloudWatch, the solutions architect can combine multiple metrics, such as CPU utilization and read IOPS, into a single alarm. This allows the company to take action only when both conditions are met, reducing false alarms and focusing on meaningful alerts.

B can help in monitoring the overall health and performance of the application. However, it does not directly address the specific requirement of triggering an action when CPU utilization and read IOPS exceed certain thresholds simultaneously.

C. Creating CloudWatch Synthetics canaries is useful for actively monitoring the application's behavior and availability. However, it does not directly address the specific requirement of monitoring CPU utilization and read IOPS to trigger an action.

D. Creating single CloudWatch metric alarms with multiple metric thresholds where possible can be an option, but it does not address the requirement of triggering an action only when both CPU utilization and read IOPS exceed their respective thresholds simultaneously.

upvoted 11 times

bishtr3 Most Recent 1 year, 3 months ago

Selected Answer: A

Rule : ALARM(CPUUtilizationTooHigh) AND ALARM(DiskReadOpsTooHigh)

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: A

Composite metric is required. multiple occurrences of the threshold breach can be configured with Cloudwatch.

upvoted 1 times

zdi561 1 year, 4 months ago

Selected Answer: D

A is not right because you do not need multiple composite alarms. You need a single one. Both A and D work if they are single

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: A

Ans A - Cloudwatch composite alarms

upvoted 2 times

huaze_lei 1 year, 9 months ago

Selected Answer: A

With CloudWatch, you can combine several alarms into one composite alarm to create a summarized, aggregated health indicator over a whole application or group of resources. Composite alarms are alarms that determine their state by monitoring the states of other alarms. You define rules to combine the status of those monitored alarms using Boolean logic.

upvoted 2 times

NikuWithPakiya 2 years, 2 months ago

composite alarms are suited for scenarios where we have to combine the alarms for different metrics or dimensions, rather than for multiple threshold of the same metric. It contradict with option A.

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: A

Composite for multiple conditions like AND/OR combinations
B: This option just made me laugh. Lol, will someone just sit and look at this dashboard?
C: CW Synthetics canaries if for API
D: Single won't monitor multiple metrics

upvoted 3 times

Modulopi 2 years, 8 months ago

Selected Answer: A

A: Composite alarms determine their states by monitoring the states of other alarms. You can use composite alarms to reduce alarm noise. For example, you can create a composite alarm where the underlying metric alarms go into ALARM when they meet specific conditions.

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: A

Composite alarms was designed to handle this scenario.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

The key reasons are:

Composite alarms allow defining alarms with multiple metrics and conditions, like high CPU AND high read IOPS in this case.
Composite alarms can avoid false positives triggered by a single metric spike.
Dashboards help visualize but won't take automated action. Synthetics tests application availability but doesn't address the metrics.
Single metric alarms with multiple thresholds can't correlate across metrics and may still trigger false positives.
Composite alarms allow acting quickly when both CPU and IOPS are high, per the stated need.

upvoted 5 times

Abrar2022 3 years ago

The composite alarm goes into ALARM state only if all conditions of the rule are met.

upvoted 3 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: A

Option A, creating Amazon CloudWatch composite alarms, is correct because it allows the solutions architect to create an alarm that is triggered only when both CPU utilization is above 50% and read IOPS on the disk are high at the same time. This meets the requirement to act as soon as possible if both conditions are met, while also reducing the number of false alarms by ensuring that the alarm is triggered only when both conditions are met.

upvoted 6 times

Buruguduystunstugudunstuy 3 years, 5 months ago

The incorrect solutions are:

In contrast, Option B, creating Amazon CloudWatch dashboards, would not directly address the requirement to trigger an alarm when both CPU utilization is high and read IOPS on the disk are high at the same time. Dashboards can be useful for visualizing metric data and identifying trends, but they do not have the capability to trigger alarms based on multiple metric thresholds.

Option C, using Amazon CloudWatch Synthetics canaries, may not be the best choice for this scenario, as canaries are used for synthetic testing rather than for monitoring live traffic. Canaries can be useful for monitoring the availability and performance of an application, but they may not be the most effective way to monitor the specific metric thresholds and conditions described in this scenario.

upvoted 4 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option D, creating single Amazon CloudWatch metric alarms with multiple metric thresholds, would not allow the solutions architect to create an alarm that is triggered only when both CPU utilization and read IOPS on the disk are high at the same time. Instead, the alarm would be triggered whenever any of the specified metric thresholds are exceeded, which may result in a higher number of false alarms.

upvoted 6 times

BENICE 3 years, 5 months ago

A is correct answer

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: A

Option A

upvoted 1 times

Qjb8m9h 3 years, 6 months ago

The AWS::CloudWatch::CompositeAlarm type creates or updates a composite alarm. When you create a composite alarm, you specify a rule expression for the alarm that takes into account the alarm states of other alarms that you have created. The composite alarm goes into ALARM state only if all conditions of the rule are met.

The alarms specified in a composite alarm's rule expression can include metric alarms and other composite alarms.Using composite alarms can reduce alarm noise.

upvoted 3 times

Question #151 Topic 1

A company wants to migrate its on-premises data center to AWS. According to the company's compliance requirements, the company can use only the ap-northeast-3 Region. Company administrators are not permitted to connect VPCs to the internet.
Which solutions will meet these requirements? (Choose two.)

A. Use AWS Control Tower to implement data residency guardrails to deny internet access and deny access to all AWS Regions except ap-northeast-3. Most Voted

B. Use rules in AWS WAF to prevent internet access. Deny access to all AWS Regions except ap-northeast-3 in the AWS account settings.

C. Use AWS Organizations to configure service control policies (SCPS) that prevent VPCs from gaining internet access. Deny access to all AWS Regions except ap-northeast-3. Most Voted

D. Create an outbound rule for the network ACL in each VPC to deny all traffic from 0.0.0.0/0. Create an IAM policy for each user to prevent the use of any AWS Region other than ap-northeast-3.

E. Use AWS Config to activate managed rules to detect and alert for internet gateways and to detect and alert for new resources deployed outside of ap-northeast-3.

Correct Answer: AC

Community vote distribution

AC (76%) 14% 5%

Comments

Six_Fingered_Jose Highly Voted 3 years, 7 months ago

Selected Answer: AC

agree with A and C
https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scps_examples_vpc.html#example_vpc_2

upvoted 21 times

cookieMr Highly Voted 2 years, 11 months ago

Selected Answer: AC

A. By using Control Tower, the company can enforce data residency guardrails and restrict internet access for VPCs and denies access to all Regions except the required ap-northeast-3 Region.

C. With Organizations, the company can configure SCPs to prevent VPCs from gaining internet access. By denying access to all Regions except ap-northeast-3, the company ensures that VPCs can only be deployed in the specified Region.

Option B is incorrect because using rules in AWS WAF alone does not address the requirement of denying access to all AWS Regions except ap-northeast-3.

Option D is incorrect because configuring outbound rules in network ACLs and IAM policies for users can help restrict traffic and access, but it does not enforce the company's requirement of denying access to all Regions except ap-northeast-3.

Option E is incorrect because using AWS Config and managed rules can help detect and alert for specific resources and configurations, but it does not directly enforce the restriction of internet access or deny access to specific Regions.

upvoted 20 times

Clouddon Most Recent 11 months, 1 week ago

Selected Answer: CE

C. Use AWS Organizations to configure service control policies (SCPs) that prevent VPCs from gaining internet access. Deny access to all AWS Region except ap-northeast-3.

Why it's correct:
Service control policies (SCPs) are the best way to enforce organization-wide restrictions. You can prevent internet access (e.g., restrict ec2:AttachInternetGateway) and explicitly deny access to AWS services or Regions other than ap-northeast-3.

E. Use AWS Config to activate managed rules to detect and alert for internet gateways and to detect and alert for new resources deployed outside of ap-northeast-3.

Why it's correct:
AWS Config lets you monitor compliance. You can use managed rules like:

vpc-internet-gateway-attached to detect unwanted internet connectivity.

cloudtrail-enabled, ec2-instance-region-check, etc., to detect deployments outside ap-northeast-3.
upvoted 1 times

iamroyalty_k 1 year, 3 months ago

Selected Answer: CE
❌ A. Control Tower does not directly prevent internet access; it only provides guardrails, but those can be bypassed in some cases.
✅ C. SCPs (Service Control Policies) in AWS Organizations provide hard restrictions at the account level, making them more enforceable than Control Tower guardrails alone.
✅ E. AWS Config ensures continuous monitoring and alerts, which Control Tower does not provide as effectively.

This ensures strong security controls, compliance enforcement, and real-time monitoring while maintaining AWS best practices.
upvoted 2 times

Skyskilo 1 year, 4 months ago

Selected Answer: CE
Option C provides centralized governance with SCPs, and Option E provides continuous monitoring and alerting for compliance. Together, these solutions meet the requirements of restricting internet access and ensuring usage of only the ap-northeast-3 region.
upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: AC
Ans A, C - Control Tower with Organisations configured. The two go together
upvoted 2 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: AC
AC for sure
upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: AC
B: Irrelevant WAF
D: This is confusing so I'll ignore it.
E: Wrong product
A: Control Tower can have residency guard rails and block internet access.
C: SCP is like a duplicate of A IMHO but it stops admins from circumventing A as Org policies cannot be overridden by admins unless they are org admins.

Too moany assumptions
upvoted 3 times

BrijMohan08 2 years, 8 months ago

Selected Answer: AC
A. Use AWS Control Tower to implement data residency guardrails to deny internet access and deny access to all AWS Regions except ap-northeast-3.
C. Use AWS Organizations to configure service control policies (SCPs) that prevent VPCs from gaining internet access. Deny access to all AWS Region except ap-northeast-3.
upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: AC
Use Control Tower to implement data residency guardrails and Service Control Policies (SCPS) to prevent VPCs from gaining internet access.
upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: AC
AWS Control Tower guardrails and AWS Organizations SCPs provide centralized, automated mechanisms to enforce no internet connectivity for VPCs and restrict Region access to only ap-northeast-3.
upvoted 4 times

Abrar2022 3 years ago

Didn't know that SCPS (Service Control Policies) could be used to deny users internet access. Good to know. Always thought it's got controlling who can and can't access AWS Services.

upvoted 5 times

hicham0101 3 years, 1 month ago

Agree with Aand C
https://aws.amazon.com/blogs/aws/new-for-aws-control-tower-region-deny-and-guardrails-to-help-you-meet-data-residency-requirements/

upvoted 2 times

yallahool 3 years, 2 months ago

I choose C and D.
For control tower, it can't be A because ap-northeast-3 doesn't support it!
Also, in the case of E, it is detection and warning, so it is difficult to prevent internet connection (although the view is a little obscure).

upvoted 1 times

michellemeloc 3 years, 1 month ago

I just check, now it's supported!!!

upvoted 3 times

notacert 3 years, 2 months ago

Selected Answer: AC

A and C

upvoted 1 times

datz 3 years, 2 months ago

Selected Answer: CD

C/D

A - CANNOT BE!!! AWS Control Tower is not available in ap-northeast-3! Check your
B- for sure no
C - SCPS (Service Control Policies)- For sure
D - Deny outbound rule to be place in prod and also IAM Policy to deny Users creating services in AP-Northeast3
E - it creates an alert, which means it happens but an alert is triggered. so I think it's not good either.

upvoted 2 times

darn 3 years, 1 month ago

False, Control Tower is in Osaka NorthEast 3
https://docs.aws.amazon.com/controltower/latest/userguide/region-how.html

upvoted 3 times

Kaireny54 3 years, 2 months ago

Selected Answer: CD

Control tower isn't available in AP-northeast-3 (only available in ap-northeast1 and 2 : https://www.aws-services.info/controltower.html)
For answer E, it creates an alert, wich means it happens but an alert is triggered. so i think it's not good either.
That's why i would go for C and D

upvoted 2 times

darn 3 years, 1 month ago

same page you posted:
ap-northeast-3 Asia Pacific (Osaka) 2023-04-20 https://aws.amazon.com/controltower

upvoted 2 times

Bmarodi 3 years ago

It's availabe now on the same tink u pasted in earlier: ap-northeast-3 Asia Pacific (Osaka) 2023-04-20.

upvoted 2 times

darn 3 years, 1 month ago

False, Control Tower is in Osaka NorthEast 3
https://docs.aws.amazon.com/controltower/latest/userguide/region-how.html

upvoted 2 times

Question #152 Topic 1

A company uses a three-tier web application to provide training to new employees. The application is accessed for only 12 hours every day. The company is using an Amazon RDS for MySQL DB instance to store information and wants to minimize costs.
What should a solutions architect do to meet these requirements?

A. Configure an IAM policy for AWS Systems Manager Session Manager. Create an IAM role for the policy. Update the trust relationship of the role. Set up automatic start and stop for the DB instance.

B. Create an Amazon ElastiCache for Redis cache cluster that gives users the ability to access the data from the cache when the DB instance is stopped. Invalidate the cache after the DB instance is started.

C. Launch an Amazon EC2 instance. Create an IAM role that grants access to Amazon RDS. Attach the role to the EC2 instance. Configure a cron job to start and stop the EC2 instance on the desired schedule.

D. Create AWS Lambda functions to start and stop the DB instance. Create Amazon EventBridge (Amazon CloudWatch Events) scheduled rules to invoke the Lambda functions. Configure the Lambda functions as event targets for the rules. Most Voted

Correct Answer: D

Community vote distribution

D (83%) Other

Comments

study_aws1 Highly Voted 3 years, 7 months ago
https://aws.amazon.com/blogs/database/schedule-amazon-rds-stop-and-start-using-aws-lambda/

It is option D. Option A could have been applicable had it been AWS Systems Manager State Manager & not AWS Systems Manager Session Manager
upvoted 40 times

123jhl0 Highly Voted 3 years, 7 months ago
Selected Answer: A
A is true for sure. "Schedule Amazon RDS stop and start using AWS Systems Manager" Steps in the documentation:
1. Configure an AWS Identity and Access Management (IAM) policy for State Manager.
2. Create an IAM role for the new policy.
3. Update the trust relationship of the role so Systems Manager can use it.
4. Set up the automatic stop with State Manager.
5. Set up the automatic start with State Manager.
https://aws.amazon.com/blogs/database/schedule-amazon-rds-stop-and-start-using-aws-systems-manager/
upvoted 9 times

Kien048 3 years, 7 months ago
Look like State manager and Session manager use for difference purpose even both in same dashboard console.
upvoted 2 times

Kien048 3 years, 7 months ago
And ofcause, D is working, so if A also right, the question is wrong.
upvoted 5 times

Bevemo 3 years, 7 months ago
Agree A, free to use state manager within limits, and don't need to code or manage lambda.
upvoted 1 times

ArielSchivo 3 years, 7 months ago
Option A refers to Session Manager, not State Manager as you pointed, so it is wrong. Option D is valid.
upvoted 13 times

NSA_Poker 2 years ago